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Graham et al.

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(54) **EDGE FLOW ELEMENT FOR ELECTROPLATING APPARATUS**

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(73) Assignee: **Lam Research Corporation**, Fremont, CA (US)

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C25D 5/08 (2006.01)
C25D 17/00 (2006.01)

(52) **U.S. Cl.**
CPC ***C25D 5/08*** (2013.01); ***C25D 17/001*** (2013.01); ***C25D 17/008*** (2013.01)

(58) **Field of Classification Search**
CPC C25D 5/08
See application file for complete search history.

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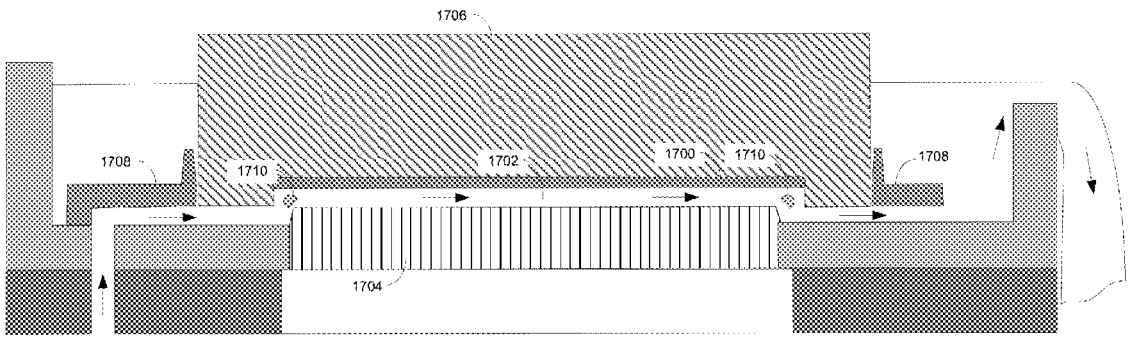
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(57) **ABSTRACT**

The embodiments herein relate to methods and apparatus for electroplating one or more materials onto a substrate. In many cases the material is a metal and the substrate is a semiconductor wafer, though the embodiments are no so limited. Typically, the embodiments herein utilize a channeled plate positioned near the substrate, creating a cross flow manifold defined on the bottom by the channeled plate, on the top by the substrate, and on the sides by a cross flow confinement ring. Also typically present is an edge flow element configured to direct electrolyte into a corner formed between the substrate and substrate holder. During plating, fluid enters the cross flow manifold both upward through the channels in the channeled plate, and laterally through a cross flow side inlet positioned on one side of the cross flow confinement ring. The flow paths combine in the cross flow manifold and exit at the cross flow exit, which is positioned opposite the cross flow inlet. These combined flow paths and the edge flow element result in improved plating uniformity, especially at the periphery of the substrate.

14 Claims, 35 Drawing Sheets



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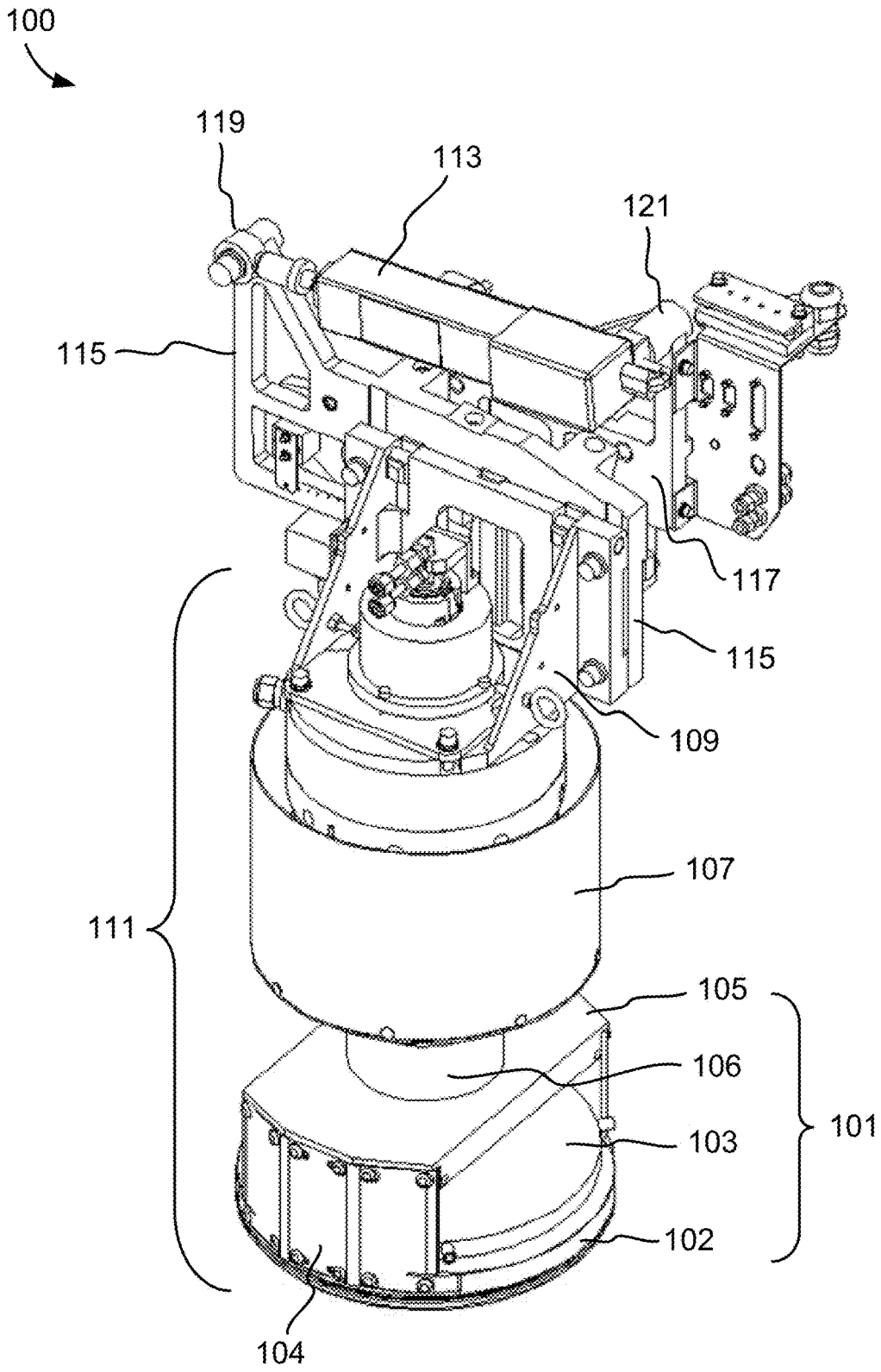


FIG. 1A

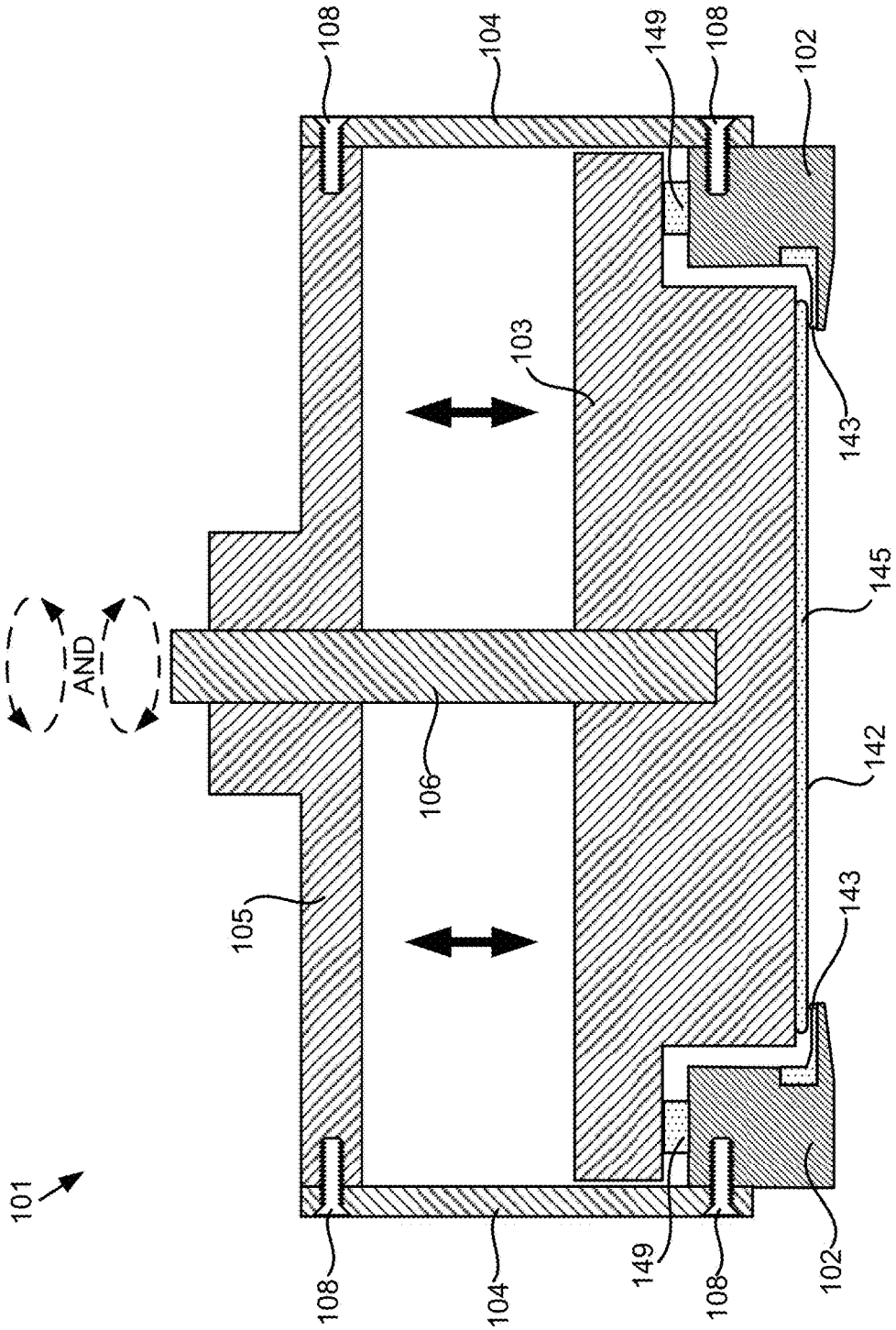

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AND
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FIG. 1B

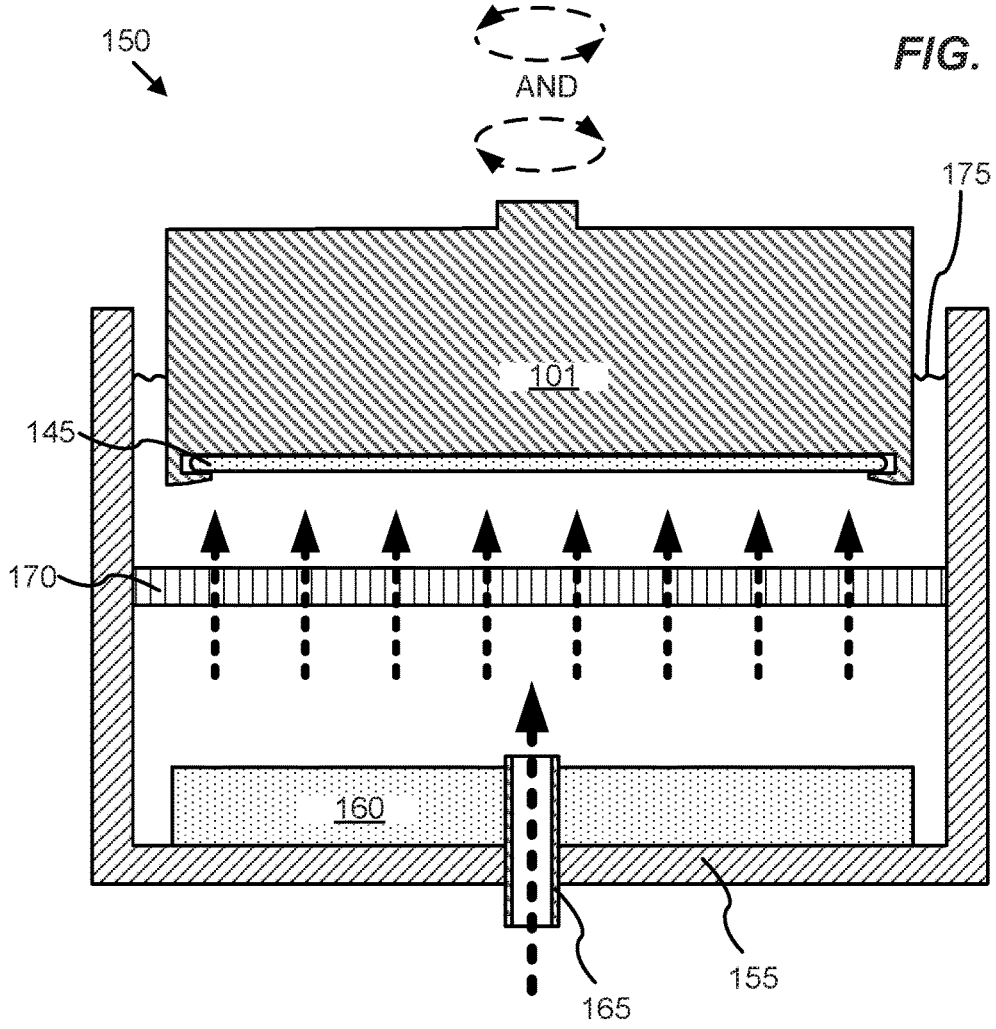


FIG. 1C

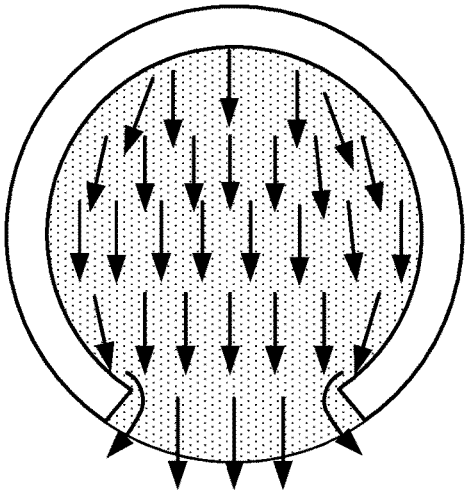


FIG. 1D

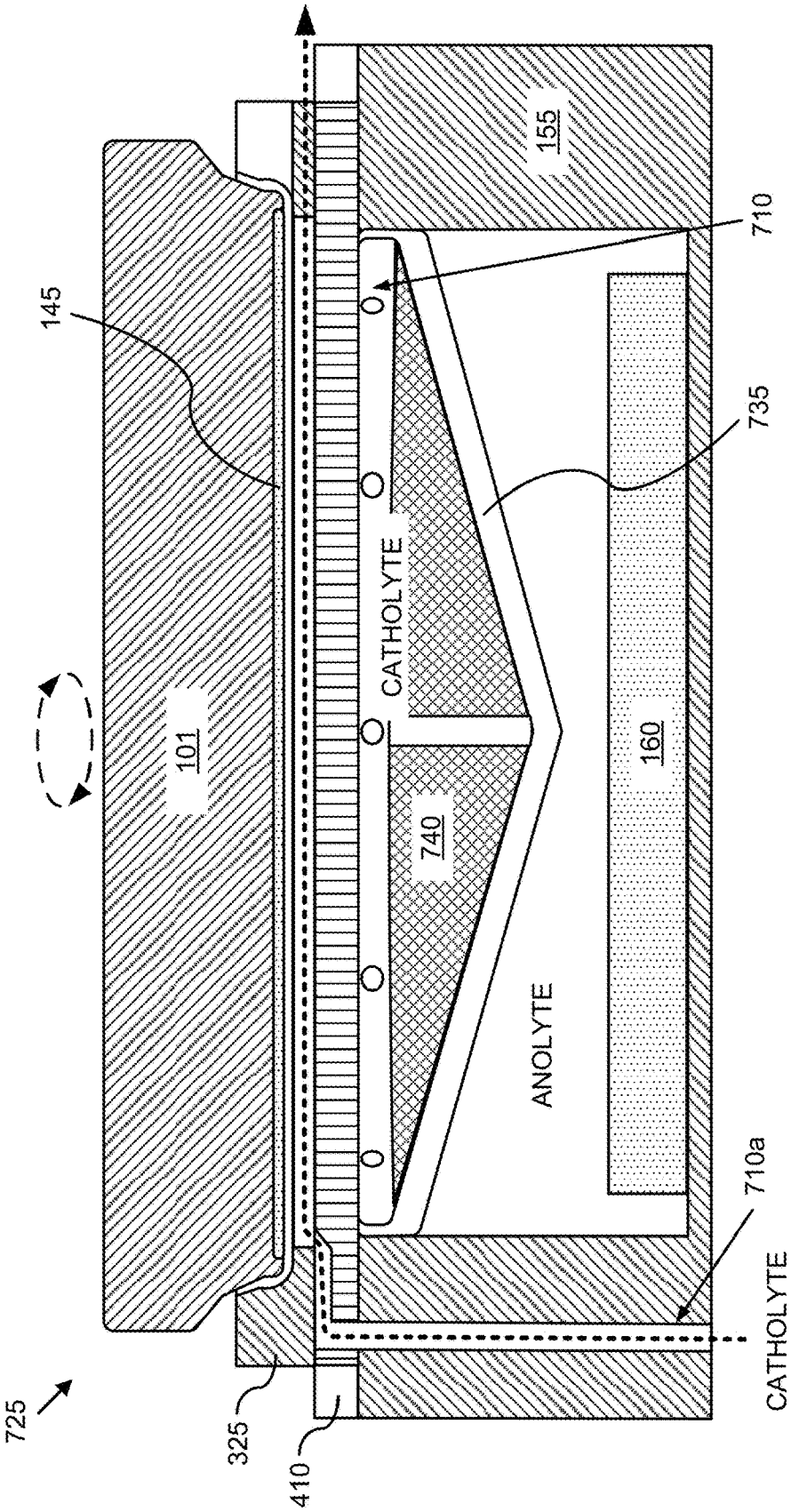

155
710
145
735
CATHOLYTE
101
160
740
ANOLYTE
710a
CATHOLYTE
725
325
410

FIG. 1E

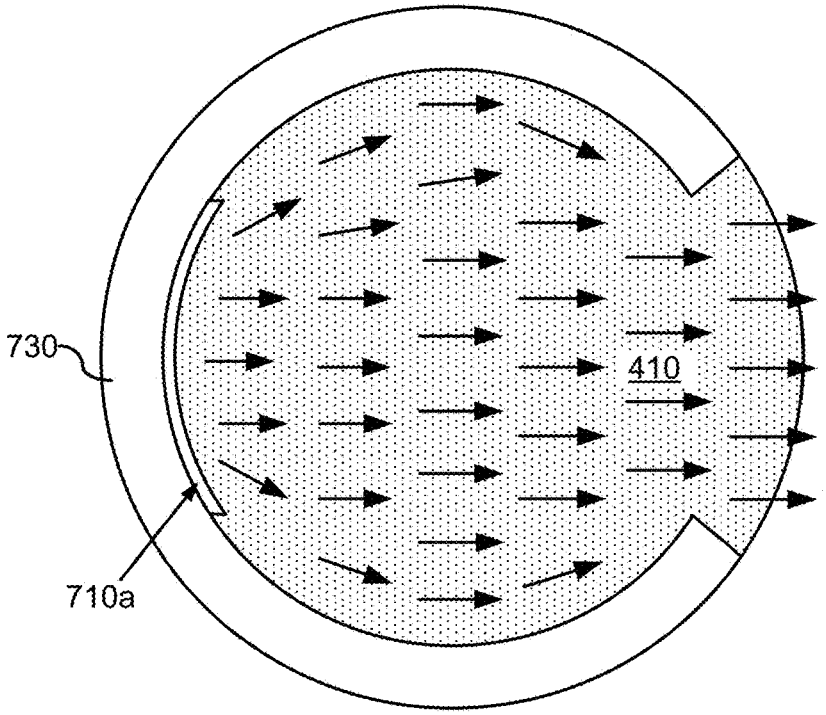


FIG. 1F

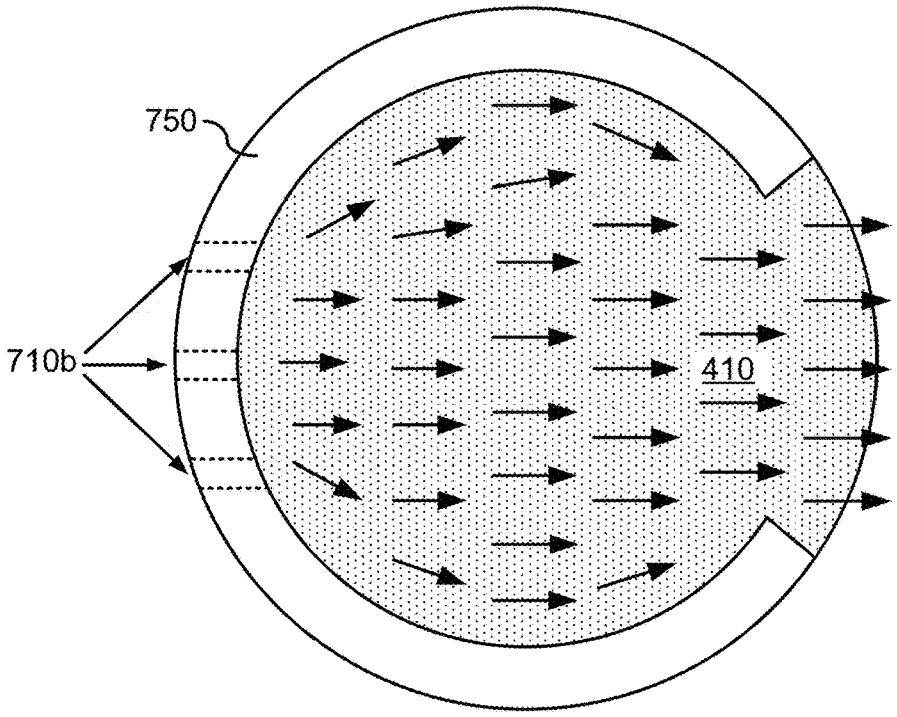


FIG. 1G

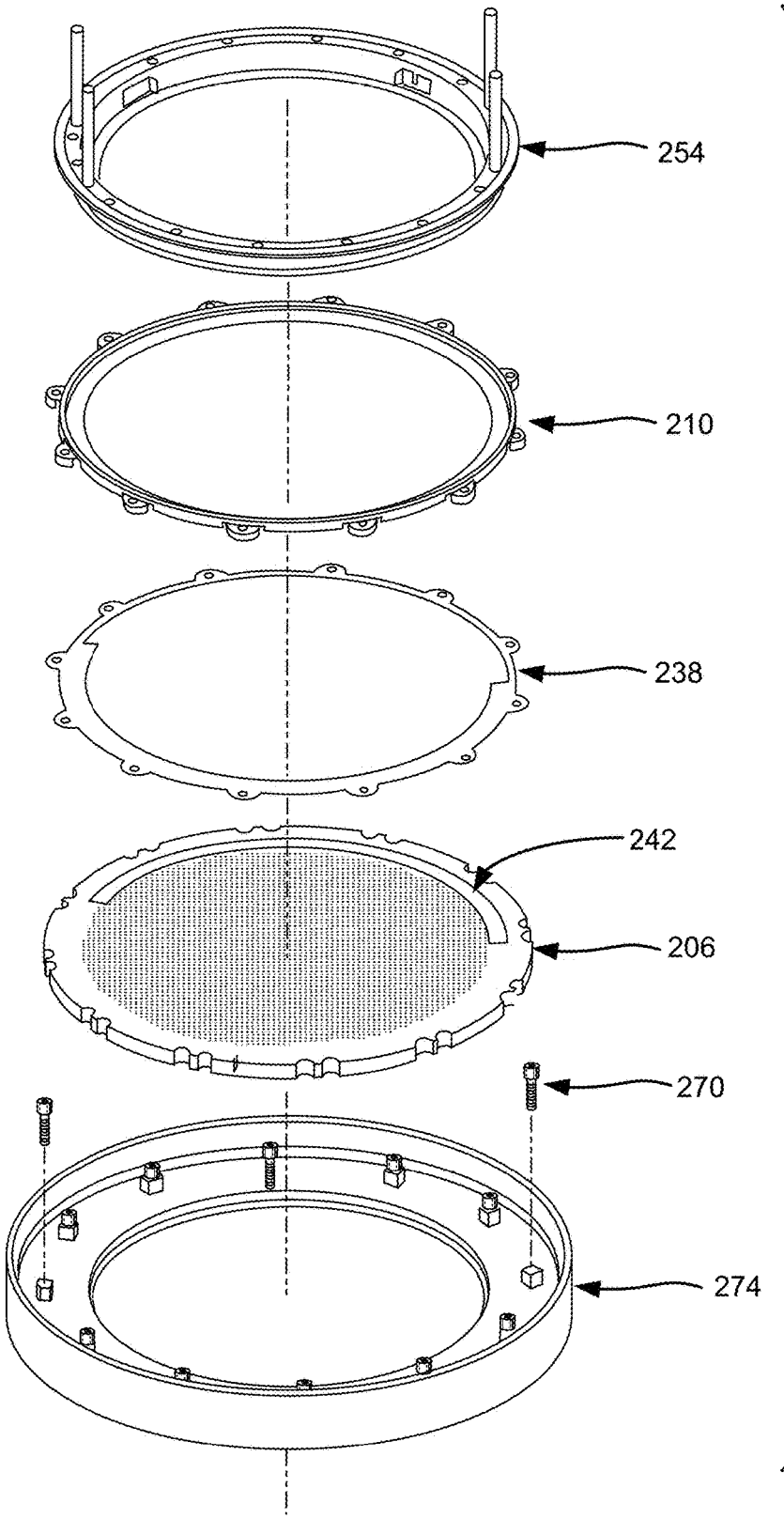


FIG. 2

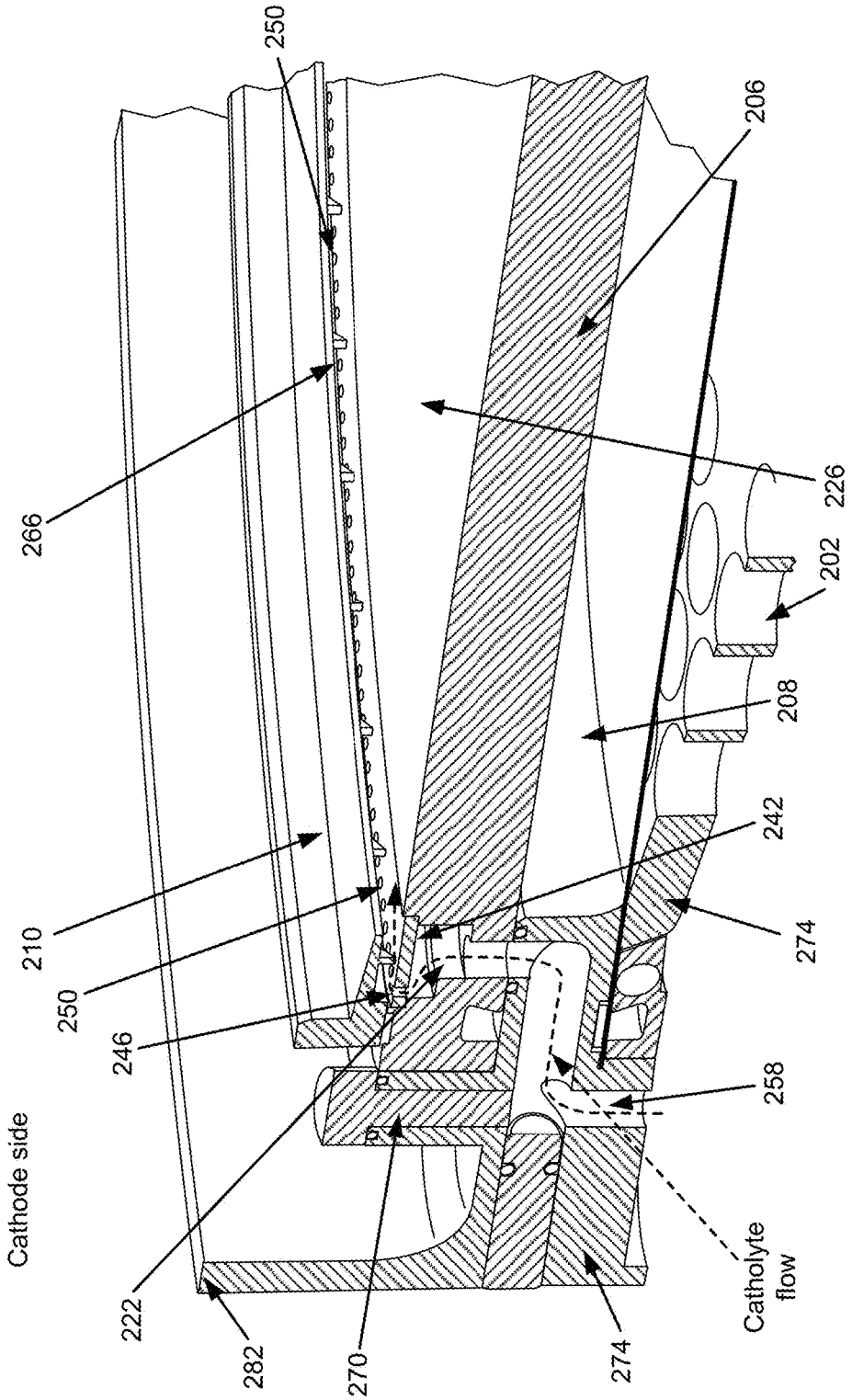

250
206
266
226
202
208
242
210
250
274
246
258
Cathode side
Catholyte flow
Anode side
222
282
270
274

FIG. 3A

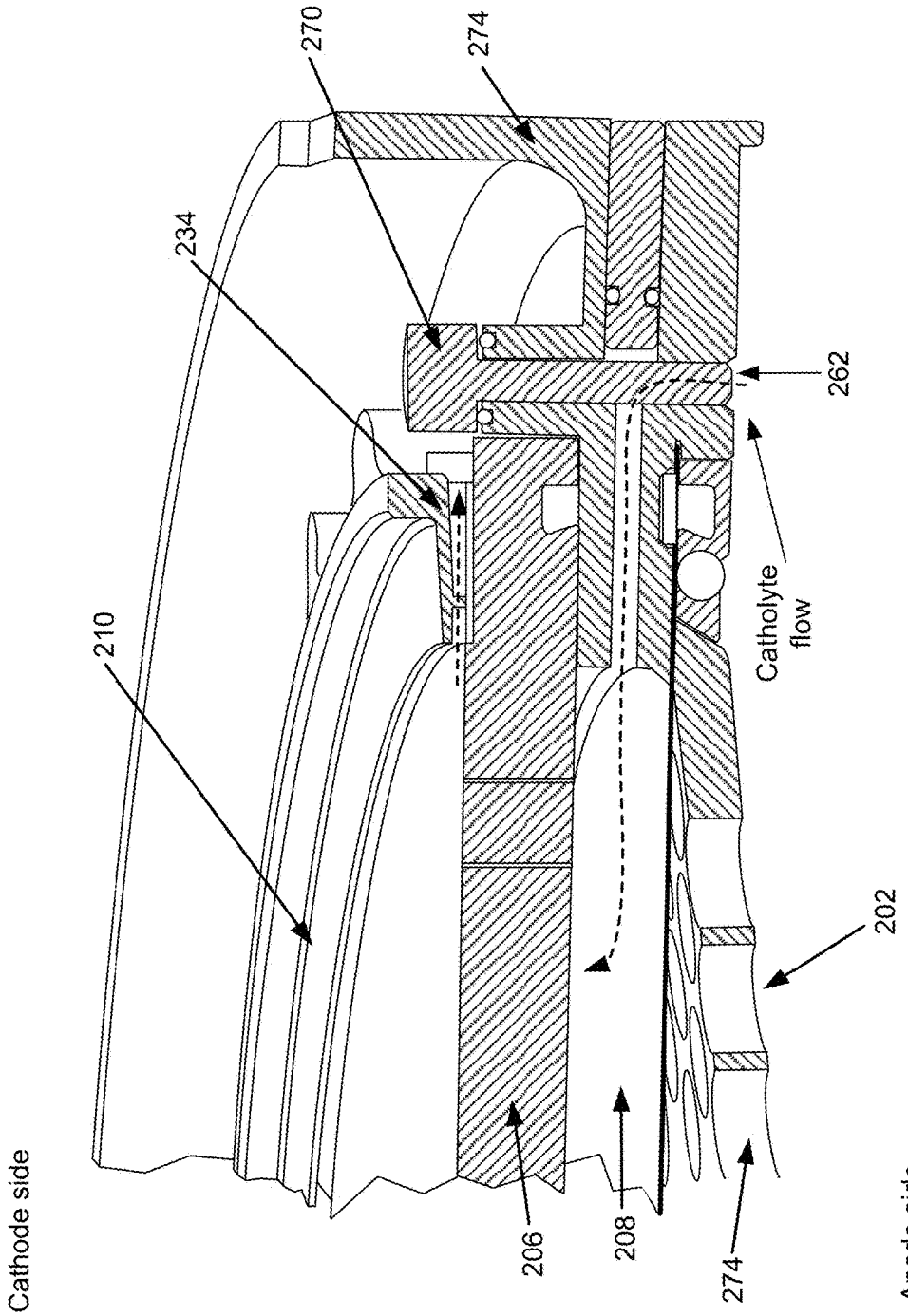

270
274
234
262
210
Catholyte flow
202
206
208
274
Cathode side
Anode side

FIG. 3B

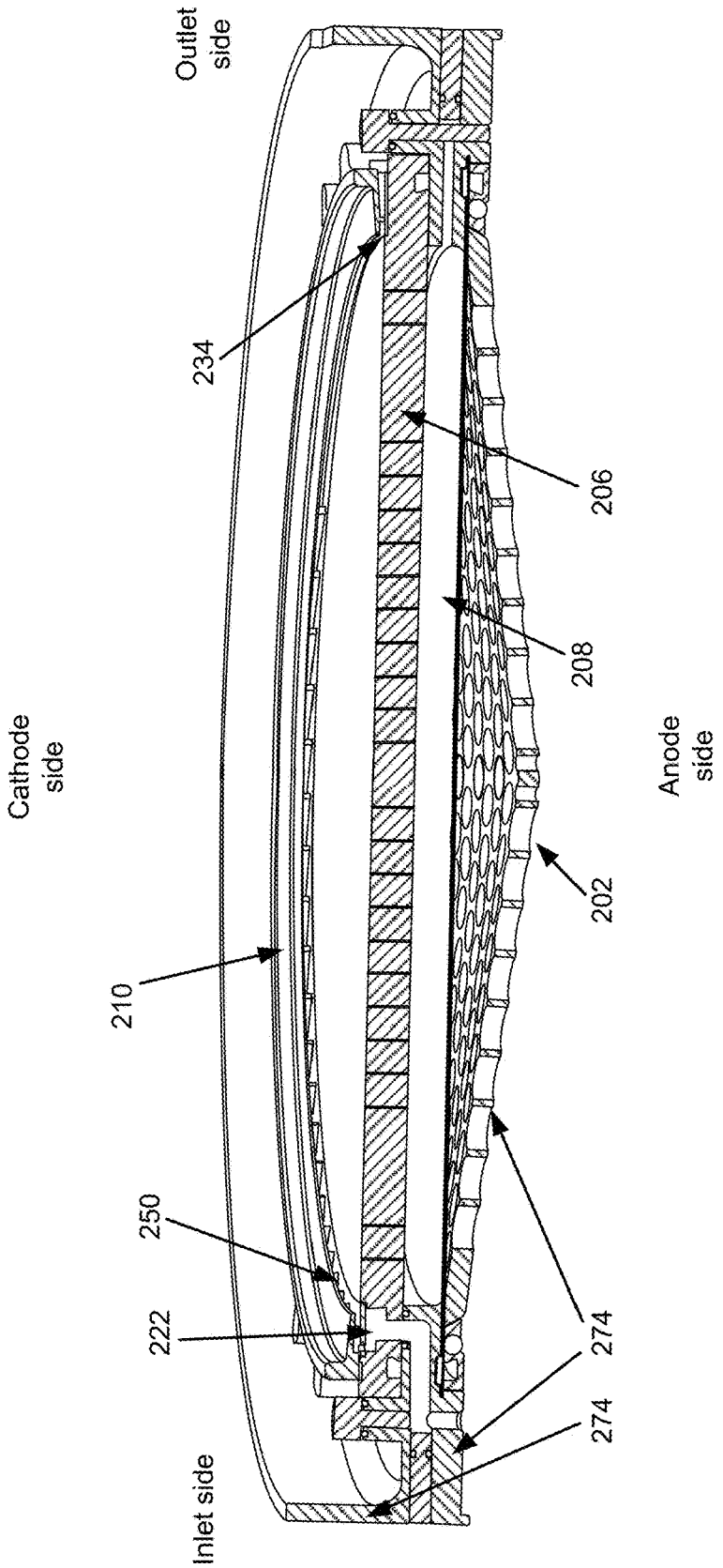

Outlet side
234
206
208
Cathode side
Anode side
202
210
250
222
274
274
Inlet side

FIG. 4

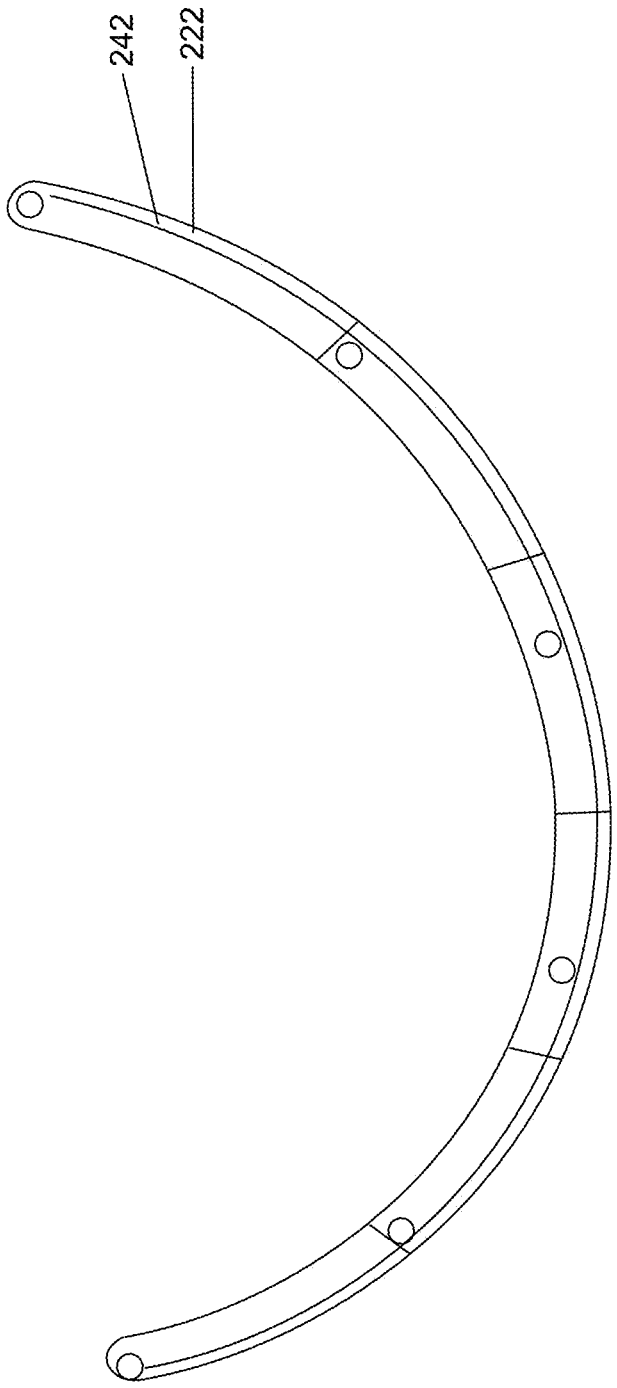


FIG. 5

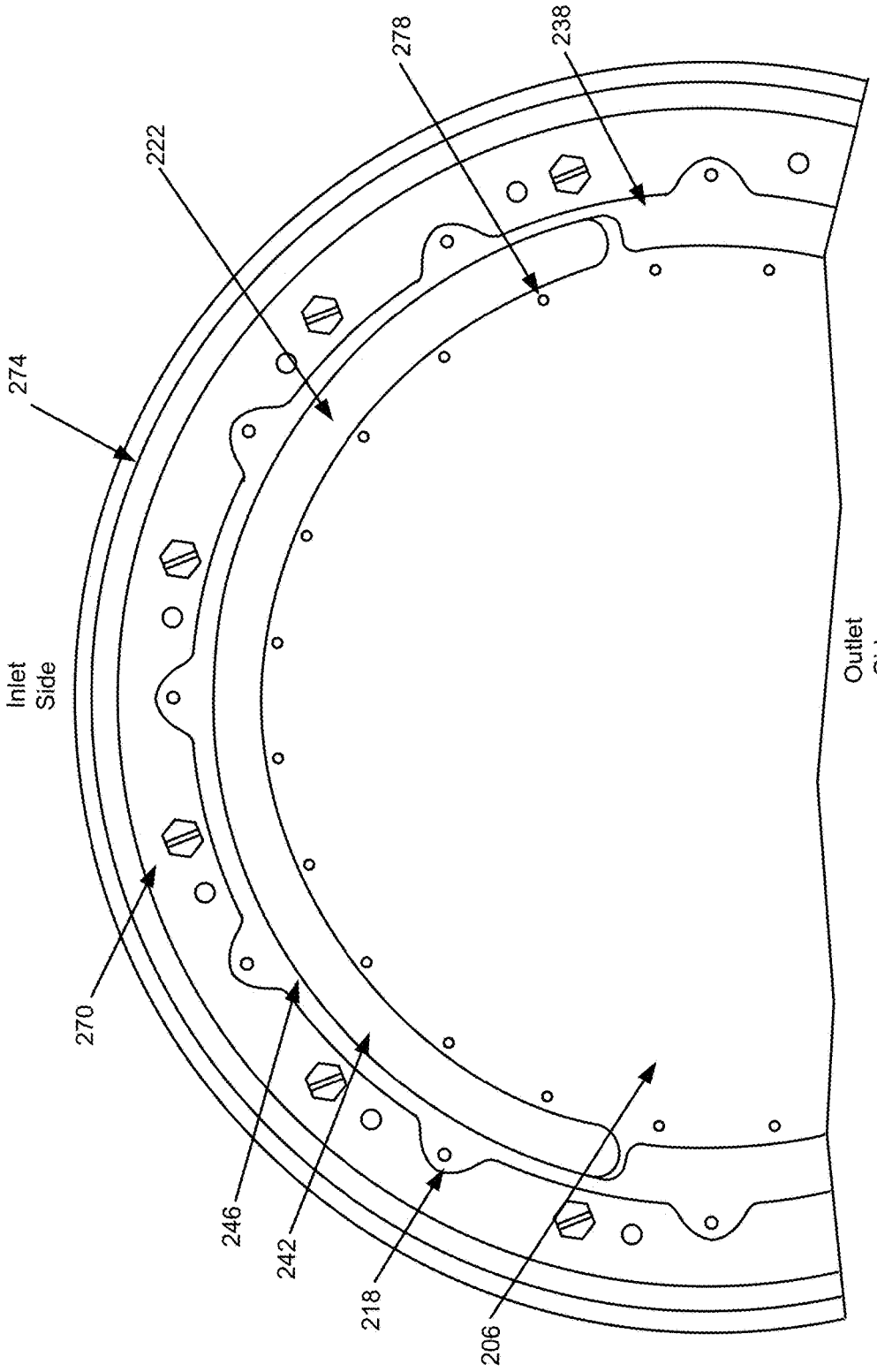


FIG. 6

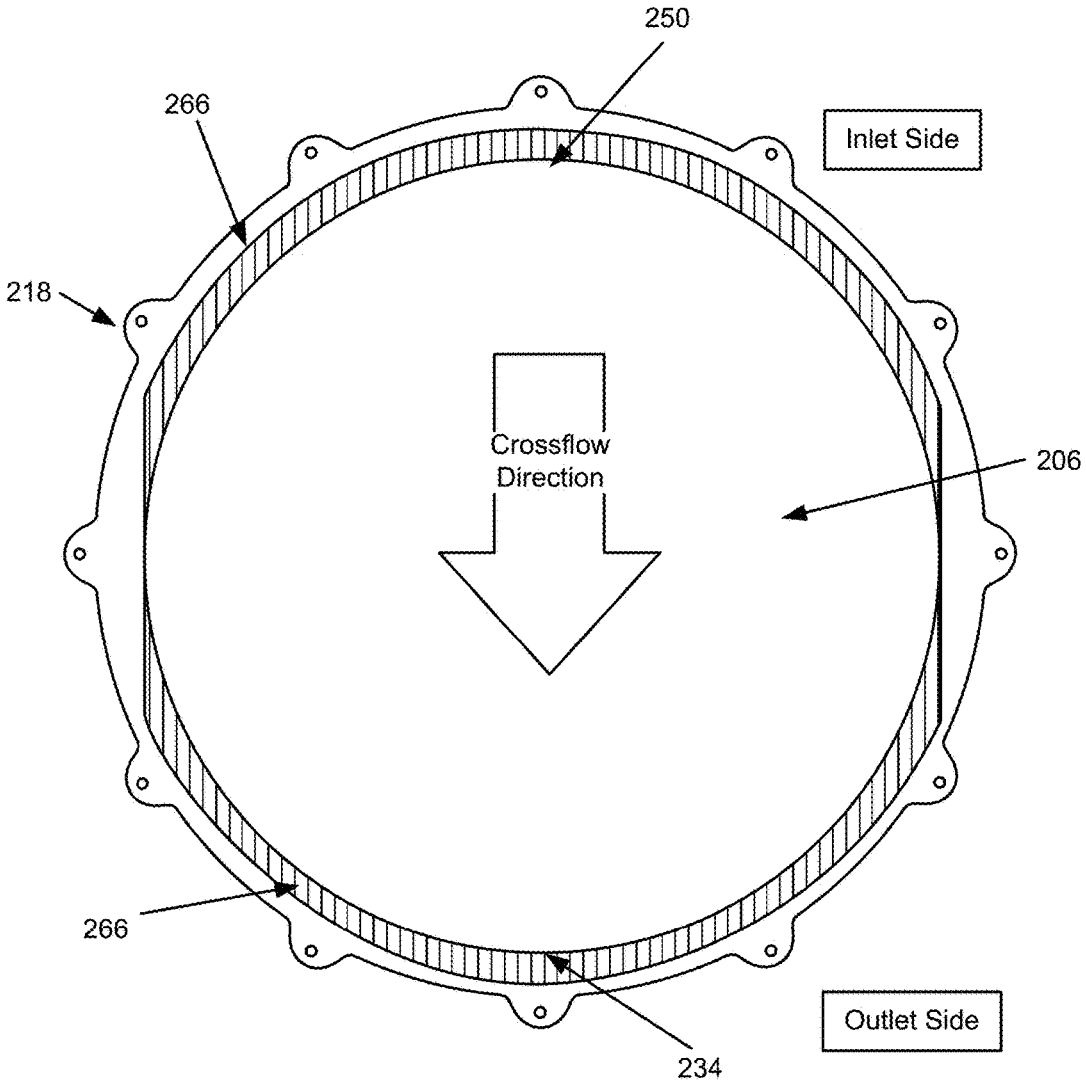

250
266
Inlet Side
218
Crossflow
Direction
206
266
Outlet Side
234

FIG. 7

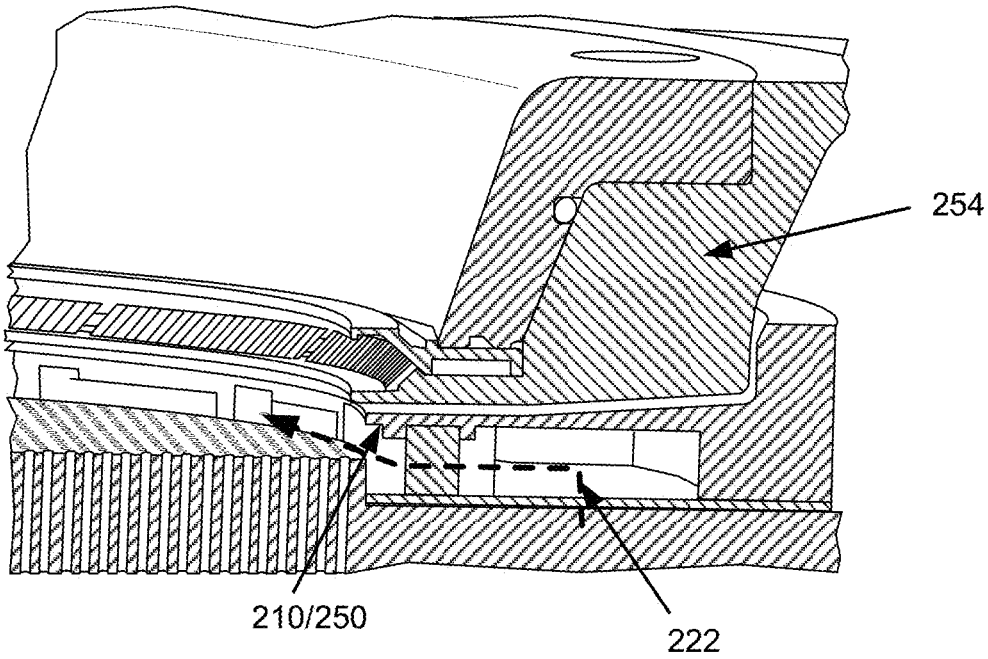


FIG. 8A

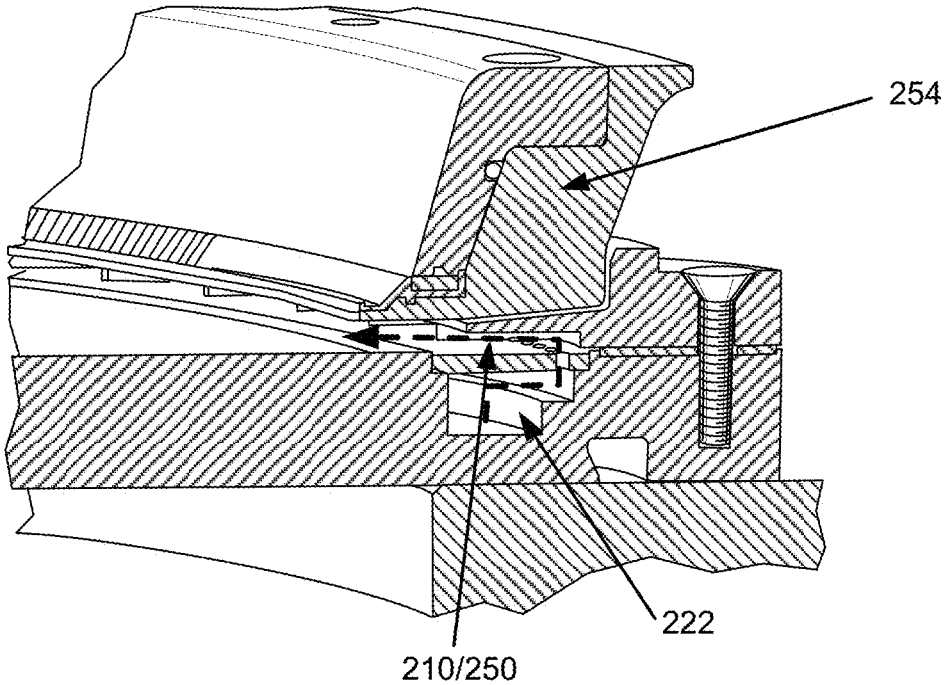


FIG. 8B

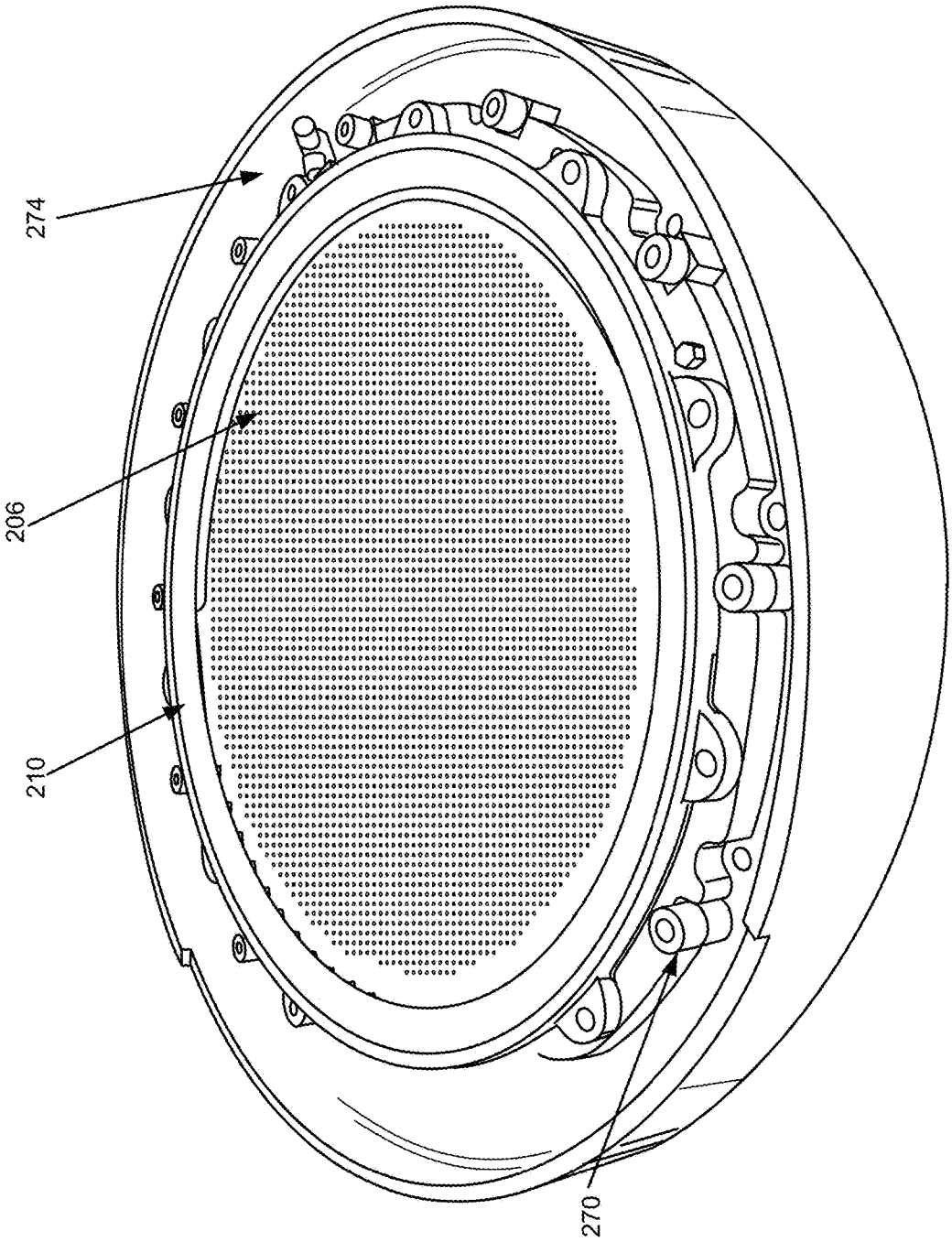


FIG. 9

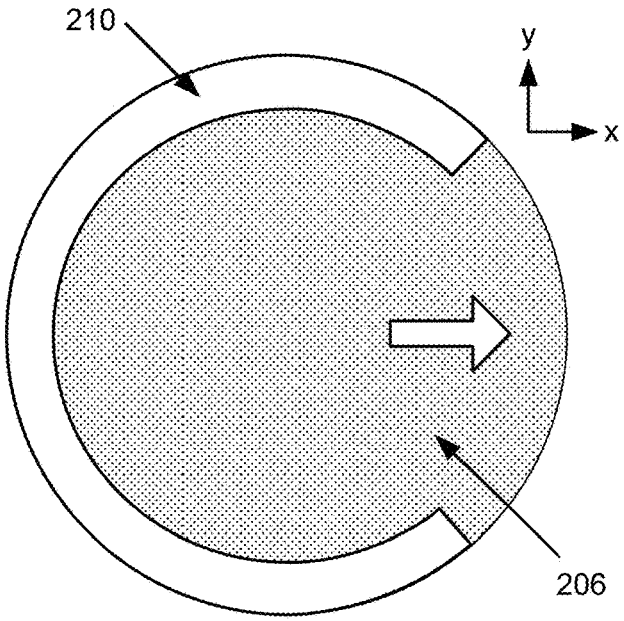


FIG. 10A

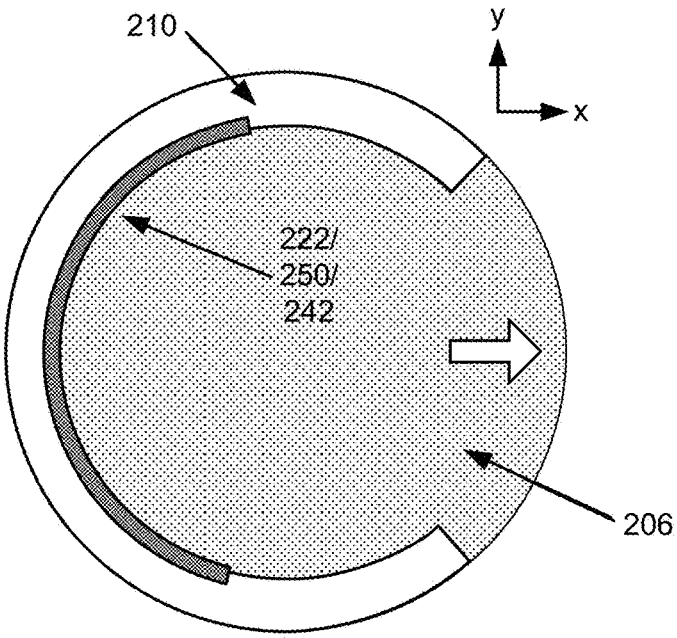


FIG. 10B

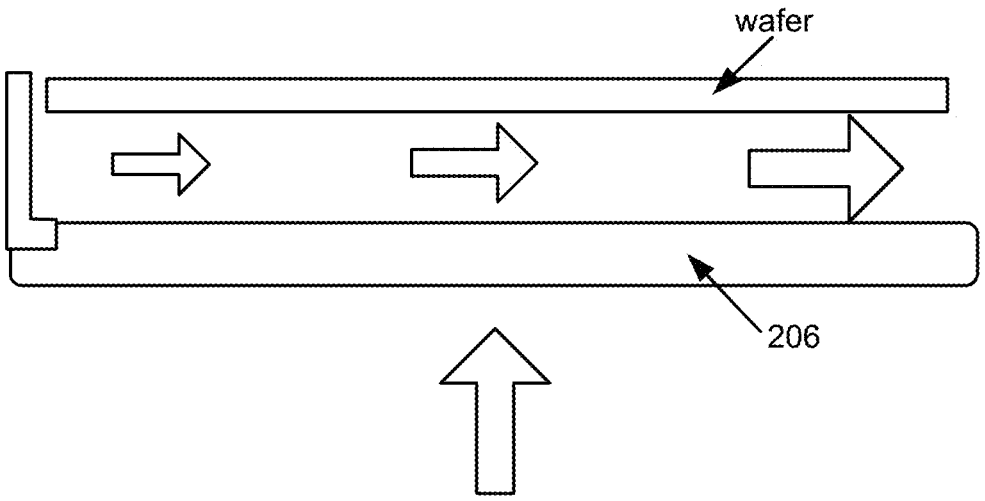


FIG. 11A

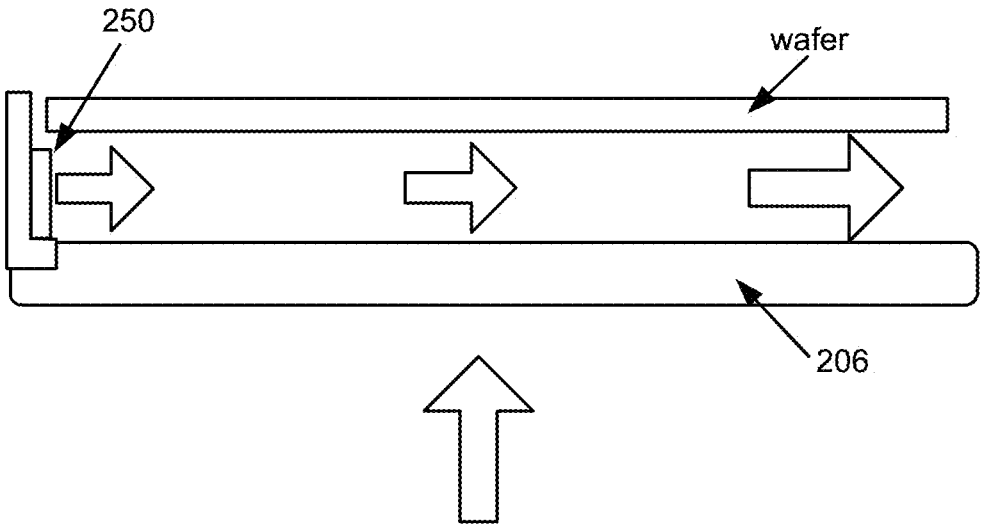


FIG. 11B

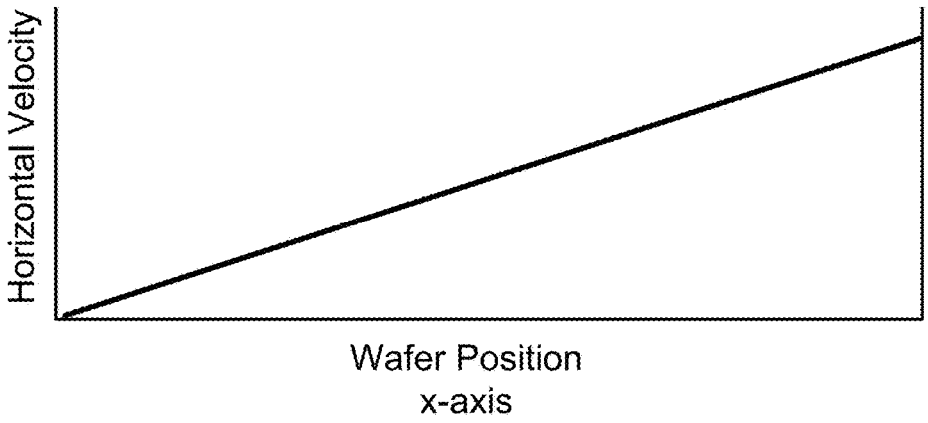


FIG. 12A

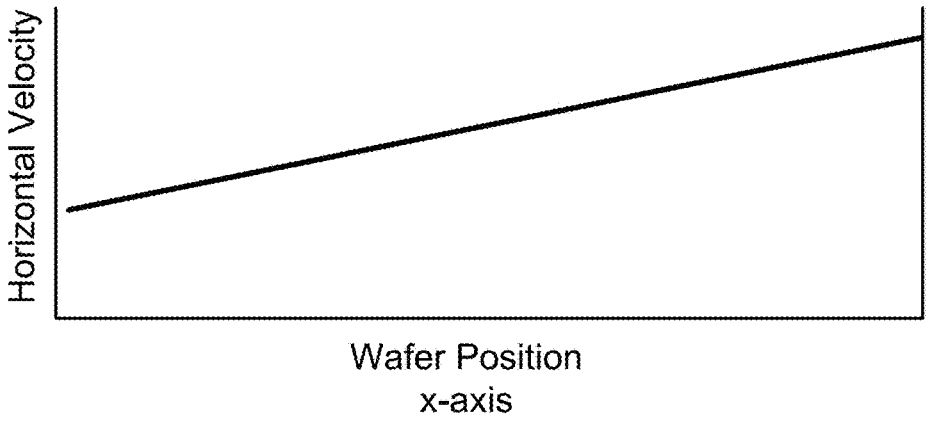


FIG. 12B

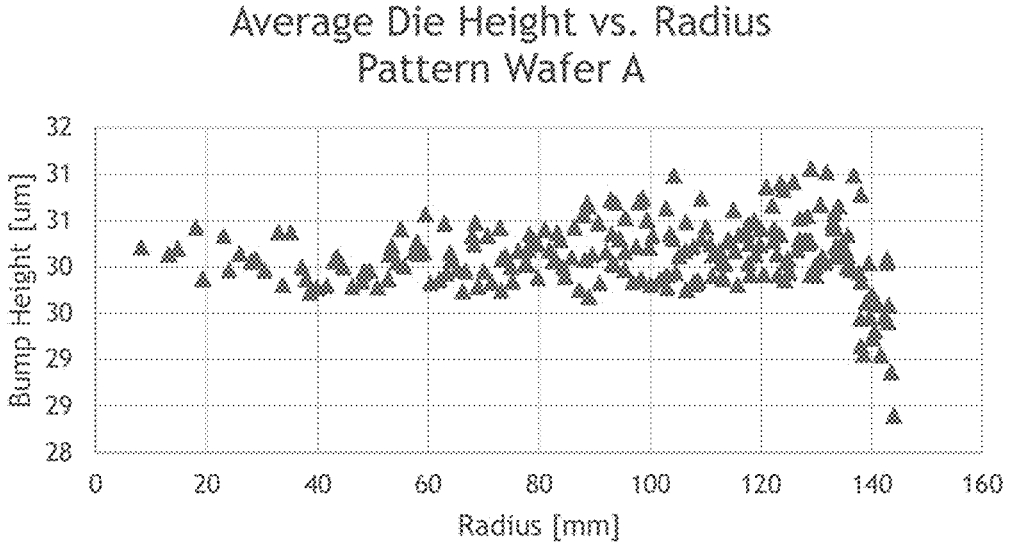


FIG. 13A

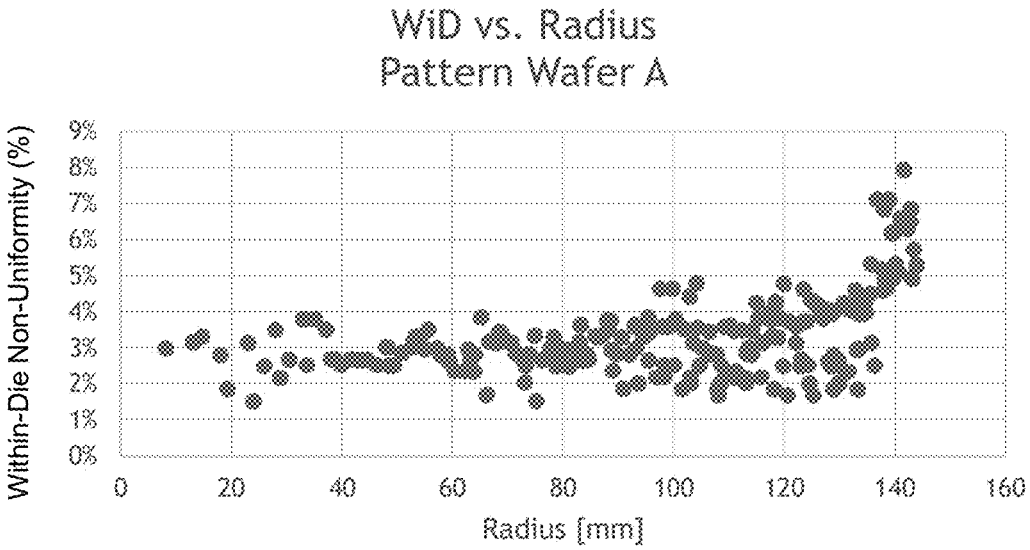


FIG. 13B

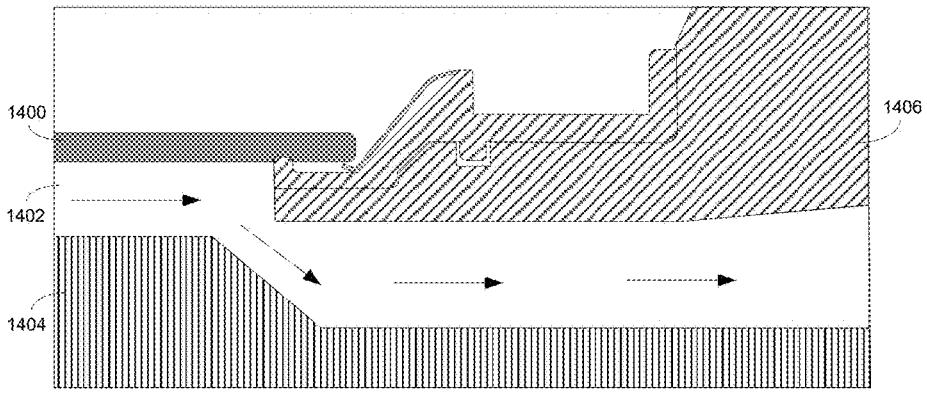


FIG. 14A

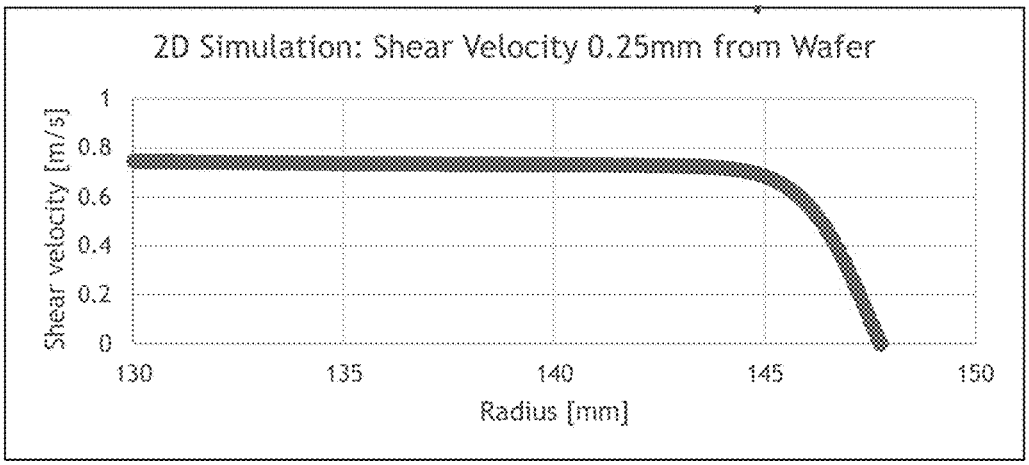


FIG. 14B

FIG. 15

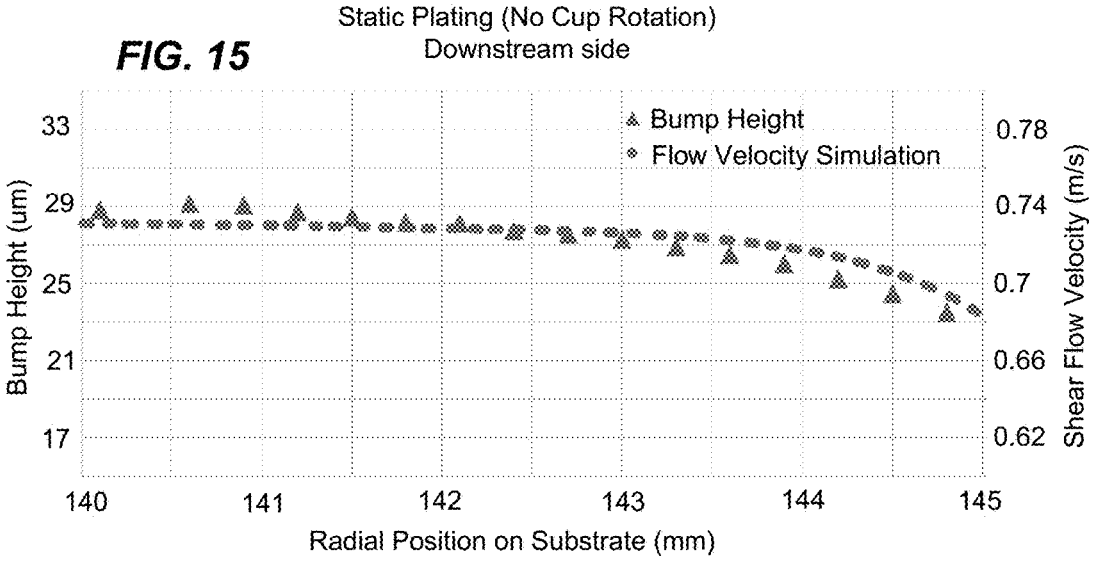


FIG. 16A

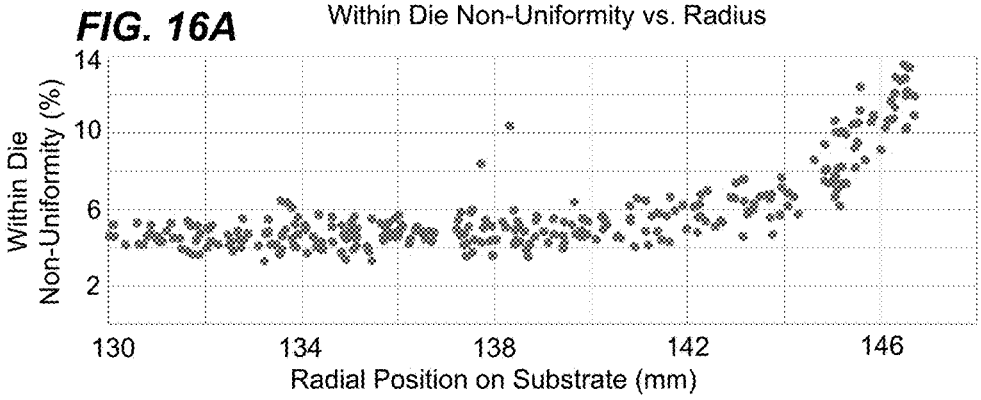
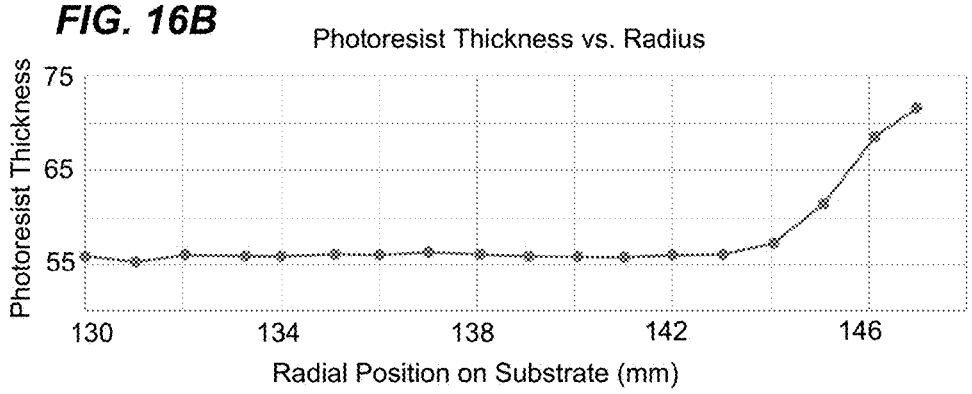


FIG. 16B



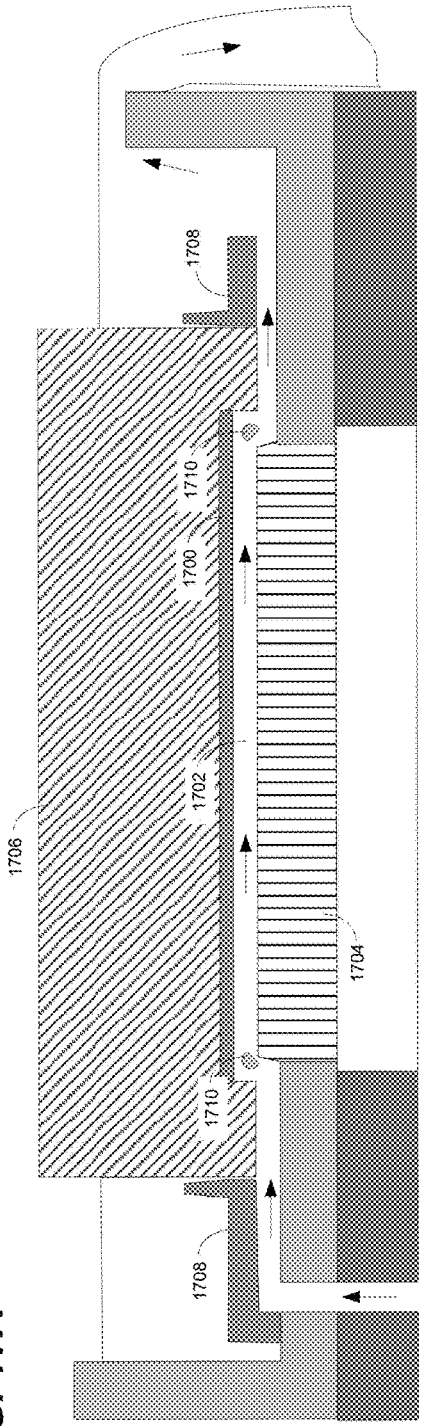


FIG. 17A

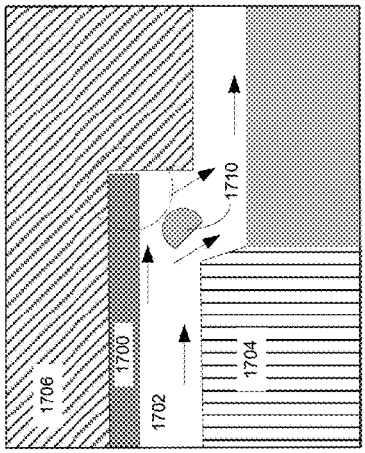


FIG. 17B

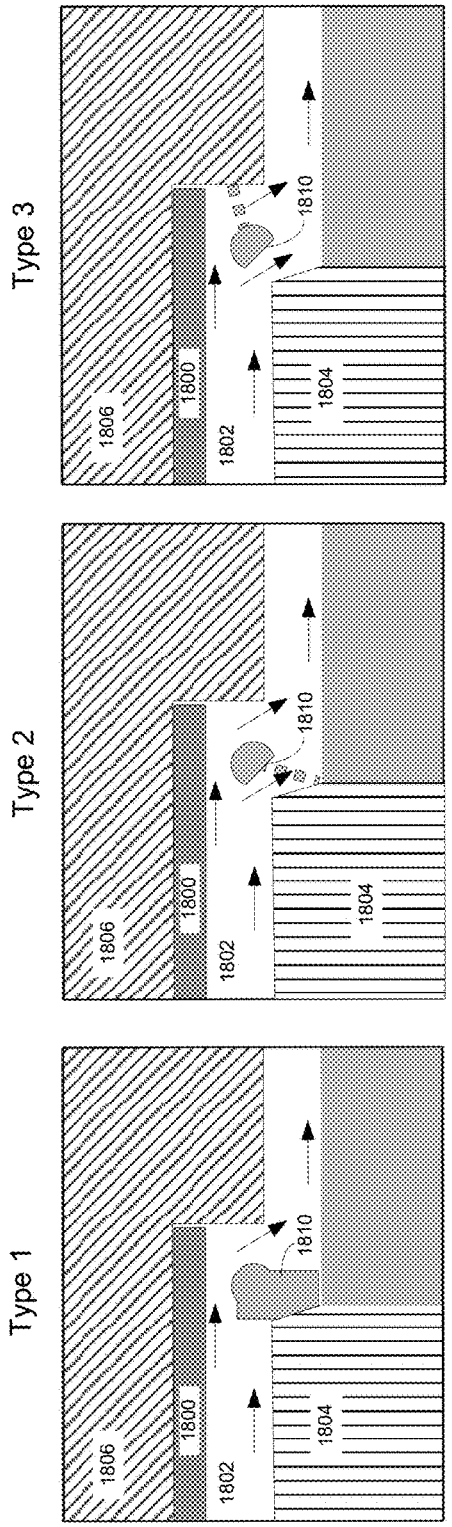


FIG. 18A

FIG. 18B

FIG. 18C

Feature of Edge Flow Element	Type 1	Type 2	Type 3
Attached to	CIRP	CIRP	Substrate Holder
Flow bypass between edge flow element and CIRP	No	Yes	Yes
Can be azimuthally asymmetric	Yes	Yes	No*

FIG. 18D

FIG. 19A

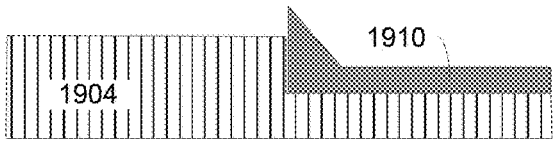


FIG. 19B

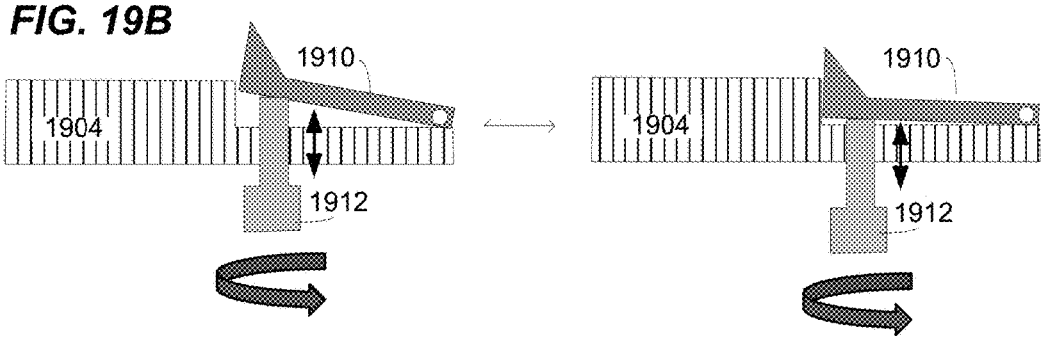


FIG. 19C

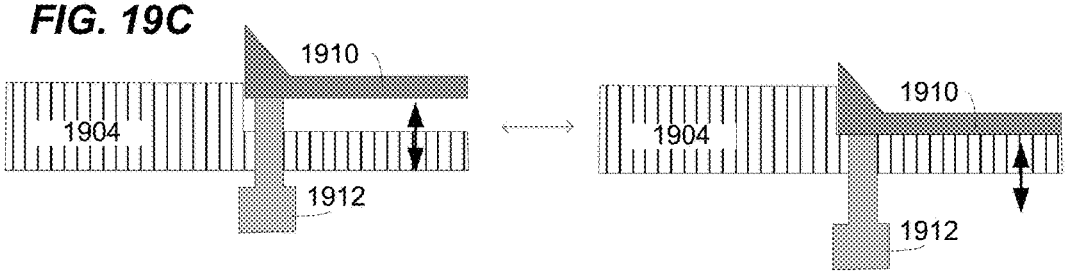


FIG. 19D

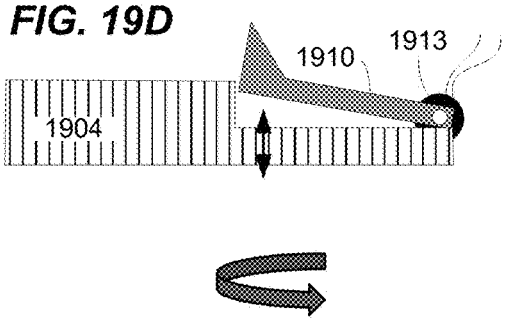
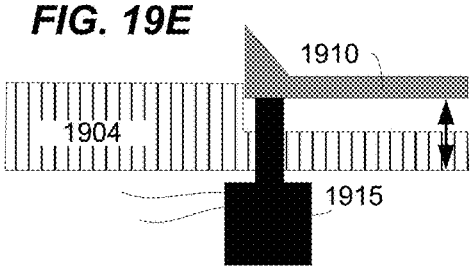


FIG. 19E



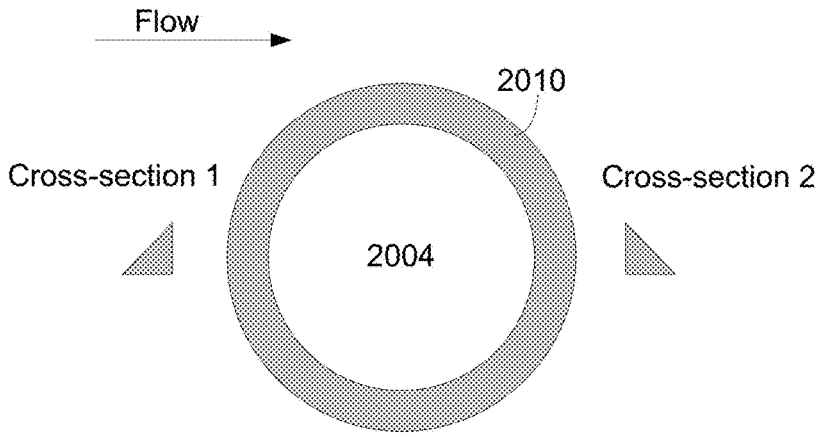


FIG. 20A

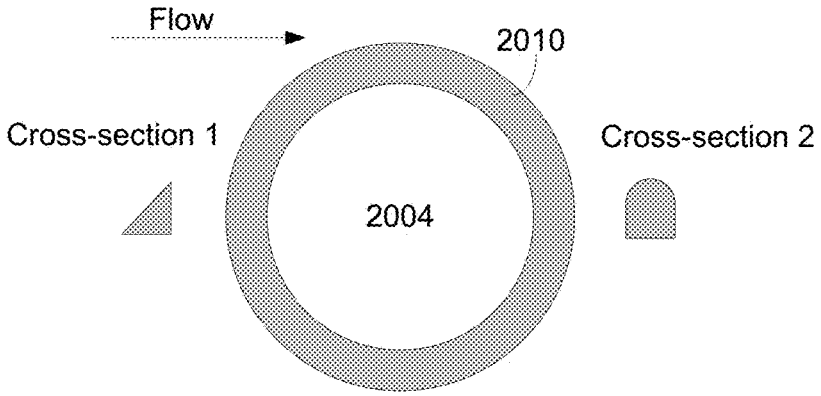


FIG. 20B

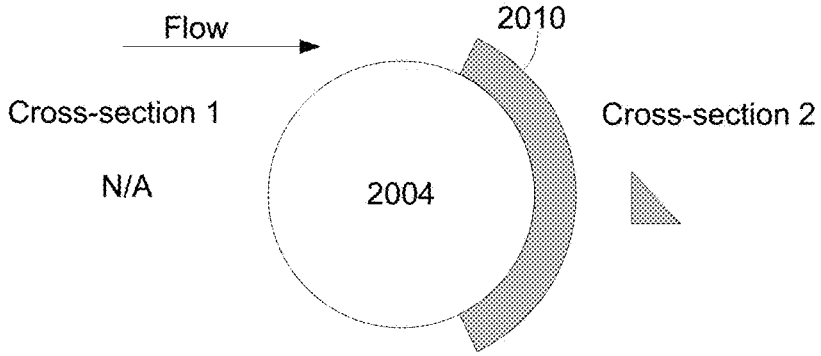


FIG. 20C

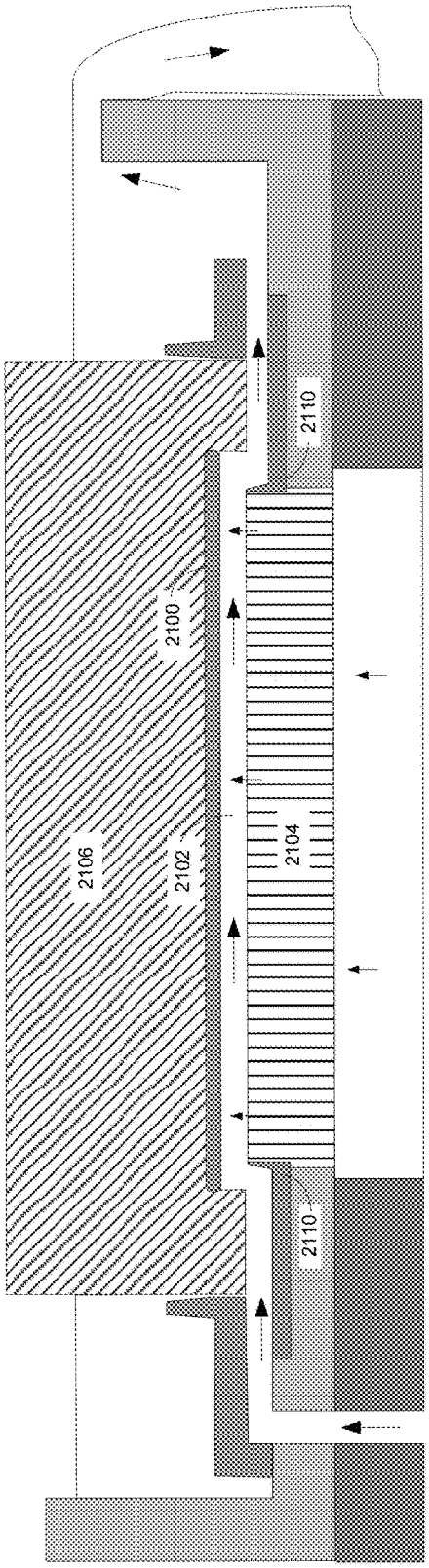


FIG. 21

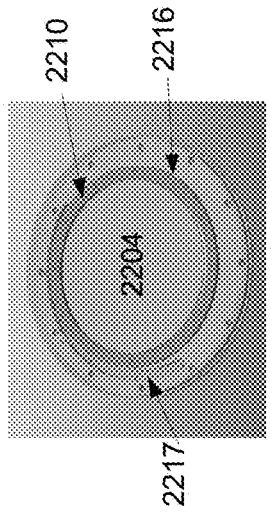


FIG. 22B

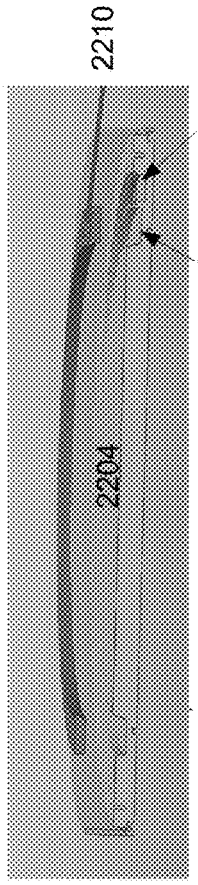


FIG. 22A

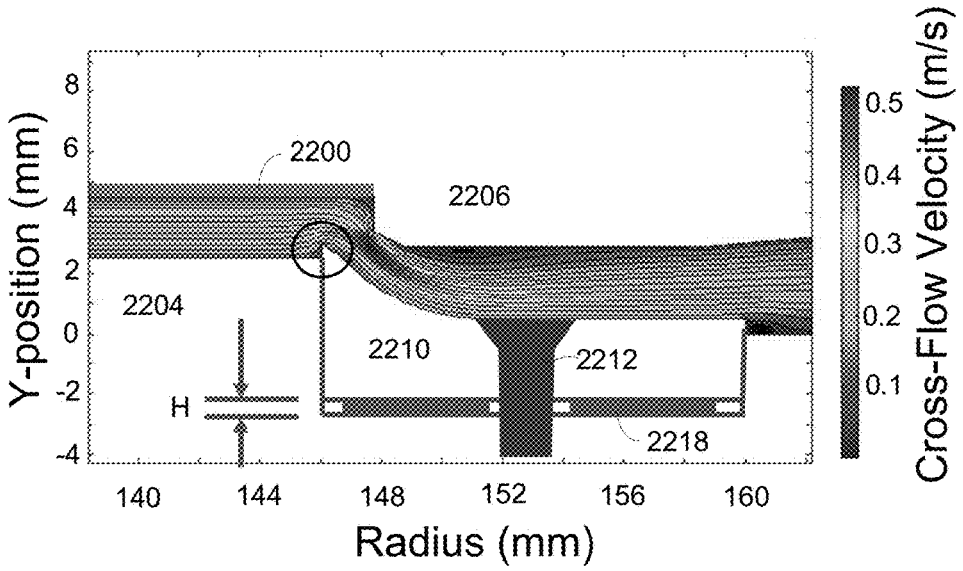


FIG. 22C

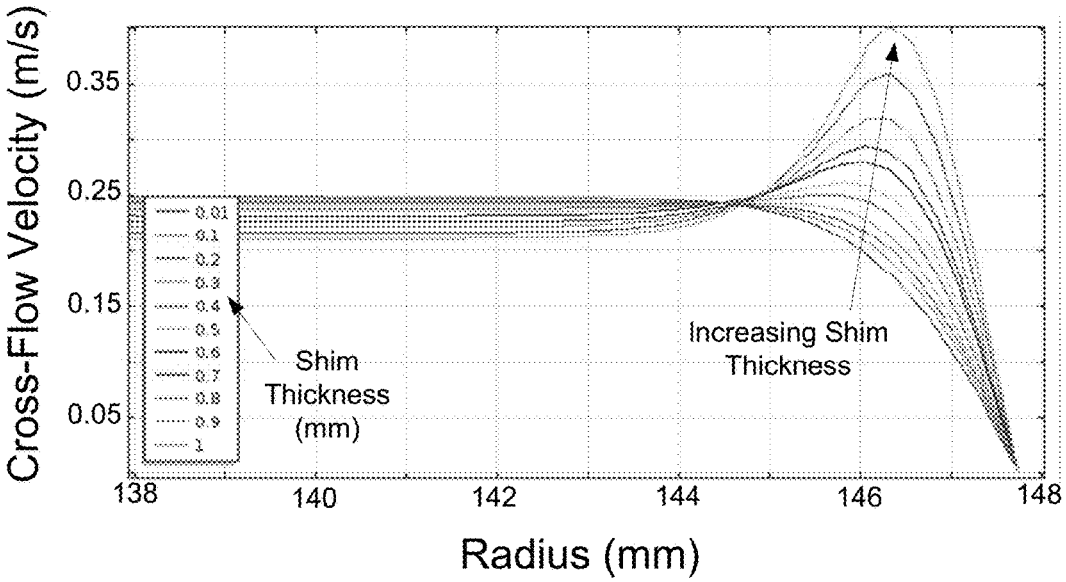


FIG. 22D

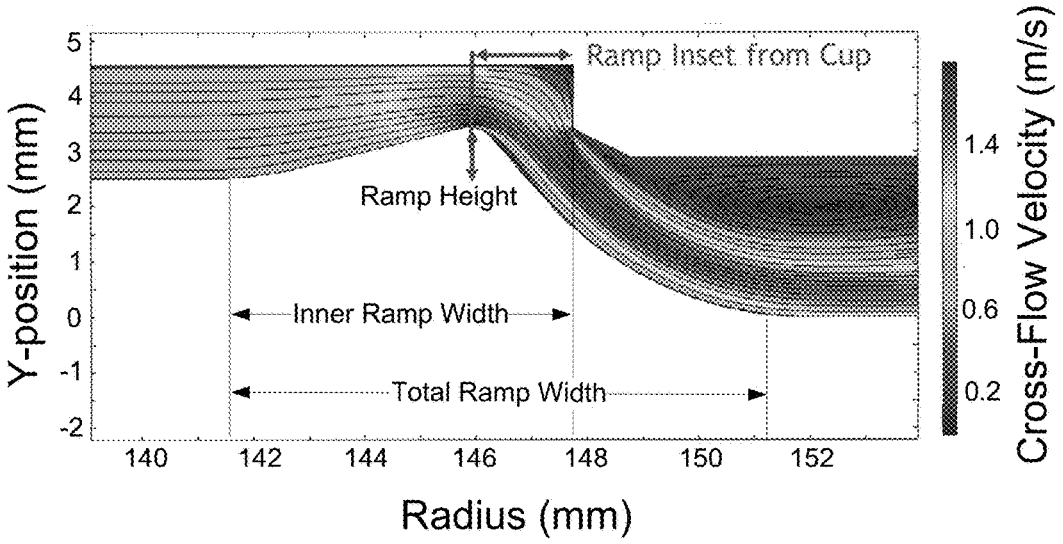


FIG. 23A

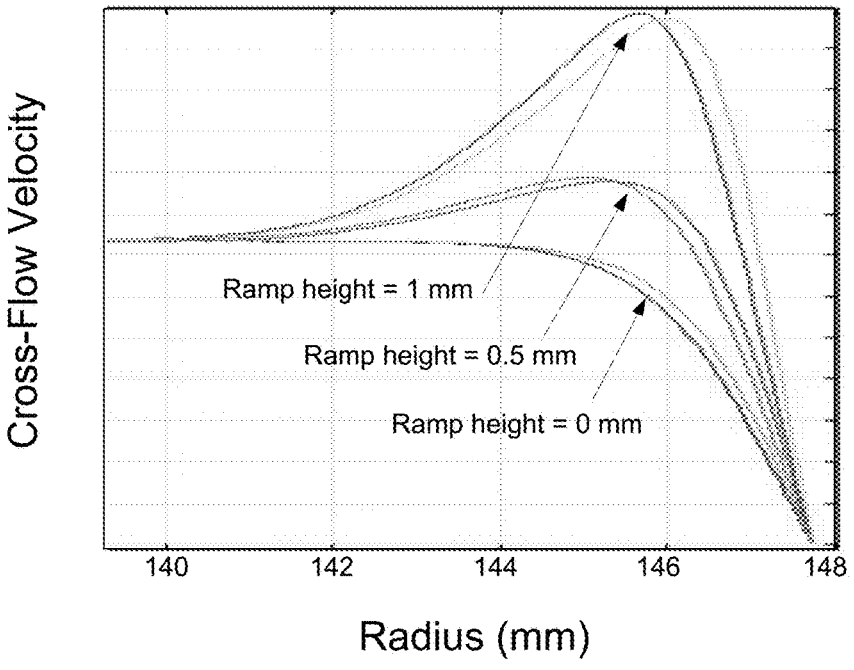


FIG. 23B

FIG. 24A

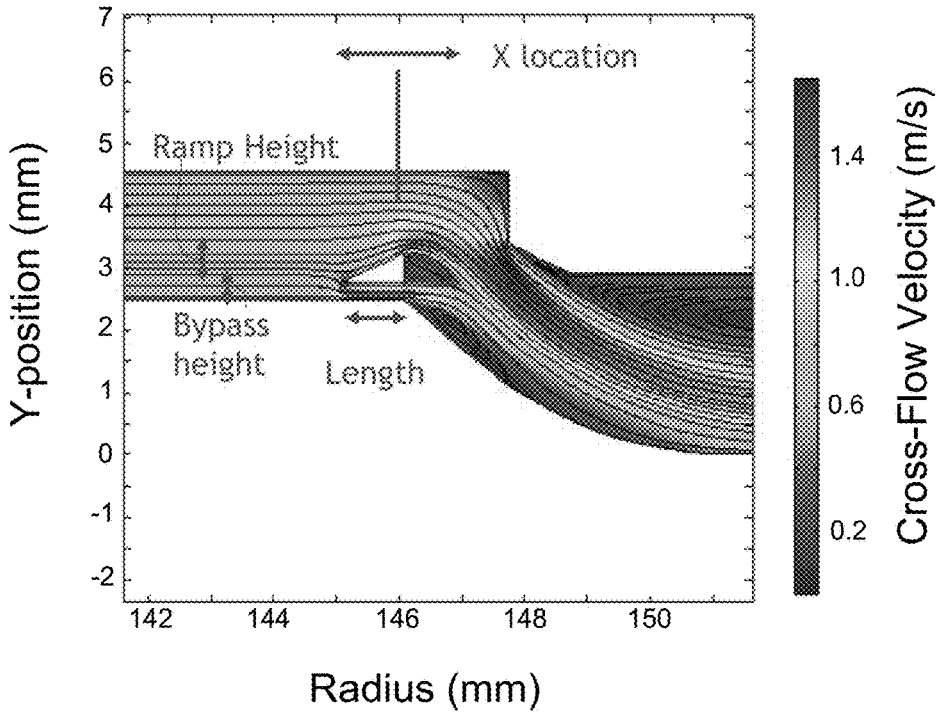
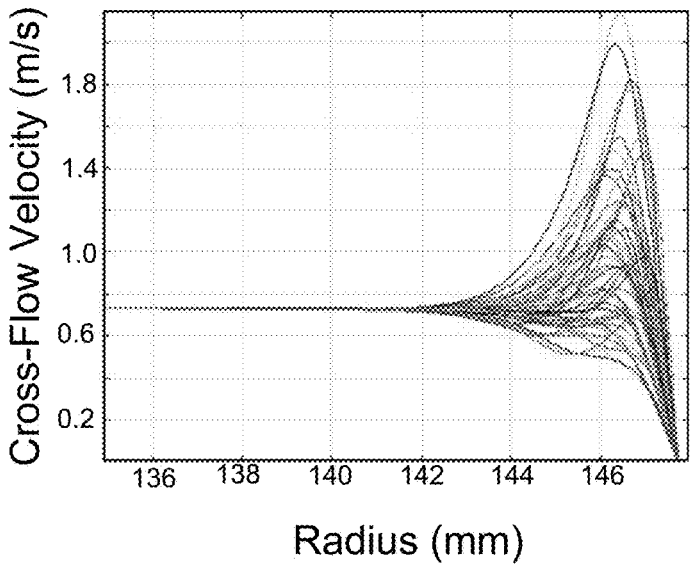


FIG. 24B



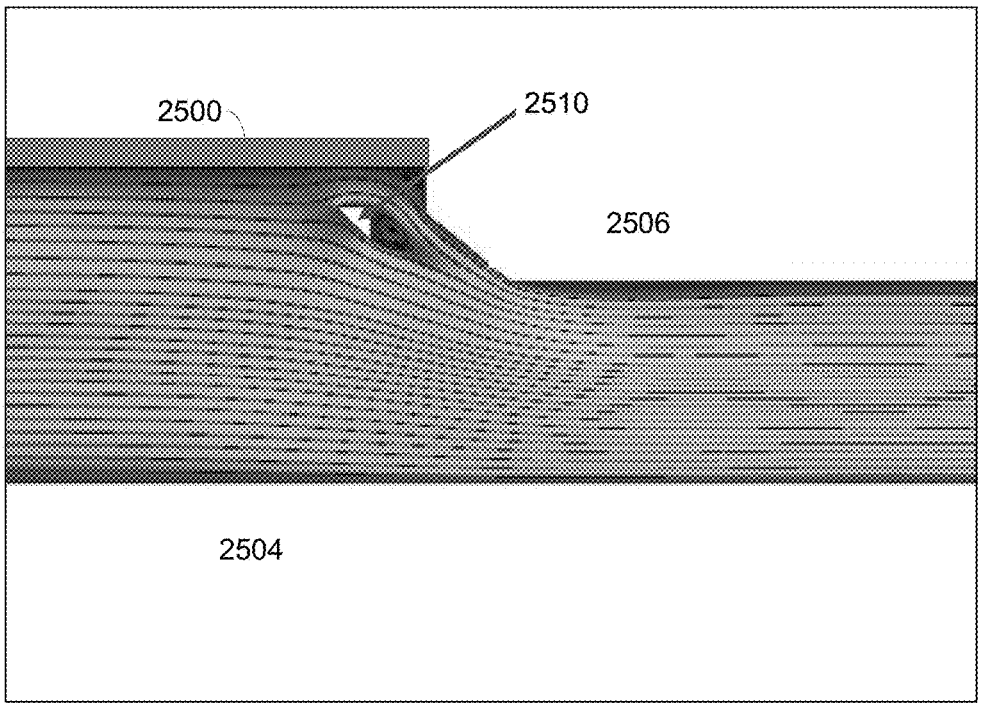


FIG. 25

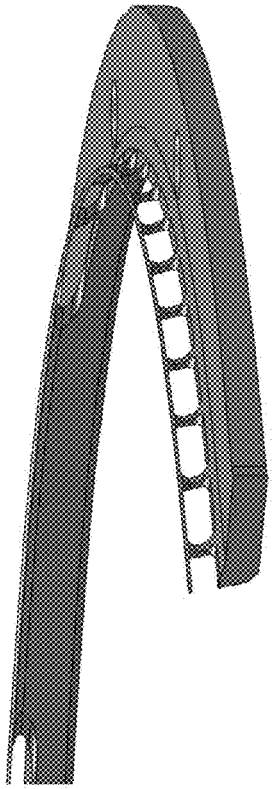


FIG. 26B

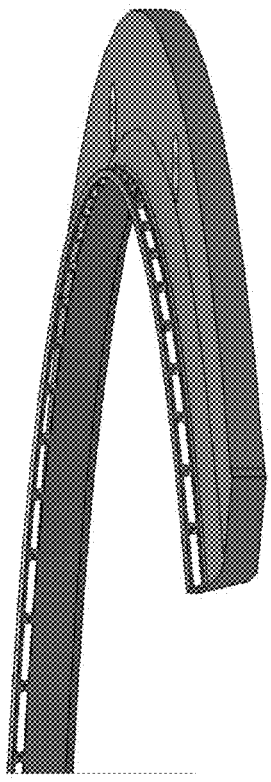


FIG. 26D

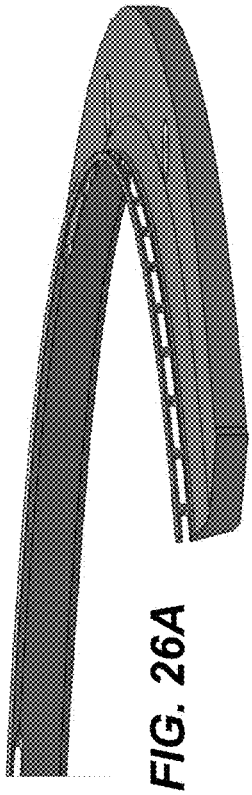


FIG. 26A

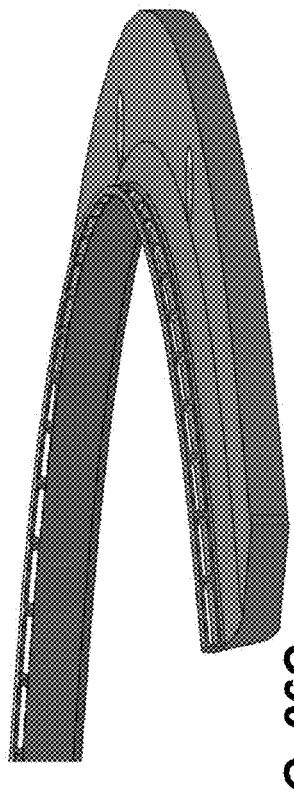


FIG. 26C

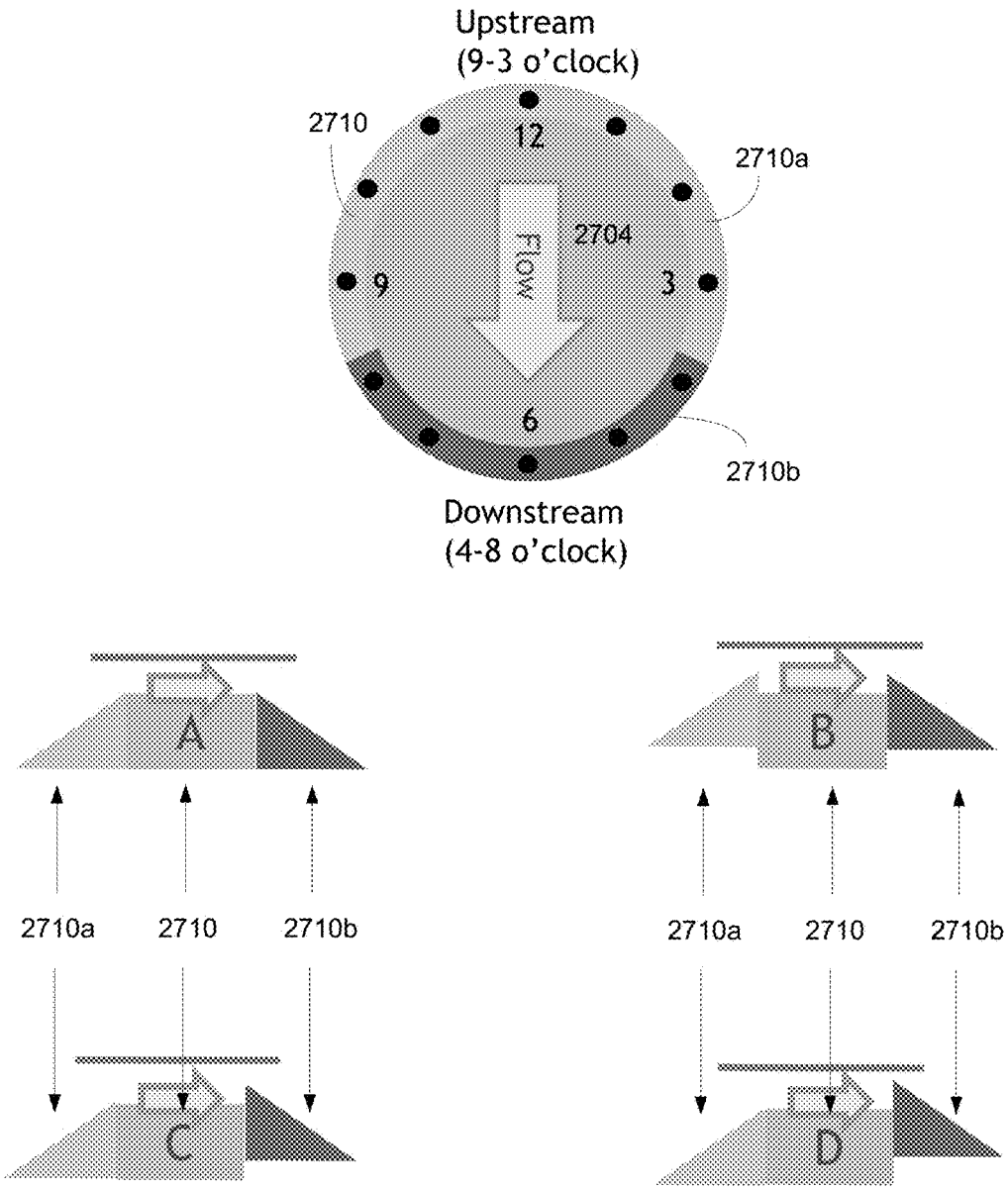

Upstream
(9-3 o'clock)
2710
12
2710a
2704
FLOW
9
3
6
2710b
Downstream
(4-8 o'clock)
A
2710a
2710
2710b
C
B
2710a
2710
2710b
D

FIG. 27A

Shim Setup	Wafer-CIRP Gap (mm)	Upstream Gap (mm)	Downstream Gap (mm)
A	2	2	2
B	2	1	1
C	2	2	1
D	2	2	0.5

FIG. 27B

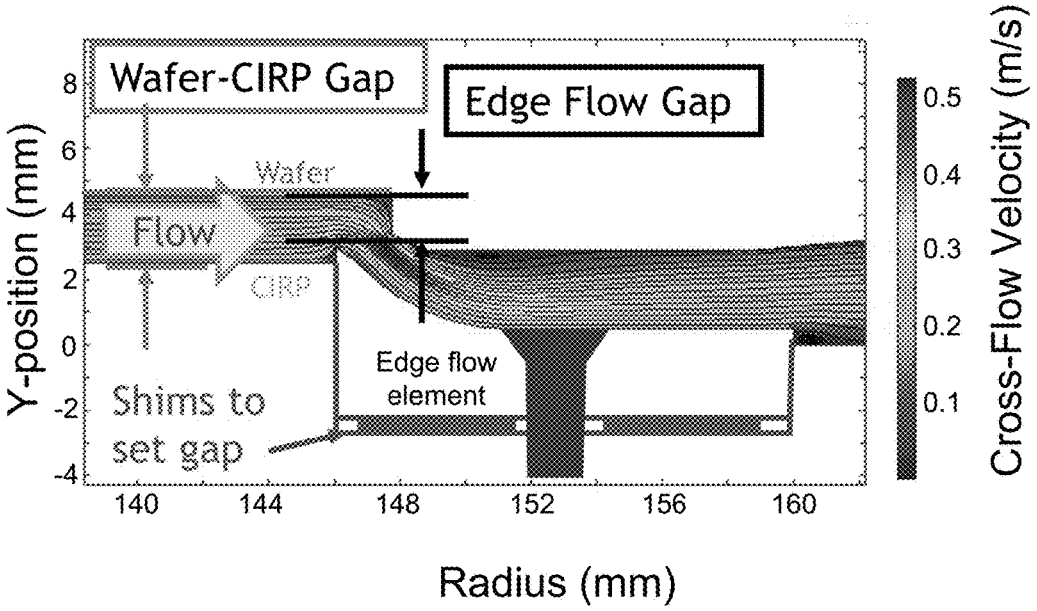


FIG. 27C

FIG. 28

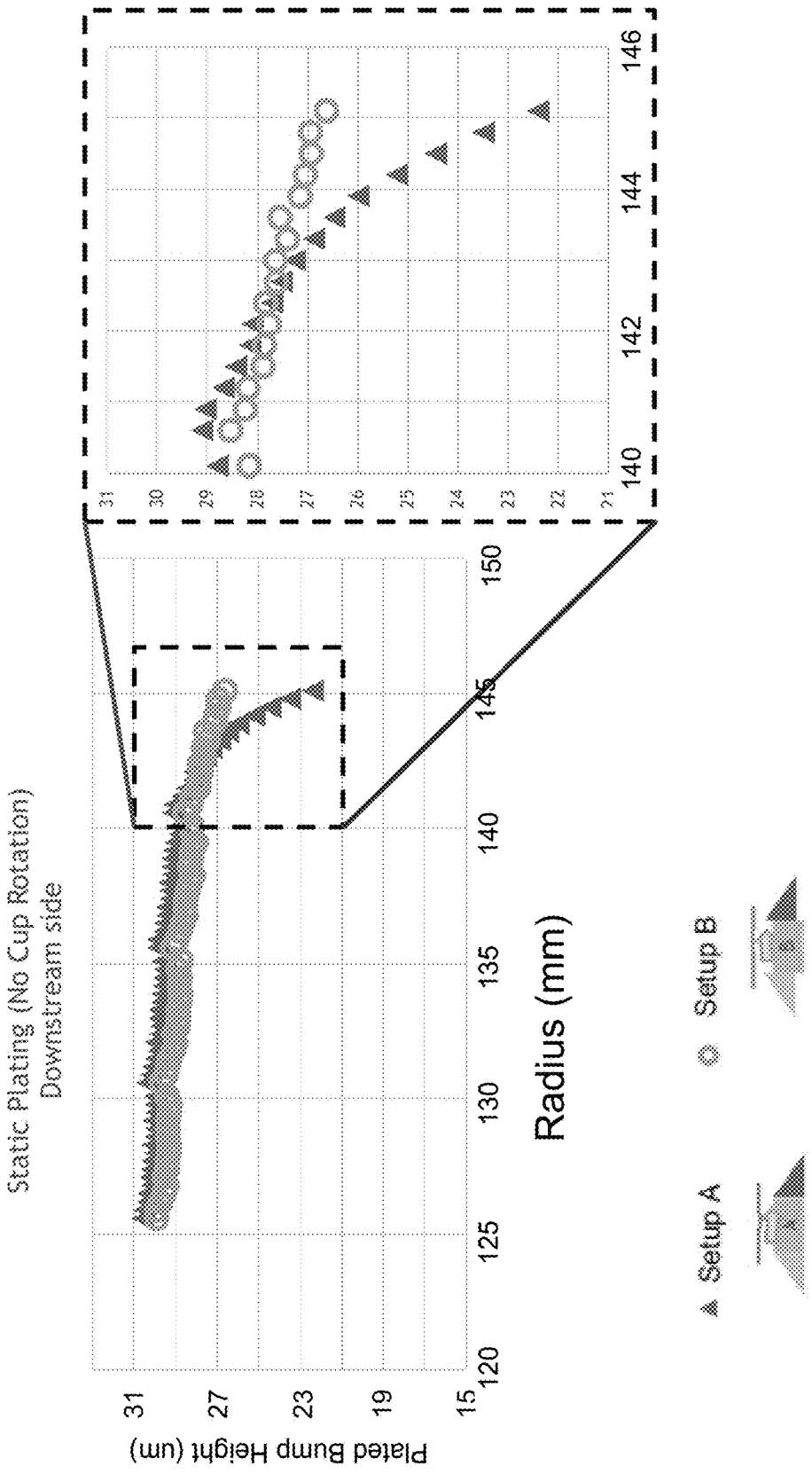
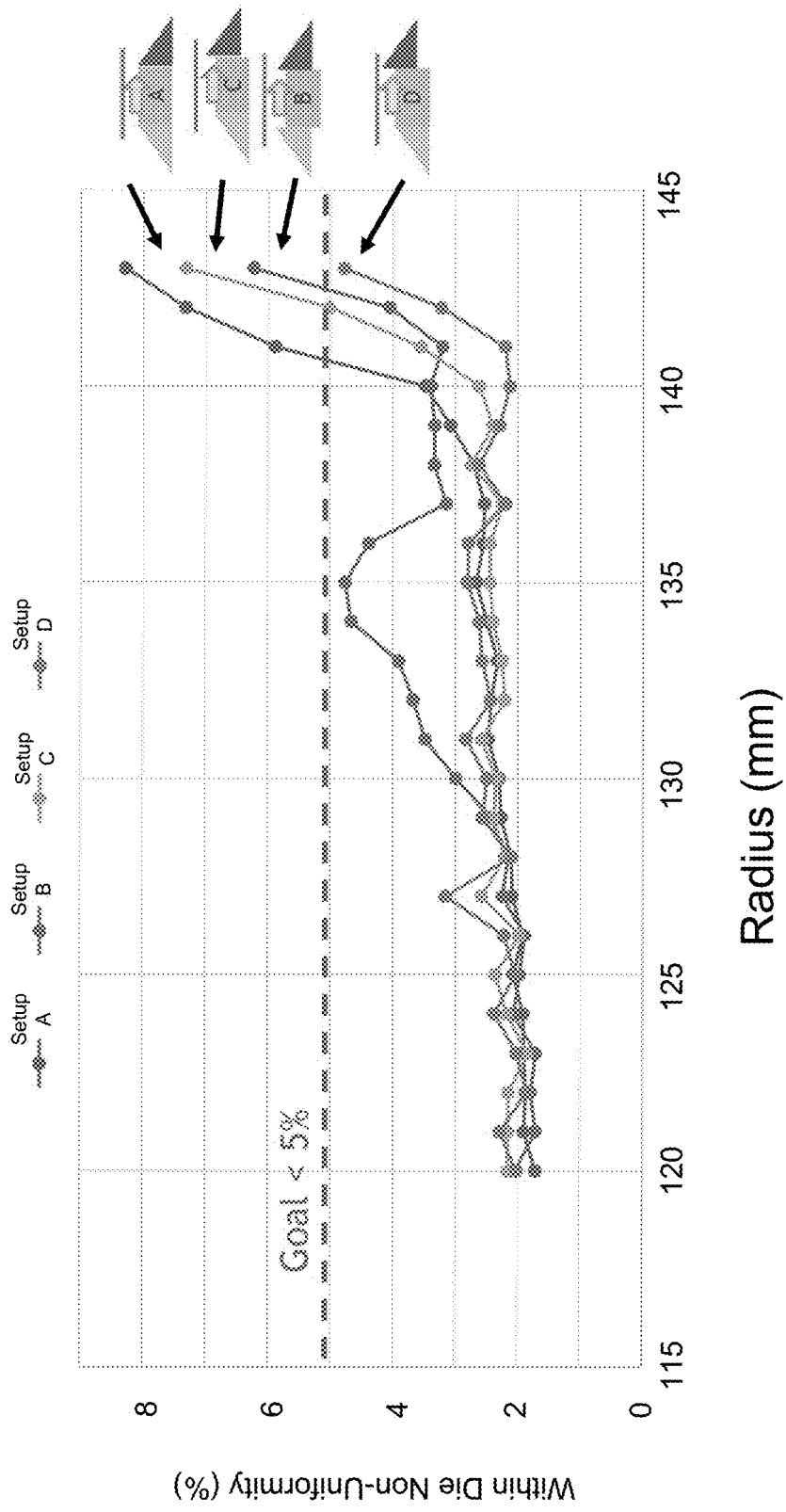

Static Plating (No Cup Rotation)
Downstream side
Plated Bump Height (um)
31
27
23
19
15
120
125
130
135
140
145
150
Radius (mm)
140
142
144
146
Setup A
Setup B

FIG. 29


Within Die Non-Uniformity (%)
0
2
4
6
8
115
120
125
130
135
140
145
Radius (mm)
Goal < 5%
Setup A
Setup B
Setup C
Setup D

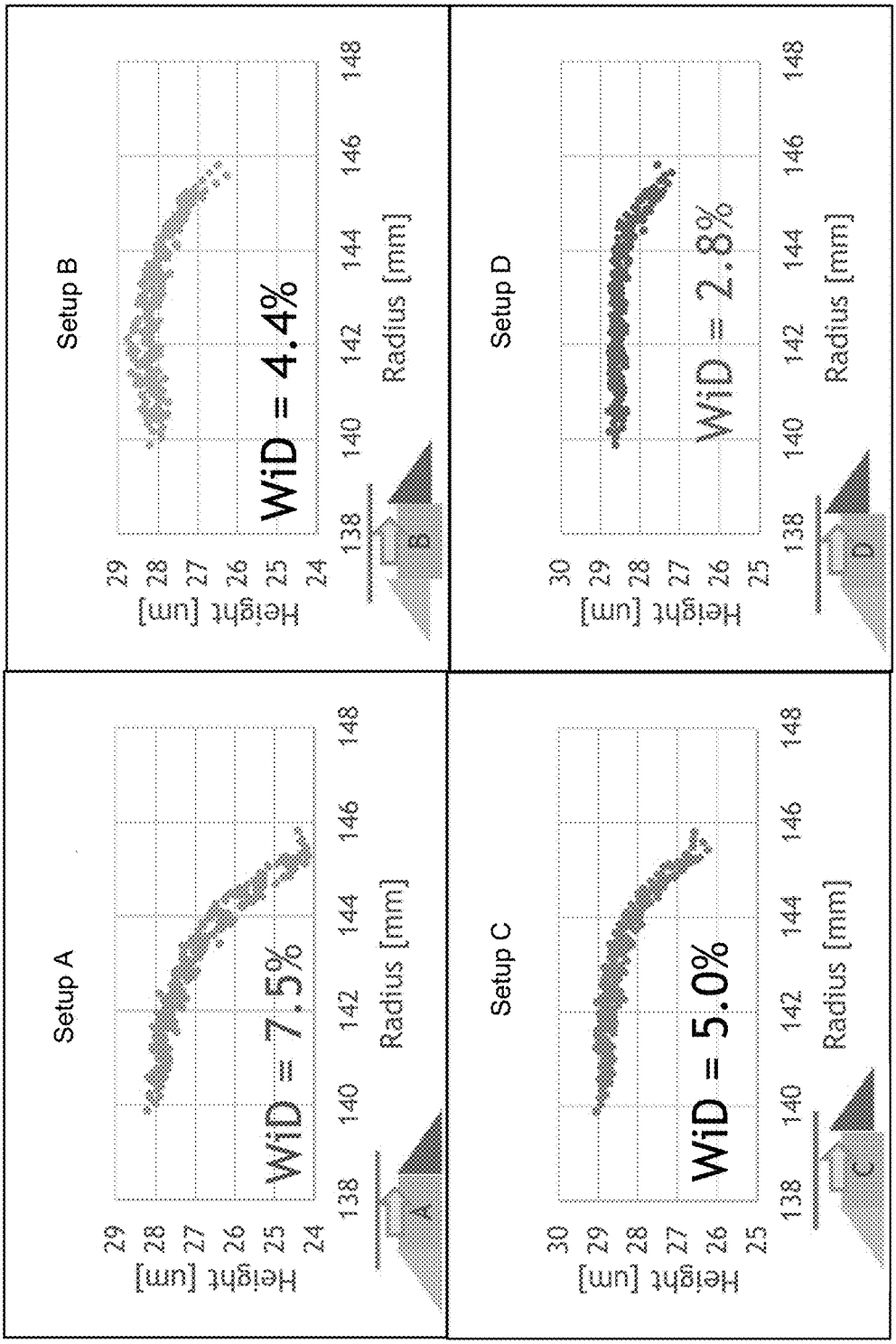

Setup A
WiD = 7.5%
Setup B
WiD = 4.4%
Setup C
WiD = 5.0%
Setup D
WiD = 2.8%
Height [um]
Radius [mm]
A
B
C
D

FIG. 30

EDGE FLOW ELEMENT FOR ELECTROPLATING APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims benefit of priority to U.S. Provisional Application No. 62/211,633, filed Aug. 28, 2015, and titled "EDGE FLOW ELEMENT FOR ELECTROPLATING APPARATUS," which is herein incorporated by reference in its entirety and for all purposes.

BACKGROUND

The disclosed embodiments relate to methods and apparatus for controlling electrolyte hydrodynamics during electroplating. More particularly, methods and apparatus described herein are particularly useful for plating metals onto semiconductor wafer substrates, such as through resist plating of small microbumping features (e.g., copper, nickel, tin and tin alloy solders) having widths less than, e.g., about 50 μm, and copper through silicon via (TSV) features.

Electrochemical deposition processes are well-established in modern integrated circuit fabrication. The transition from aluminum to copper metal line interconnections in the early years of the twenty-first century drove a need for increasingly sophisticated electrodeposition processes and plating tools. Much of the sophistication evolved in response to the need for ever smaller current carrying lines in device metallization layers. These copper lines are formed by electroplating the metal into very thin, high-aspect ratio trenches and vias in a methodology commonly referred to as "damascene" processing (pre-passivation metalization).

Electrochemical deposition is now poised to fill a commercial need for sophisticated packaging and multichip interconnection technologies known generally and colloquially as wafer level packaging (WLP) and through silicon via (TSV) electrical connection technology. These technologies present their own very significant challenges due in part to the generally larger feature sizes (compared to Front End of Line (FEOL) interconnects) and high aspect ratios.

Depending on the type and application of the packaging features (e.g., through chip connecting TSV, interconnection redistribution wiring, or chip to board or chip bonding, such as flip-chip pillars), plated features are usually, in current technology, greater than about 2 micrometers and are typically about 5-100 micrometers in their principal dimension (for example, copper pillars may be about 50 micrometers). For some on-chip structures such as power busses, the feature to be plated may be larger than 100 micrometers. The aspect ratios of the WLP features are typically about 1:1 (height to width) or lower, though they can range as high as perhaps about 2:1 or so, while TSV structures can have very high aspect ratios (e.g., in the neighborhood of about 20:1).

With the shrinking of WLP structure sizes from 100-200 um to less than 50 um comes a unique set of problems because at this scale, the hydrodynamic and mass transfer boundary layers are nearly equivalent. For prior generations with larger features, the transport of fluid and mass into a feature was carried by the general penetration of the flow fields into the features, but with smaller features, the formation of flow eddies and stagnation can inhibit both the rate and uniformity of mass transport within the growing feature. Therefore, new methods of creating uniform mass transfer within smaller "microbump" and TSV features are required.

Further, the time constant τ(the 1D diffusion equilibration time constant) for a purely diffusion process scales with feature depth L and the diffusion constant D as

$$\tau = \frac{L^2}{2D} \text{ (sec).}$$

Assuming an average-reasonable value for the diffusion coefficient of a metal ion (e.g., 5×10^{-6} cm^2/sec), a relatively large FEOL 0.3 um deep damascene feature would have a time constant of only about 0.1 msec, but a 50 um deep TSV of WLP bump would have a time constant of several seconds.

Not only feature size, but also plating speed differentiates WLP and TSV applications from damascene applications. For many WLP applications, depending on the metal being plated (e.g., copper, nickel, gold, silver solders, etc.), there is a balance between the manufacturing and cost requirements on the one hand and the technical requirements and technical difficulty on the other hand (e.g., goals of capital productivity with wafer pattern variability and on wafer requirements like within die and within feature targets). For copper, this balance is usually achieved at a rate of at least about 2 micrometers/minute, and typically at least about 3-4 micrometers/minute or more. For tin plating, a plating rate of greater than about 3 um/min, and for some applications at least about 7 micrometers/minute may be required. For nickel and strike gold (e.g., low concentration gold flash film layers), the plating rates may be between about 0.1 to 1 um/min. At these metal-relative higher plating rate regimes, efficient mass transfer of metal ions in the electrolyte to the plating surface is important.

In certain embodiments, plating must be conducted in a highly uniform manner over the entire face of a wafer to achieve good plating uniformity WIthin a Wafer (WIW), WIthin and among all the features of a particular Die (WID), and also WIthin the individual Features themselves (WIF). The high plating rates of WLP and TSV applications present challenges with respect to uniformity of the electrodeposited layer. For various WLP applications, plating must exhibit at most about 5% half range variation radially along the wafer surface (referred to as WIW non-uniformity, measured on a single feature type in a die at multiple locations across the wafer's diameter). A similar equally challenging requirement is the uniform deposition (thickness and shape) of various features of either different sizes (e.g. feature diameters) or feature density (e.g. an isolated or embedded feature in the middle of an array of the chip die). This performance specification is generally referred to as the WID non-uniformity. WID non-uniformity is measured as the local variability (e.g. <5% half range) of the various features types as described above versus the average feature height or other dimension within a given wafer die at that particular die location on the wafer (e.g. at the mid radius, center or edge).

A final challenging requirement is the general control of the within feature shape. Without proper flow and mass transfer convection control, after plating a line or pillar can end up being sloped in either a convex, flat or concave fashion in two or three dimensions (e.g. a saddle or a domed shape), with a flat profile generally, though not always, preferred. While meeting these challenges, WLP applications must compete with conventional, potentially less expensive pick and place serial routing operations. Still further, electrochemical deposition for WLP applications

may involve plating various non-copper metals such as solders like lead, tin, tin-silver, and other underbump metallization materials, such as nickel, gold, palladium, and various alloys of these, some of which include copper. Plating of tin-silver near eutectic alloys is an example of a plating technique for an alloy that is plated as a lead free solder alternative to lead-tin eutectic solder.

SUMMARY

Certain embodiments herein relate to methods and apparatus for electroplating one or more materials onto a substrate. In many cases the material is a metal and the substrate is a semiconductor wafer, though the embodiments are no so limited. Typically, the embodiments herein utilize a channeled ionically resistive plate (CIRP) positioned near the substrate, creating a cross flow manifold defined on the bottom by the CIRP, and on the top by the substrate. During plating, fluid enters the cross flow manifold both upward through the channels in the CIRP, and laterally through a cross flow side inlet positioned proximate one side of the substrate. The flow paths combine in the cross flow manifold and exit at the cross flow exit, which is positioned opposite the cross flow inlet. In various embodiments, an edge flow element may be used to direct flow near the periphery of the substrate. The edge flow element may be integral with the CIRP or with a substrate holder, or it may be separate. The edge flow element promotes a relatively higher degree of shear flow near the edge of the substrate, where the substrate contacts the substrate holder, than would otherwise be accomplished without the edge flow element. This increased shear flow near the periphery of the substrate results in more uniform plating results.

In one aspect of the embodiments herein, an electroplating apparatus is provided, the apparatus including: (a) an electroplating chamber configured to contain an electrolyte and an anode while electroplating metal onto a substantially planar substrate; (b) a substrate holder configured to hold a substantially planar substrate such that a plating face of the substrate is separated from the anode during electroplating, where when the substrate is positioned in the substrate holder, a corner forms at the interface between the substrate and substrate holder, the corner defined on top by the plating face of the substrate and on the side by the substrate holder; (c) an ionically resistive element including a substrate-facing surface that is separated from the plating face of the substrate by a gap of about 10 mm or less, where the ionically resistive element is at least coextensive with the plating face of the substrate during electroplating, the ionically resistive element adapted to provide ionic transport through the element during electroplating; (d) an inlet to the gap for introducing electrolyte to the gap; (e) an outlet to the gap for receiving electrolyte flowing in the gap; and (f) an edge flow element configured to direct electrolyte into the corner at the interface between the substrate and the substrate holder, the edge flow element being arc-shaped or ring-shaped and positioned proximate a periphery of the substrate and at least partially radially inside of the corner at the interface between the substrate and the substrate holder, where the inlet and outlet are positioned proximate azimuthally opposing perimeter locations on the plating face of the substrate during electroplating, and where the inlet and outlet are adapted to generate cross-flowing electrolyte in the gap to create or maintain a shearing force on the plating face of the substrate during electroplating.

In certain implementations, the edge flow element is configured to attach to the ionically resistive element and/or to the substrate holder. In some embodiments, the edge flow element is integral with the ionically resistive element and includes a raised portion proximate the periphery of the ionically resistive element, the raised portion being raised with respect to a height of a remaining portion of the substrate-facing surface of the ionically resistive element, the remaining portion of the substrate-facing surface being positioned radially interior of the raised portion.

In a number of embodiments, the ionically resistive element includes a groove into which the edge flow element is installed. In some such cases, the apparatus further includes one or more shims positioned between the ionically resistive element and the edge flow element. The one or more shims may result in the edge flow element being positioned in a manner that is azimuthally asymmetric.

In certain implementations, the edge flow element is azimuthally asymmetric with respect to one or more of (a) position (b) shape, and/or (c) presence or shape of flow bypass passages. In certain embodiments, the azimuthal asymmetry may be positioned at a certain location. For example, in some cases the edge flow element includes at least a first portion and a second portion, the portions being defined based on an azimuthal asymmetry in the edge flow element, where the first portion is centered near the inlet to the gap or the outlet to the gap.

The edge flow element can have a variety of shapes and features. In various implementations, the edge flow element includes flow bypass passages that allow electrolyte to flow through the edge flow element. In some embodiments, flow bypass passages may allow electrolyte to flow between an upper edge of the edge flow element and the ionically resistive element. In these or other cases, the flow bypass passages may allow electrolyte to flow between a lower edge of the edge flow element and the substrate holder. In some cases, the edge flow element is ring-shaped. In other cases, the edge flow element may be arc-shaped.

The edge flow element may be adjustable in one or more respects. For instance, a position of the edge flow element with respect to the ionically resistive element may be adjustable. In some such cases, the apparatus further includes shims and/or screws for adjusting the position of the edge flow element with respect to a position of the ionically resistive element. In various embodiments, the edge flow element may be raised and/or lowered with respect to a plane formed by the ionically resistive element. Such adjustment can affect the flow pattern of electrolyte near the interface between the substrate and substrate holder, thereby achieving a large degree of tunability. In certain embodiments, the apparatus further includes an actuator for adjusting the position of the edge flow element with respect to a position of the ionically resistive element, where the actuator permits the position of the edge flow element to be adjusted during electroplating.

In another aspect of the disclosed embodiments, an edge flow element for use in electroplating is provided, the edge flow element including: an element configured to mate with an ionically resistive element and/or a substrate holder in an electroplating apparatus, the element being ring-shaped or arc-shaped, the element including an electrically insulating material, where when installed in the electroplating apparatus having a substrate therein, the element is positioned, at least partially, radially interior of an inner edge of the substrate holder, and where during electroplating, the element directs fluid into a corner formed at the interface between the substrate and the substrate holder, the corner being defined on its top by the substrate and on its side by the substrate holder.

In certain implementations, the edge flow element is azimuthally asymmetric. In some embodiments, the edge flow element further includes flow bypass passages through which electrolyte can flow during electroplating.

In a further aspect of the disclosed embodiments, a method for electroplating a substrate is provided, the method including: (a) receiving a substantially planar substrate in a substrate holder, where a plating face of the substrate is exposed, and where the substrate holder is configured to hold the substrate such that the plating face of the substrate is separated from an anode during electroplating; (b) immersing the substrate in electrolyte, where a gap of about 10 mm or less is formed between the plating face of the substrate and an upper surface of an ionically resistive element, where the ionically resistive element is at least coextensive with the plating face of the substrate, and where the ionically resistive element is adapted to provide ionic transport through the ionically resistive element during electroplating; (c) flowing electrolyte in contact with the substrate in the substrate holder (i) from a side inlet, into the gap, over and/or under an edge flow element, and out a side outlet, and (ii) from below the ionically resistive element, through the ionically resistive element, into the gap, and out the side outlet, where the inlet and outlet are positioned proximate azimuthally opposed perimeter locations on the plating face of the substrate, and where the inlet and outlet are designed or configured to generate cross-flowing electrolyte in the gap during electroplating; (d) rotating the substrate holder; and (e) electroplating material onto the plating face of the substrate while flowing the electrolyte as in (c), where the edge flow element is configured to direct electrolyte into a corner that forms between the substrate and the substrate holder, the corner defined on its top by the plating face of the substrate and on its side by the inner edge of the substrate holder.

In some embodiments, the edge flow element is azimuthally asymmetric. The edge flow element may, in certain cases, include flow bypass passages that allow electrolyte to flow through the edge flow element. In some embodiments, a position of the edge flow element may be adjusted during electroplating.

These and other features will be described below with reference to the associated drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**A shows a perspective view of a substrate holding and positioning apparatus for electrochemically treating semiconductor wafers.

FIG. **1**B depicts a cross-sectional view of a portion of a substrate holding assembly including a cone and cup.

FIG. **1**C depicts a simplified view of an electroplating cell that may be used in practicing the embodiments herein.

FIG. **1**D-**1**G illustrate various electroplating apparatus embodiments that may be used to enhance cross flow across the face of a substrate, along with top views of the flow dynamics achieved when practicing these embodiments.

FIG. **2** illustrates an exploded view of various parts of an electroplating apparatus typically present in the cathode chamber in accordance with certain embodiments disclosed herein.

FIG. **3**A shows a close-up view of a cross flow side inlet and surrounding hardware in accordance with certain embodiments herein.

FIG. **3**B shows a close-up view of a cross flow outlet, a CIRP manifold inlet, and surrounding hardware in accordance with various disclosed embodiments.

FIG. **4** depicts a cross-sectional view of various parts of the electroplating apparatus shown in FIGS. **3**A-**3**B.

FIG. **5** shows a cross flow injection manifold and showerhead split into 6 individual segments according to certain embodiments.

FIG. **6** shows a top view of a CIRP and associated hardware according to an embodiment herein, focusing especially on the inlet side of the cross flow.

FIG. **7** illustrates a simplified top view of a CIRP and associated hardware showing both the inlet and outlet sides of the cross flow manifold according to various disclosed embodiments.

FIGS. **8**A-**8**B depict an initial (**8**A) and revised (**8**B) design of a cross flow inlet region according to certain embodiments.

FIG. **9** shows an embodiment of a CIRP partially covered by a flow confinement ring and supported by a frame.

FIG. **10**A shows a simplified top view of a CIRP and flow confinement ring where no side inlet is used.

FIG. **10**B shows a simplified top view of a CIRP, flow confinement ring, and cross flow side inlet according to various embodiments disclosed herein.

FIGS. **11**A-**11**B illustrate the cross flow through the cross flow manifold for the apparatus shown in FIGS. **10**A-**10**B, respectively.

FIGS. **12**A-**12**B are graphs showing the horizontal cross flow velocity during plating vs. wafer position for the apparatus shown in FIGS. **10**A-**10**B, respectively.

FIGS. **13**A and **13**B present experimental results showing bump height vs. radial position on the substrate, illustrating problems related to a low plating rate near the periphery of the substrate.

FIG. **14**A depicts a cross-sectional view of a portion of an electroplating apparatus.

FIG. **14**B shows modeling results related to the flow through the apparatus depicted in FIG. **14**A.

FIG. **15** depicts modeling results related to shear flow velocity vs. radial position on the substrate and experimental results related to bump height vs. radial position on the substrate, showing a lower degree of plating near the periphery of the substrate.

FIGS. **16**A and **16**B show experimental results related to within-die thickness non-uniformity (FIG. **16**A) and photoresist thickness (FIG. **16**B) at different radial positions on the substrate.

FIGS. **17**A and **17**B depicts a cross-sectional view of an electroplating apparatus according to one embodiment where an edge flow element is used.

FIGS. **18**A-**18**C illustrates three types of attachment configurations for installing an edge flow element in an electroplating apparatus according to various embodiments.

FIG. **18**D presents a table describing certain features of the edge flow elements shown in FIGS. **18**A-**18**C.

FIGS. **19**A-**19**E illustrate methods for adjusting an edge flow element in an electroplating apparatus.

FIGS. **20**A-**20**C illustrate several types of edge flow elements that may be used according to various embodiments, some of which are azimuthally asymmetric.

FIG. **21** illustrates a cross-sectional view of an electroplating cell according to certain embodiments where an edge flow element and top flow insert are used.

FIGS. **22**A and **22**B depicts a channeled ionically resistive plate (CIRP) having a groove therein, into which an edge flow element is installed.

FIGS. **22**C and **22**D depict modeling results describing the flow velocity near the edge of the substrate for various shim thicknesses.

FIGS. **23**A and **23**B present modeling results related to an electroplating apparatus having an edge flow element that has a ramp shape, according to certain embodiments.

FIGS. **24**A, **24**B, and **25** present modeling results related to electroplating apparatus having edge flow elements that include different types of flow bypass passages according to certain embodiments.

FIGS. **26**A-**26**D illustrates several examples of an edge flow element, each having flow bypass passages therein.

FIGS. **27**A-**27**C describe an experimental setup used to generate the results shown in FIGS. **28**-**30**.

FIGS. **28**-**30** present experimental results related to plated bump height (FIGS. **28** and **30**) or within-die thickness non-uniformity (FIG. **29**) vs. radial position on the substrate, for the experimental setups described in relation to FIGS. **27**A-**27**C.

DETAILED DESCRIPTION

In this application, the terms "semiconductor wafer," "wafer," "substrate," "wafer substrate," and "partially fabricated integrated circuit" are used interchangeably. One of ordinary skill in the art would understand that the term "partially fabricated integrated circuit" can refer to a silicon wafer during any of many stages of integrated circuit fabrication thereon. The following detailed description assumes the invention is implemented on a wafer. Oftentimes, semiconductor wafers have a diameter of 200, 300 or 450 mm. However, the invention is not so limited. The work piece may be of various shapes, sizes, and materials. In addition to semiconductor wafers, other work pieces that may take advantage of this invention include various articles such as printed circuit boards and the like.

In the following description, numerous specific details are set forth in order to provide a thorough understanding of the presented embodiments. The disclosed embodiments may be practiced without some or all of these specific details. In other instances, well-known process operations have not been described in detail to not unnecessarily obscure the disclosed embodiments. While the disclosed embodiments will be described in conjunction with the specific embodiments, it will be understood that it is not intended to limit the disclosed embodiments.

Described herein are apparatus and methods for electroplating one or more metals onto a substrate. Embodiments are described generally where the substrate is a semiconductor wafer; however the invention is not so limited.

Disclosed embodiments include electroplating apparatus configured for, and methods including, control of electrolyte hydrodynamics during plating so that highly uniform plating layers are obtained. In specific implementations, the disclosed embodiments employ methods and apparatus that create combinations of impinging flow (flow directed at or perpendicular to the work piece surface) and shear flow (sometimes referred to as "cross flow" or flow with velocity parallel to the work piece surface).

One embodiment is an electroplating apparatus including the following features: (a) a plating chamber configured to contain an electrolyte and an anode while electroplating metal onto a substantially planar substrate; (b) a substrate holder configured to hold the substantially planar substrate such that a plating face of the substrate is separated from the anode during electroplating, where when the substrate is positioned in the substrate holder, a corner forms at the interface between the substrate and substrate holder, the corner defined on top by the plating face of the substrate and on the side by the substrate holder; (c) a channeled ionically resistive element including a substrate-facing surface that is substantially parallel to and separated from a plating face of the substrate during electroplating, the channeled ionically resistive element including a plurality of non-communicating channels, where the non-communicating channels allow for transport of the electrolyte through the element during electroplating; (d) a mechanism for creating and/or applying a shearing force (cross flow) to the electrolyte flowing at the plating face of the substrate; and (e) a mechanism for promoting shear flow near the periphery of the substrate, proximate a substrate/substrate holder interface. Though the wafer is substantially planar, it also typically has one or more microscopic trenches and may have one or more portions of the surface masked from electrolyte exposure. In various embodiments, the apparatus also includes a mechanism for rotating the substrate and/or the channeled ionically resistive element while flowing electrolyte in the electroplating cell in the direction of the substrate plating face.

In certain implementations, the mechanism for applying cross flow is an inlet with, for example, appropriate flow directing and distributing means on or proximate to the periphery of the channeled ionically resistive element. The inlet directs cross flowing catholyte along the substrate-facing surface of the channeled ionically resistive element. The inlet is azimuthally asymmetric, partially following the circumference of the channeled ionically resistive element, and having one or more gaps, and defining a cross flow injection manifold between the channeled ionically resistive element and the substantially planar substrate during electroplating. Other elements are optionally provided for working in concert with the cross flow injection manifold. These may include a cross flow injection flow distribution showerhead and a cross flow confinement ring, which are further described below in conjunction with the figures.

In certain implementations, the mechanism for promoting shear flow near the periphery of the substrate is an edge flow element. The edge flow element may be an integral part of a channeled ionically resistive plate or substrate holder in some cases. In other cases, the edge flow element may be a separate piece that interfaces with the channeled ionically resistive plate or with the substrate holder. In some cases where the edge flow element is a separate piece, a variety of differently shaped edge flow elements may be separately provided to allow the flow distribution near the edge of a substrate to be tuned for a given application. In various cases the edge flow element may be azimuthally asymmetric. Further details regarding the edge flow element are presented below.

In certain embodiments, the apparatus is configured to enable flow of electrolyte in the direction towards or perpendicular to a substrate plating face to produce an average flow velocity of at least about 3 cm/s (e.g., at least about 5 cm/s or at least about 10 cm/s) exiting the holes of the channeled ionically resistive element during electroplating. In certain embodiments, the apparatus is configured to operate under conditions that produce an average transverse electrolyte velocity of about 3 cm/sec or greater (e.g., about 5 cm/s or greater, about 10 cm/s or greater, about 15 cm/s or greater, or about 20 cm/s or greater) across the center point of the plating face of the substrate. These flow rates (i.e., the flow rate exiting the holes of the ionically resistive element and the flow rate across the plating face of the substrate) are in certain embodiments appropriate in an electroplating cell employing an overall electrolyte flow rate of about 20 L/min and an approximately 12 inch diameter substrate. The embodiments herein may be practiced with various substrate sizes. In some cases, the substrate has a diameter of about

200 mm, about 300 mm, or about 450 mm. Further, the embodiments herein may be practiced at a wide variety of overall flow rates. In certain implementations, the overall electrolyte flow rate is between about 1-60 L/min, between about 6-60 L/min, between about 5-25 L/min, or between about 15-25 L/min. The flow rates achieved during plating may be limited by certain hardware constraints, such as the size and capacity of the pump being used. One of skill in the art would understand that the flow rates cited herein may be higher when the disclosed techniques are practiced with larger pumps.

In some embodiments, the electroplating apparatus contains separated anode and cathode chambers in which there are different electrolyte compositions, electrolyte circulation loops, and/or hydrodynamics in each of two chambers. An ionically permeable membrane may be employed to inhibit direct convective transport (movement of mass by flow) of one or more components between the chambers and maintain a desired separation between the chambers. The membrane may block bulk electrolyte flow and exclude transport of certain species such as organic additives while permitting transport of ions such as cations. In some embodiments, the membrane contains DuPont's NAFION™ or a related ionically selective polymer. In other cases, the membrane does not include an ion exchange material, and instead includes a micro-porous material. Conventionally, the electrolyte in the cathode chamber is referred to as "catholyte" and the electrolyte in the anode chamber is referred to as "anolyte." Frequently, the anolyte and catholyte have different compositions, with the anolyte containing little or no plating additives (e.g., accelerator, suppressor, and/or leveler) and the catholyte containing significant concentrations of such additives. The concentration of metal ions and acids also often differs between the two chambers. An example of an electroplating apparatus containing a separated anode chamber is described in U.S. Pat. No. 6,527,920, filed Nov. 3, 2000; U.S. Pat. No. 6,821,407, filed Aug. 27, 2002, and U.S. Pat. No. 8,262,871, filed Dec. 17, 2009 each of which is incorporated herein by reference in its entirety.

In some embodiments, the anode membrane need not include an ion exchange material. In some examples, the membrane is made from a micro-porous material such as polyethersulfone manufactured by Koch Membrane of Wilmington, Mass. This membrane type is most notably applicable for inert anode applications such as tin-silver plating and gold plating, but may also be used for soluble anode applications such as nickel plating.

In certain embodiments, and as described more fully elsewhere herein, catholyte is injected into a manifold region, referred to hereafter as the "CIRP manifold region", in which electrolyte is fed, accumulates, and then is distributed and passes substantially uniformly through the various non-communication channels of the CIRP directly towards the wafer surface.

In the following discussion, when referring to top and bottom features (or similar terms such as upper and lower features, etc.) or elements of the disclosed embodiments, the terms top and bottom are simply used for convenience and represent only a single frame of reference or implementation of the invention. Other configurations are possible, such as those in which the top and bottom components are reversed with respect to gravity and/or the top and bottom components become the left and right or right and left components.

While some aspects described herein may be employed in various types of plating apparatus, for simplicity and clarity, most of the examples will concern wafer-face-down, "fountain" plating apparatus. In such apparatus, the work piece to plated (typically a semiconductor wafer in the examples presented herein) generally has a substantially horizontal orientation (which may in some cases vary by a few degrees from true horizontal for some part of, or during the entire plating process) and may be powered to rotate during plating, yielding a generally vertically upward electrolyte convection pattern. Integration of the impinging flow mass from the center to the edge of the wafer, as well as the inherent higher angular velocity of a rotating wafer at its edge relative to its center, creates a radially increasing sheering (wafer parallel) flow velocity. One example of a member of the fountain plating class of cells/apparatus is the Sabre® Electroplating System produced by and available from Novellus Systems, Inc. of San Jose, Calif. Additionally, fountain electroplating systems are described in, e.g., U.S. Pat. No. 6,800,187, filed Aug. 10, 2001 and U.S. Pat. No. 8,308,931, filed Nov. 7, 2008, which are incorporated herein by reference in their entireties.

The substrate to be plated is generally planar or substantially planar. As used herein, a substrate having features such as trenches, vias, photoresist patterns and the like is considered to be substantially planar. Often these features are on the microscopic scale, though this is not necessarily always the case. In many embodiments, one or more portions of the surface of the substrate may be masked from exposure to the electrolyte.

The following description of FIGS. **1**A and **1**B provides a general non-limiting context to assist in understanding the apparatus and methods described herein. FIG. **1**A provides a perspective view of a wafer holding and positioning apparatus **100** for electrochemically treating semiconductor wafers. Apparatus **100** includes wafer engaging components (sometimes referred to herein as "clamshell" components). The actual clamshell includes a cup **102** and a cone **103** that enables pressure to be applied between the wafer and the seal, thereby securing the wafer in the cup.

Cup **102** is supported by struts **104**, which are connected to a top plate **105**. This assembly (**102-105**), collectively assembly **101**, is driven by a motor **107**, via a spindle **106**. Motor **107** is attached to a mounting bracket **109**. Spindle **106** transmits torque to a wafer (not shown in this figure) to allow rotation during plating. An air cylinder (not shown) within spindle **106** also provides vertical force between the cup and cone **103** to create a seal between the wafer and a sealing member (lipseal) housed within the cup. For the purposes of this discussion, the assembly including components **102-109** is collectively referred to as a wafer holder **111**. Note however, that the concept of a "wafer holder" extends generally to various combinations and sub-combinations of components that engage a wafer and allow its movement and positioning.

A tilting assembly including a first plate **115**, that is slidably connected to a second plate **117**, is connected to mounting bracket **109**. A drive cylinder **113** is connected both to plate **115** and plate **117** at pivot joints **119** and **121**, respectively. Thus, drive cylinder **113** provides force for sliding plate **115** (and thus wafer holder **111**) across plate **117**. The distal end of wafer holder **111** (i.e. mounting bracket **109**) is moved along an arced path (not shown) which defines the contact region between plates **115** and **117**, and thus the proximal end of wafer holder **111** (i.e. cup and cone assembly) is tilted upon a virtual pivot. This allows for angled entry of a wafer into a plating bath.

The entire apparatus **100** is lifted vertically either up or down to immerse the proximal end of wafer holder **111** into a plating solution via another actuator (not shown). Thus, a two-component positioning mechanism provides both ver-

tical movement along a trajectory perpendicular to an electrolyte and a tilting movement allowing deviation from a horizontal orientation (parallel to electrolyte surface) for the wafer (angled-wafer immersion capability). A more detailed description of the movement capabilities and associated hardware of apparatus **100** is described in U.S. Pat. No. 6,551,487 filed May 31, 2001 and issued Apr. 22, 2003, which is herein incorporated by reference in its entirety.

Note that apparatus **100** is typically used with a particular plating cell having a plating chamber which houses an anode (e.g., a copper anode or a non-metal inert anode) and electrolyte. The plating cell may also include plumbing or plumbing connections for circulating electrolyte through the plating cell—and against the work piece being plated. It may also include membranes or other separators designed to maintain different electrolyte chemistries in an anode compartment and a cathode compartment. In one embodiment, one membrane is employed to define an anode chamber, which contains electrolyte that is substantially free of suppressors, accelerators, or other organic plating additives, or in another embodiment, where the inorganic plating composition of the anolyte and catholyte are substantially different. Means of transferring anolyte to the catholyte or to the main plating bath by physical means (e.g. direct pumping including values, or an overflow trough) may optionally also be supplied.

The following description provides more detail of the cup and cone assembly of the clamshell. FIG. **1**B depicts a portion, **101**, of assembly **100**, including cone **103** and cup **102** in cross-section format. Note that this figure is not meant to be a true depiction of a cup and cone product assembly, but rather a stylized depiction for discussion purposes. Cup **102** is supported by top plate **105** via struts **104**, which are attached via screws **108**. Generally, cup **102** provides a support upon which wafer **145** rests. It includes an opening through which electrolyte from a plating cell can contact the wafer. Note that wafer **145** has a front side **142**, which is where plating occurs. The periphery of wafer **145** rests on the cup **102**. The cone **103** presses down on the back side of the wafer to hold it in place during plating.

To load a wafer into **101**, cone **103** is lifted from its depicted position via spindle **106** until cone **103** touches top plate **105**. From this position, a gap is created between the cup and the cone into which wafer **145** can be inserted, and thus loaded into the cup. Then cone **103** is lowered to engage the wafer against the periphery of cup **102** as depicted, and mate to a set of electrical contacts (not shown in **1**B) radially beyond the lip seal **143** along the wafer's outer periphery.

Spindle **106** transmits both vertical force for causing cone **103** to engage a wafer **145** and torque for rotating assembly **101**. These transmitted forces are indicated by the arrows in FIG. **1**B. Note that wafer plating typically occurs while the wafer is rotating (as indicated by the dashed arrows at the top of FIG. **1**B).

Cup **102** has a compressible lip seal **143**, which forms a fluid-tight seal when cone **103** engages wafer **145**. The vertical force from the cone and wafer compresses lip seal **143** to form the fluid tight seal. The lip seal prevents electrolyte from contacting the backside of wafer **145** (where it could introduce contaminating species such as copper or tin ions directly into silicon) and from contacting sensitive components of apparatus **101**. There may also be seals located between the interface of the cup and the wafer which form fluid-tight seals to further protect the backside of wafer **145** (not shown).

Cone **103** also includes a seal **149**. As shown, seal **149** is located near the edge of cone **103** and an upper region of the cup when engaged. This also protects the backside of wafer **145** from any electrolyte that might enter the clamshell from above the cup. Seal **149** may be affixed to the cone or the cup, and may be a single seal or a multi-component seal.

Upon initiation of plating, cone **103** is raised above cup **102** and wafer **145** is introduced to assembly **102**. When the wafer is initially introduced into cup **102**—typically by a robot arm—its front side, **142**, rests lightly on lip seal **143**. During plating the assembly **101** rotates in order to aid in achieving uniform plating. In subsequent figures, assembly **101** is depicted in a more simplistic format and in relation to components for controlling the hydrodynamics of electrolyte at the wafer plating surface **142** during plating. Thus, an overview of mass transfer and fluid shear at the work piece follows.

As depicted in FIG. **1**C, a plating apparatus **150** includes a plating cell **155** which houses anode **160**. In this example, electrolyte **175** is flowed into cell **155** centrally through an opening in anode **160**, and the electrolyte passes through a channeled ionically resistive element **170** having vertically oriented (non-intersecting) through holes through which electrolyte flows and then impinges on wafer **145**, which is held in, positioned and moved by, wafer holder **101**. Channeled ionically resistive elements such as **170** provide uniform impinging flow upon the wafer plating surface. In accordance with certain embodiments described herein, apparatus utilizing such channeled ionically resistive elements are configured and/or operated in a manner that facilitates high rate and high uniformity plating across the face of the wafer, including plating under high deposition rate regimes such as for WLP and TSV applications. Any or all of the various embodiments described can be implemented in the context of Damascene as well as TSV and WLP applications.

FIGS. **1**D-**1**G relate to certain techniques that may be used to encourage cross flow across the face of a substrate being plated. Various techniques described in relation to these figures present alternative strategies for encouraging cross flow. As such, certain elements described in these figures are optional, and are not present in all embodiments.

In some embodiments, electrolyte flow ports are configured to aid transverse flow, alone or in combination with a flow shaping plate and a flow diverter as described herein. Various embodiments are described below in relation to a combination with a flow shaping plate and a flow diverter, but the invention is not so limited. Note that in certain embodiments it is believed that the magnitude of the electrolyte flow vectors across the wafer surface are larger proximate the vent or gap and progressively smaller across the wafer surface, being smallest at the interior of the pseudo chamber furthest from the vent or gap. As depicted in FIG. **1**D, by using appropriately configured electrolyte flow ports, the magnitude of these transverse flow vectors is more uniform across the wafer surface.

Some embodiments include electrolyte inlet flow ports configured for transverse flow enhancement in conjunction with flow shaping plate and flow diverter assemblies. FIG. **1**E depicts a cross-section of components of a plating apparatus, **725**, for plating copper onto a wafer, **145**, which is held, positioned and rotated by wafer holder **101**. Apparatus **725** includes a plating cell, **155**, which is dual chamber cell, having an anode chamber with a copper anode, **160**, and anolyte. The anode chamber and cathode chamber are separated by a cationic membrane **740** which is supported by a support member **735**. Plating apparatus **725** includes a flow shaping plate, **410**, as described herein. A flow diverter, **325**, is on top of flow shaping plate **410**, and aides in creating

transverse shear flow as described herein. Catholyte is introduced into the cathode chamber (above membrane **740**) via flow ports **710**. From flow ports **710**, catholyte passes through flow plate **410** as described herein and produces impinging flow onto the plating surface of wafer **145**. In addition to catholyte flow ports **710**, an additional flow port, **710***a*, introduces catholyte at its exit at a position distal to the vent or gap of flow diverter **325**. In this example, flow port **710***a*'s exit is formed as a channel in flow shaping plate **410**. The functional result is that catholyte flow is introduced directly into the pseudo chamber formed between the flow plate and the wafer plating surface in order to enhance transverse flow across the wafer surface and thereby normalize the flow vectors across the wafer (and flow plate **410**).

FIG. **1**F depicts a flow diagram depicting the flow port **710***a* (from FIG. **1**E). As seen in FIG. **1**F, flow port **710***a*'s exit spans 90 degrees of the inner circumference of flow diverter **730**. One of ordinary skill in the art would appreciate that the dimensions, configuration and location of port **710***a* may vary without escaping the scope of the invention. One of skill in the art would also appreciate that equivalent configurations would include having the catholyte exit from a port or channel in flow diverter **325** and/or in combination with a channel such as depicted in FIG. **1**E (in flow plate **410**). Other embodiments include one or more ports in the (lower) side wall of a flow diverter, i.e. that side wall nearest the flow shaping plate top surface, where the one or more ports are located in a portion of the flow diverter opposite the vent or gap. FIG. **1**G depicts a flow diverter, **730**, assembled with a flow shaping plate **410**, where flow diverter **730** has catholyte flow ports, **710***b*, that supply electrolyte from the flow diverter opposite the gap of the flow diverter. Flow ports such as **710***a* and **710***b* may supply electrolyte at any angle relative to the wafer plating surface or the flow shaping plate top surface. The one or more flow ports can deliver impinging flow to the wafer surface and/or transverse (shear) flow.

In one embodiment, for example as described in relation to FIGS. **1**E-**1**G, a flow shaping plate as described herein is used in conjunction with a flow diverter, where a flow port configured for enhanced transverse flow (as described herein) is also used with the flow plate/flow diverter assembly. In one embodiment the flow shaping plate has non-uniform hole distribution, in one embodiment, a spiral hole pattern.

Terminology and Flow Paths

Numerous figures are provided to further illustrate and explain the embodiments disclosed herein. The figures include, among other things, various drawings of the structural elements and flow paths associated with a disclosed electroplating apparatus. These elements are given certain names/reference numbers, which are used consistently in describing FIGS. **2** through **22**A-**22**B.

The following embodiments assume, for the most part, that electroplating apparatus includes a separate anode chamber. The described features are contained in a cathode chamber, which includes a membrane frame **274** and membrane **202** that separate the anode chamber from the cathode chamber. Any number of possible anode and anode chamber configurations may be employed. In the following embodiments, the catholyte contained in the cathode chamber is largely located either in a cross flow manifold **226** or in the channeled ionically resistive plate manifold **208** or in channels **258** and **262** for delivering catholyte to these two separate manifolds.

Much of the focus in the following description is on controlling the catholyte in the cross flow manifold **226**. The catholyte enters the cross flow manifold **226** through two separate entry points: (1) the channels in the channeled ionically resistive plate **206** and (2) cross flow initiating structure **250**. The catholyte arriving in the cross flow manifold **226** via the channels in the CIRP **206** is directed toward the face of the work piece, typically in a substantially perpendicular direction. Such channel delivered catholyte may form small jets that impinge on the face of the work piece, which is typically rotating slowly (e.g., between about 1 to 30 rpm) with respect to the channeled plate. The catholyte arriving in the cross flow manifold **226** via the cross flow initiating structure **250** is, in contrast, directed substantially parallel to the face of the work piece.

As indicated in the discussion above, a "channeled ionically resistive plate" **206** (or "channeled ionically resistive element" or "CIRP") is positioned between the working electrode (the wafer or substrate) and the counter electrode (the anode) during plating, in order to shape the electric field and control electrolyte flow characteristics. Various figures herein show the relative position of the channeled ionically resistive plate **206** with respect to other structural features of the disclosed apparatus. One example of such an ionically resistive element **206** is described in U.S. Pat. No. 8,308,931, filed Nov. 7, 2008, which was previously incorporated by reference herein in its entirety. The channeled ionically resistive plate described therein is suitable to improve radial plating uniformity on wafer surfaces such as those containing relatively low conductivity or those containing very thin resistive seed layers. Further aspects of certain embodiments of the channeled element are described below.

A "membrane frame" **274** (sometimes referred to as an anode membrane frame in other documents) is a structural element employed in some embodiments to support a membrane **202** that separates an anode chamber from a cathode chamber. It may have other features relevant to certain embodiments disclosed herein. Particularly, with reference to the embodiments of the figures, it may include flow channels **258** and **262** for delivering catholyte toward a cross flow manifold **226** and showerhead **242** configured to deliver cross flowing catholyte to the cross flow manifold **226**. The membrane frame **274** may also contain a cell weir wall **282**, which is useful in determining and regulating the uppermost level of the catholyte. Various figures herein depict the membrane frame **274** in the context of other structural features associated with the disclosed cross flow apparatus.

Turning to FIG. **2**, the membrane frame **274** is a rigid structural member for holding a membrane **202** that is typically an ion exchange membrane responsible for separating an anode chamber from a cathode chamber. As explained, the anode chamber may contain electrolyte of a first composition while the cathode chamber contains electrolyte of a second composition. The membrane frame **274** may also include a plurality of fluidic adjustment rods **270** (sometimes referred to as flow constricting elements) which may be used to help control fluid delivery to the channeled ionically resistive element **206**. The membrane frame **274** defines the bottom-most portion of the cathode chamber and the uppermost portion of the anode chamber. The described components are all located on the work piece side of an electrochemical plating cell above the anode chamber and the anode chamber membrane **202**. They can all be viewed as being part of a cathode chamber. It should be understood, however, that certain implementations of a cross flow injec-

tion apparatus do not employ a separated anode chamber, and hence a membrane frame **274** is not essential.

Located generally between the work piece and the membrane frame **274** is the channeled ionically resistive plate **206**, as well as a cross flow ring gasket **238** and wafer cross flow confinement ring **210**, which may each be affixed to the channeled ionically resistive plate **206**. More specifically, the cross flow ring gasket **238** may be positioned directly atop the CIRP **206**, and the wafer cross flow confinement ring **210** may be positioned over the cross flow ring gasket **238** and affixed to a top surface of the channeled ionically resistive plate **206**, effectively sandwiching the gasket **238**. Various figures herein show the cross flow confinement ring **210** arranged with respect to the channeled ionically resistive plate **206**.

The upper most relevant structural feature of the present disclosure, as shown in FIG. **2**, is a work piece or wafer holder. In certain embodiments, the work piece holder may be a cup **254**, which is commonly used in cone and cup clamshell type designs such as the design embodied in Novellus Systems' Sabre® electroplating tool mentioned above. FIGS. **2** and **8**A-**8**B, for example, show the relative orientation of the cup **254** with respect to other elements of the apparatus.

In various embodiments, an edge flow element (not shown in FIG. **2**) may be provided. The edge flow element may be provided at a location that is generally above and/or within a channeled ionically resistive plate **206**, and under the cup **254**. The edge flow element is further described below.

FIG. **3**A shows a close-up cross sectional view of a cross flow inlet side according to an embodiment disclosed herein. FIG. **3**B shows a close-up cross sectional view of the cross flow outlet side according to an embodiment herein. FIG. **4** shows a cross-sectional view of a plating apparatus showing both the inlet and outlet sides, in accordance with certain embodiments herein. During a plating process, catholyte fills and occupies the region between the top of the membrane **202** on the membrane frame **274** and the membrane frame weir wall **282**. This catholyte region can be subdivided into three sub-regions: 1) a channeled ionically resistive plate manifold region **208** below the CIRP **206** and (for designs employing an anode chamber cationic membrane) above the separated-anode-chambers cationic-membrane **202** (this element is also sometimes referred to as a lower manifold region **208**), 2) the cross flow manifold region **226**, between the wafer and the upper surface of the CIRP **206**, and 3) an upper cell region or "electrolyte containment region", outside of the clamshell/cup **254** and inside the cell weir wall **282** (which is a physical part of the membrane frame **274**). When the wafer is not immersed and the clamshell/cup **254** is not in the down position, the second region and third region are combined into one region.

Region (2) above, between the top of the channeled ionically resistive plate **206** and the bottom of the workpiece when installed in the workpiece holder **254** contains catholyte and is referred to as the "cross flow manifold" **226**. In some embodiments, catholyte enters the cathode chamber via a single inlet port. In other embodiments, catholyte enters the cathode chamber through one or more ports located elsewhere in the plating cell. In some cases, there is a single inlet for the bath of the cell, peripheral to the anode chamber and cut out of the anode chamber cell walls. This inlet connects to a central catholyte inlet manifold at the base of the cell and anode chamber. In certain disclosed embodiments, that main catholyte manifold chamber feeds a plurality of catholyte chamber inlet holes (e.g., 12 catholyte chamber inlet holes). In various cases, these catholyte chamber inlet holes are divided into two groups: one group which feeds catholyte to a cross flow injection manifold **222**, and a second group which feeds catholyte to the CIRP manifold **208**. FIG. **3**B shows a cross section of a single inlet hole feeding the CIRP manifold **208** through channel **262**. The dotted line indicates the path of fluid flow.

The separation of catholyte into two different flow paths or streams occurs at the base of the cell in the central catholyte inlet manifold (not shown). That manifold is fed by a single pipe connected to the base of the cell. From the main catholyte manifold, the flow of catholyte separates into two streams: 6 of the 12 feeder holes, located on one side of the cell, lead to source the CIRP manifold region **208** and eventually supply the impinging catholyte flow through the CIRP's various microchannels. The other 6 holes also feed from the central catholyte inlet manifold, but then lead to the cross flow injection manifold **222**, which then feeds the cross flow shower head's **242** distribution holes **246** (which may number more than 100). After leaving the cross flow shower head holes **246**, the catholyte's flow direction changes from (a) normal to the wafer to (b) parallel to the wafer. This change in flow occurs as the flow impinges upon and is confined by a surface in the cross flow confinement ring **210** inlet cavity **250**. Finally, upon entering the cross flow manifold region **226**, the two catholyte flows, initially separated at the base of the cell in the central catholyte inlet manifold, are rejoined.

In the embodiments shown in the figures, a fraction of the catholyte entering the cathode chamber is provided directly to the channeled ionically resistive plate manifold **208** and a portion is provided directly to the cross flow injection manifold **222**. At least some, and often but not always all of the catholyte delivered to the channeled ionically resistive plate manifold **208** and then to the CIRP lower surface passes through the various microchannels in the plate **206** and reaches the cross flow manifold **226**. The catholyte entering the cross flow manifold **226** through the channels in the channeled ionically resistive plate **206** enters the cross flow manifold as substantially vertically directed jets (in some embodiments the channels are made at an angle, so they are not perfectly normal to the surface of the wafer, e.g., the angle of the jet may be up to about 45 degrees with respect to the wafer surface normal). The portion of the catholyte that enters the cross flow injection manifold **222** is delivered directly to the cross flow manifold **226** where it enters as a horizontally oriented cross flow below the wafer. On its way to the cross flow manifold **226**, the cross flowing catholyte passes through the cross flow injection manifold **222** and the cross flow shower head plate **242** (which, e.g., contains about 139 distributed holes **246** having a diameter of about 0.048"), and is then redirected from a vertically upwards flow to a flow parallel to the wafer surface by the actions/geometry of the cross-flow-confinement-ring's **210** entrance cavity **250**.

The absolute angles of the cross flow and the jets need not be exactly horizontal or exactly vertical or even oriented at exactly 90° with one another. In general, however, the cross flow of catholyte in the cross flow manifold **226** is generally along the direction of the work piece surface and the direction of the jets of catholyte emanating from the top surface of the microchanneled ionically resistive plate **206** generally flow towards/perpendicular to the surface of the work piece.

As mentioned, the catholyte entering the cathode chamber is divided between (i) catholyte that flows from the channeled ionically resistive plate manifold **208**, through the

channels in the CIRP **206** and then into the cross flow manifold **226** and (ii) catholyte that flows into the cross flow injection manifold **222**, through the holes **246** in the showerhead **242**, and then into the cross flow manifold **226**. The flow directly entering from the cross flow injection manifold region **222** may enter via the cross flow confinement ring entrance ports, sometimes referred to as cross flow side inlets **250**, and emanate parallel to the wafer and from one side of the cell. In contrast, the jets of fluid entering the cross flow manifold region **226** via the microchannels of the CIRP **206** enter from below the wafer and below the cross flow manifold **226**, and the jetting fluid is diverted (redirected) within the cross flow manifold **226** to flow parallel to the wafer and towards the cross flow confinement ring exit port **234**, sometimes also referred to as the cross flow outlet or outlet.

In some embodiments, the fluid entering the cathode chamber is directed into multiple channels **258** and **262** distributed around the periphery of the cathode chamber portion of the electroplating cell chamber (often a peripheral wall). In a specific embodiment, there are 12 such channels contained in the wall of the cathode chamber.

The channels in the cathode chamber walls may connect to corresponding "cross flow feed channels" in the membrane frame. Some of these feed channels **262** deliver catholyte directly to the channeled ionically resistive plate manifold **208**. As mentioned, the catholyte provided to this manifold subsequently passes through the small vertically oriented channels of the channeled ionically resistive plate **206** and enters the cross flow manifold **226** as jets of catholyte.

As mentioned, in an embodiment depicted in the figures, catholyte feeds the "CIRP manifold chamber" **208** through 6 of the 12 catholyte feeder lines/tubes. Those 6 main tubes or lines **262** feeding the CIRP manifold **208** reside below the cross flow confinement ring's exit cavity **234** (where the fluid passes out of the cross flow manifold region **226** below the wafer), and opposite all the cross flow manifold components (cross flow injection manifold **222**, showerhead **242**, and confinement ring entrance cavity **250**).

As depicted in various figures, some cross flow feed channels **258** in the membrane frame lead directly to the cross flow injection manifold **222** (e.g., 6 of 12). These cross flow feed channels **258** start at the base of the anode chamber of the cell and then pass through matching channels of the membrane frame **274** and then connect with corresponding cross flow feed channels **258** on the lower portion of the channeled ionically resistive plate **206**. See FIG. **3**A, for example.

In a specific embodiment, there are six separate feed channels **258** for delivering catholyte directly to the cross flow injection manifold **222** and then to the cross flow manifold **226**. In order to effect cross flow in the cross flow manifold **226**, these channels **258** exit into the cross flow manifold **226** in an azimuthally non-uniform manner. Specifically, they enter the cross flow manifold **226** at a particular side or azimuthal region of the cross flow manifold **226**. In a specific embodiment depicted in FIG. **3**A, the fluid paths **258** for directly delivering catholyte to the cross flow injection manifold **222** pass through four separate elements before reaching the cross flow injection manifold **222**: (1) dedicated channels in the cell's anode chamber wall, (2) dedicated channels in the membrane frame **274**, (3) dedicated channels the channeled ionically resistive element **206** (i.e., not the 1-D channels used for delivering catholyte from the CIRP manifold **208** to the cross flow manifold **226**), and finally, (4) fluid paths in the wafer cross flow confinement ring **210**.

As mentioned, the portions of the flow paths passing through the membrane frame **274** and feeding the cross flow injection manifold **222** are referred to as cross flow feed channels **258** in the membrane frame. The portions of the flow paths passing through the microchanneled ionically resistive plate **206** and feeding the CIRP manifold are referred to as cross flow feed channels **262** feeding the channeled ionically resistive plate manifold **208**, or CIRP manifold feed channels **262**. In other words, the term "cross flow feed channel" includes both the catholyte feed channels **258** feeding the cross flow injection manifold **222** and the catholyte feed channels **262** feeding the CIRP manifold **208**. One difference between these flows **258** and **262** was noted above: the direction of the flow through the CIRP **206** is initially directed at the wafer and is then turned parallel to the wafer due to the presence of the wafer and the cross flow confinement ring **210**, whereas the cross flow portion coming from the cross flow injection manifold **222** and out through the cross flow confinement ring entrance ports **250** starts substantially parallel to the wafer. While not wishing to be held to any particular model or theory, this combination and mixing of impinging and parallel flow is believed to facilitate substantially improved flow penetration within a recessed/embedded feature and thereby improve the mass transfer. By creating a spatially uniform convective flow field under the wafer and rotating the wafer, each feature, and each die, exhibits a nearly identical flow pattern over the course of the rotation and the plating process.

The flow path within the channeled ionically resistive plate **206** that does not pass through the plate's microchannels (instead entering the cross flow manifold **226** as flow parallel to the face of the wafer) begins in a vertically upward direction as it passes through the cross flow feed channel **258** in the plate **206**, and then enters a cross flow injection manifold **222** formed within the body of the channeled ionically resistive plate **206**. The cross flow injection manifold **222** is an azimuthal cavity which may be a dug out channel within the plate **206** that can distribute the fluid from the various individual feed channels **258** (e.g., from each of the individual 6 cross flow feed channels) to the various multiple flow distribution holes **246** of the cross flow shower head plate **242**. This cross flow injection manifold **222** is located along an angular section of the peripheral or edge region of the channeled ionically resistive plate **206**. See for example FIGS. **3**A and **4**-**6**. In certain embodiments, the cross flow injection manifold **222** forms a C-shaped structure over an angle of about 90 to 180° of the plate's perimeter region. In certain embodiments, the angular extent of the cross flow injection manifold **222** is about 120 to about 170°, and in a more specific embodiment is between about 140 and 150°. In these or other embodiments, the angular extent of the cross flow injection manifold **222** is at least about 90°. In many implementations, the showerhead **242** spans approximately the same angular extent as the cross flow injection manifold **222**. Further, the overall inlet structure **250** (which in many cases includes one or more of the cross flow injection manifold **222**, the showerhead **242**, the showerhead holes **246**, and an opening in the cross flow confinement ring) may span these same angular extents.

In some embodiments, the cross flow in the injection manifold **222** forms a continuous fluidically coupled cavity within the channeled ionically resistive plate **206**. In this case all of the cross flow feed channels **258** feeding the cross flow injection manifold (e.g., all 6) exit into one continuous

and connected cross flow injection manifold chamber. In other embodiments, the cross flow injection manifold **222** and/or the cross flow showerhead **242** are divided into two or more angularly distinct and completely or partially separated segments, as shown in FIG. **5** (which shows 6 separated segments). In some embodiments, the number of angularly separated segments is between about 1-12, or between about 4-6. In a specific embodiment, each of these angularly distinct segments is fluidically coupled to a separate cross flow feed channel **258** disposed in the channeled ionically resistive plate **206**. Thus, for example, there may be six angularly distinct and separated subregions within the cross flow injection manifold **222**. In certain embodiments, each of these distinct subregions of the cross flow injection manifold **222** has the same volume and/or the same angular extent.

In many cases, catholyte exits the cross flow injection manifold **222** and passes through a cross flow showerhead plate **242** having many angularly separated catholyte outlet ports (holes) **246**. See for example FIGS. **2**, **3**A-**3**B and **6**. In certain embodiments, the cross flow showerhead plate **242** is integrated into the channeled ionically resistive plate **206**, as shown in FIG. **6** for example. In some embodiments the showerhead plate **242** is glued, bolted, or otherwise affixed to the top of the cross flow injection manifold **222** of the channeled ionically resistive plate **206**. In certain embodiments, the top surface of the cross flow showerhead **242** is flush with or slightly elevated above a plane or top surface of the channeled ionically resistive plate **206**. In this manner, catholyte flowing through the cross flow injection manifold **222** may initially travel vertically upward through the showerhead holes **246** and then laterally under the cross flow confinement ring **210** and into the cross flow manifold **226** such that the catholyte enters the cross flow manifold **226** in a direction that is substantially parallel with the top face of the channeled ionically resistive plate. In other embodiments, the showerhead **242** may be oriented such that catholyte exiting the showerhead holes **246** is already traveling in a wafer-parallel direction.

In a specific embodiment, the cross flow showerhead **242** has **139** angularly separated catholyte outlet holes **246**. More generally, any number of holes that reasonably establish uniform cross flow within the cross flow manifold **226** may be employed. In certain embodiments, there are between about 50 and about 300 such catholyte outlet holes **246** in the cross flow showerhead **242**. In certain embodiments, there are between about 100 and 200 such holes. In certain embodiments, there are between about 120 and 160 such holes. Generally, the size of the individual ports or holes **246** can range from about 0.020" to 0.10", more specifically from about 0.03" to 0.06" in diameter.

In certain embodiments, these holes **246** are disposed along the entire angular extent of the cross flow showerhead **242** in an angularly uniform manner (i.e. the spacing between the holes **246** is determined by a fixed angle between the cell center and two adjacent holes). See for example FIGS. **3**A and **7**. In other embodiments, the holes **246** are distributed along the angular extent in an angularly non-uniform manner. In further embodiments, the angularly non-uniform hole distribution is nevertheless a linearly ("x" direction") uniform distribution. Put another way, in this latter case, the hole distribution is such that the holes are spaced equally far apart if projected onto an axis perpendicular to the direction of cross flow (this axis is the "x" direction). Each hole **246** is positioned at the same radial distance from the cell center, and is spaced the same distance in the "x" direction from adjacent holes. The net effect of having these angularly non-uniform holes **246** is that the overall cross flow pattern is much more uniform. These two types of arrangements for the cross flow shower head holes **246** are examined further in the Experimental section, below. See FIG. **22**B and the associated discussion below.

In certain embodiments, the direction of the catholyte exiting the cross flow showerhead **242** is further controlled by a wafer cross flow confinement ring **210**. In certain embodiments, this ring **210** extends over the full circumference of the channeled ionically resistive plate **206**. In certain embodiments, a cross section of the cross flow confinement ring **210** has an L-shape, as shown in FIGS. **3**A and **4**. In certain embodiments, the wafer cross flow confinement ring **210** contains a series of flow directing elements such as directional fins **266** in fluidic communication with the outlet holes **246** of the cross flow showerhead **242**. More specifically, the directional fins **266** define largely segregated fluid passages under an upper surface of the wafer cross flow confinement ring **210** and between adjacent directional fins **266**. In some cases, the purpose of the fins **266** is to redirect and confine flow exiting from the cross flow showerhead holes **246** from an otherwise radially inward direction to a "left to right" flow trajectory (left being the inlet side **250** of the cross flow, right being the outlet side **234**). This helps to establish a substantially linear cross flow pattern. The catholyte exiting the holes **246** of the cross flow showerhead **242** is directed by the directional fins **266** along a flow streamline caused by the orientation of the directional fins **266**. In certain embodiments, all the directional fins **266** of the wafer cross flow confinement ring **210** are parallel to one another. This parallel arrangement helps to establish a uniform cross flow direction within the cross flow manifold **226**. In various embodiments, the directional fins **266** of the wafer cross flow confinement ring **210** are disposed both along the inlet **250** and outlet **234** side of the cross flow manifold **226**. This is illustrated in the top view of FIG. **7**, for example.

As indicated, catholyte flowing in the cross flow manifold **226** passes from an inlet region **250** of the wafer cross flow confinement ring **210** to an outlet side **234** of the ring **210**, as shown in FIGS. **3**B and **4**. At the outlet side **234**, in certain embodiments, there are multiple directional fins **266** that may be parallel to and may align with the directional fins **266** on the inlet side. The cross flow passes through channels created by the directional fins **266** on the outlet side **234** and then ultimately and directly out of the cross flow manifold **226**. The flow then passes into another region of the cathode chamber generally radially outwards and beyond the wafer holder **254** and cross flow confinement ring **210**, with fluid collected and temporarily retained by the upper weir wall **282** of the membrane frame before flowing over the weir **282** for collection and recirculation. It should therefore be understood that the figures (e.g., FIGS. **3**A, **3**B and **4**) show only a partial path of the entire circuit of catholyte entering and exiting the cross flow manifold. Note that, in the embodiment depicted in FIGS. **3**B and **4**, for example, fluid exiting from the cross flow manifold **226** does not pass through small holes or back through channels analogous to the feed channels **258** on the inlet side, but rather passes outward in a generally parallel-to-the wafer direction as it is accumulated in the aforementioned accumulation region.

FIG. **6** shows a top view of the cross flow manifold **226** depicting an embedded cross flow injection manifold **222** within the channeled ionically resistive plate **206**, along with the showerhead **242** and 139 outlet holes **246**. All six fluidic adjustment rods **270** for the cross flow injection manifold flow are also shown. The cross flow confinement ring **210** is not installed in this depiction, but the outline of the cross

flow confinement ring sealing gasket **238**, which seals between the cross flow confinement ring **210** and the upper surface of the CIRP **206**, is shown. Other elements which are shown in FIG. **6** include the cross flow confinement ring fasteners **218**, membrane frame **274**, and screw holes **278** on the anode side of the CIRP **206** (which may be used for a cathodic shielding insert, for example).

In some embodiments, the geometry of the cross flow confinement ring outlet **234** may be tuned in order to further optimize the cross flow pattern. For example, a case in which the cross flow pattern diverges to the edge of the confinement ring **210** may be corrected by reducing the open area in the outer regions of the cross flow confinement ring outlet **234**. In certain embodiments, the outlet manifold **234** may include separated sections or ports, much like the cross flow injection manifold **222**. In some embodiments, the number of outlet sections is between about 1-12, or between about 4-6. The ports are azimuthally separated, occupying different (usually adjacent) positions along the outlet manifold **234**. The relative flow rates through each of the ports may be independently controlled in some cases. This control may be achieved, for example, by using control rods **270** similar to the control rods described in relation to the inlet flow. In another embodiment, the flow through the different sections of the outlet can be controlled by the geometry of the outlet manifold. For example, an outlet manifold that has less open area near each side edge and more open area near the center would result in a solution flow pattern where more flow exits near the center of the outlet and less flow exits near the edges of the outlet. Other methods of controlling the relative flow rates through the ports in the outlet manifold **234** may be used as well (e.g., pumps, etc.).

As mentioned, bulk catholyte entering the catholyte chamber is directed separately into the cross flow injection manifold **222** and the channeled ionically resistive plate manifold **208** through multiple channels **258** and **262**, e.g., 12 separate channels. In certain embodiments, the flows through these individual channels **258** and **262** are independently controlled from one another by an appropriate mechanism. In some embodiments, this mechanism involves separate pumps for delivering fluid into the individual channels. In other embodiments, a single pump is used to feed a main catholyte manifold, and various flow restriction elements that are adjustable may be provided in one or more of the channels feeding the flow path provided so as to modulate the relative flows between the various channels **258** and **262** and between the cross flow injection manifold **222** and CIRP manifold **208** regions and/or along the angular periphery of the cell. In various embodiments depicted in the figures, one or more fluidic adjustment rods **270** (sometimes also referred to as flow control elements) are deployed in the channels where independent control is provided. In the depicted embodiments, the fluidic adjustment rod **270** provides an annular space in which catholyte is constricted during its flow toward the cross flow injection manifold **222** or the channeled ionically resistive plate manifold **208**. In a fully retracted state, the fluidic adjustment rod **270** provides essentially no resistance to flow. In a fully engaged state, the fluidic adjustment rod **270** provides maximal resistance to flow, and in some implementations stops all flow through the channel. In intermediate states or positions, the rod **270** allows intermediate levels of constriction of the flow as fluid flows through a restricted annular space between the channel's inner diameter and the fluid adjustment rod's outer diameter.

In some embodiments, the adjustment of the fluidic adjustment rods **270** allows the operator or controller of the electroplating cell to favor flow to either the cross flow injection manifold **222** or to the channeled ionically resistive plate manifold **208**. In certain embodiments, independent adjustment of the fluidics adjustment rods **270** in the channels **258** that deliver catholyte directly to the cross flow injection manifold **222** allows the operator or controller to control the azimuthal component of fluid flow into the cross flow manifold **226**. The effect of these adjustments are discussed further in the Experimental section below.

FIGS. **8**A-**8**B show cross sectional views of a cross flow injection manifold **222** and corresponding cross flow inlet **250** relative to a plating cup **254**. The position of the cross flow inlet **250** is defined, at least in part, by the position of the cross flow confinement ring **210**. Specifically, the inlet **250** may be considered to begin where the cross flow confinement ring **210** terminates. Note that in the case of an initial design, seen in FIG. **8**A, the confinement ring **210** termination point (and inlet **250** commencement point) was under the edge of the wafer, whereas in a revised design, seen in FIG. **8**B, the termination/commencement point is under the plating cup and further radially outward from the wafer edge, as compared to the initial design. Also, the cross flow injection manifold **222** in the earlier design had a step in the cross flow ring cavity (where the generally leftward arrow begins rising upwards) which potentially formed some unwanted turbulence near that point of fluid entry into the cross flow manifold region **226**. In some cases, an edge flow element (not shown) may be present proximate the periphery of the substrate and/or the periphery of the channeled ionically resistive plate. The edge flow element may be present proximate the inlet **250** and/or proximate the outlet (not shown in FIGS. **8**A and **8**B). The edge flow element may be used to direct electrolyte into a corner that forms between the plating face of the substrate and the edge of the cup **254**, thereby counteracting an otherwise relatively low cross-flow in this region.

The disclosed apparatus may be configured to perform the methods described herein. A suitable apparatus includes hardware as described and shown herein and one or more controllers having instructions for controlling process operations in accordance with the present invention. The apparatus will include one or more controllers for controlling, inter alia, the positioning of the wafer in the cup **254** and cone, the positioning of the wafer with respect to the channeled ionically resistive plate **206**, the rotation of the wafer, the delivery of catholyte into the cross flow manifold **226**, delivery of catholyte into the CIRP manifold **208**, delivery of catholyte into the cross flow injection manifold **222**, the resistance/position of the fluidic adjustment rods **270**, the delivery of current to the anode and wafer and any other electrodes, the mixing of electrolyte components, the timing of electrolyte delivery, inlet pressure, plating cell pressure, plating cell temperature, wafer temperature, position of an edge flow element, and other parameters of a particular process performed by a process tool.

A system controller will typically include one or more memory devices and one or more processors configured to execute the instructions so that the apparatus will perform a method in accordance with the present invention. The processor may include a central processing unit (CPU) or computer, analog and/or digital input/output connections, stepper motor controller boards, and other like components. Machine-readable media containing instructions for controlling process operations in accordance with the present invention may be coupled to the system controller. Instructions for implementing appropriate control operations are executed on the processor. These instructions may be stored

on the memory devices associated with the controller or they may be provided over a network. In certain embodiments, the system controller executes system control software . . . .

System control software may be configured in any suitable way. For example, various process tool component subroutines or control objects may be written to control operation of the process tool components necessary to carry out various process tool processes. System control software may be coded in any suitable computer readable programming language.

In some embodiments, system control software includes input/output control (IOC) sequencing instructions for controlling the various parameters described above. For example, each phase of an electroplating process may include one or more instructions for execution by the system controller. The instructions for setting process conditions for an immersion process phase may be included in a corresponding immersion recipe phase. In some embodiments, the electroplating recipe phases may be sequentially arranged, so that all instructions for an electroplating process phase are executed concurrently with that process phase.

Other computer software and/or programs may be employed in some embodiments. Examples of programs or sections of programs for this purpose include a substrate positioning program, an electrolyte composition control program, a pressure control program, a heater control program, and a potential/current power supply control program.

In some cases, the controllers control one or more of the following functions: wafer immersion (translation, tilt, rotation), fluid transfer between tanks, etc. The wafer immersion may be controlled by, for example, directing the wafer lift assembly, wafer tilt assembly and wafer rotation assembly to move as desired. The controller may control the fluid transfer between tanks by, for example, directing certain valves to be opened or closed and certain pumps to turn on and off. The controllers may control these aspects based on sensor output (e.g., when current, current density, potential, pressure, etc. reach a certain threshold), the timing of an operation (e.g., opening valves at certain times in a process) or based on received instructions from a user.

The apparatus/process described hereinabove may be used in conjunction with lithographic patterning tools or processes, for example, for the fabrication or manufacture of semiconductor devices, displays, LEDs, photovoltaic panels and the like. Typically, though not necessarily, such tools/processes will be used or conducted together in a common fabrication facility. Lithographic patterning of a film typically comprises some or all of the following steps, each step enabled with a number of possible tools: (1) application of photoresist on a workpiece, i.e., substrate, using a spin-on or spray-on tool; (2) curing of photoresist using a hot plate or furnace or UV curing tool; (3) exposing the photoresist to visible or UV or x-ray light with a tool such as a wafer stepper; (4) developing the resist so as to selectively remove resist and thereby pattern it using a tool such as a wet bench; (5) transferring the resist pattern into an underlying film or workpiece by using a dry or plasma-assisted etching tool; and (6) removing the resist using a tool such as an RF or microwave plasma resist stripper.

Features of a Channeled Ionically Resistive Element

Electrical Function

In certain embodiments, the channeled ionically resistive element **206** approximates a nearly constant and uniform current source in the proximity of the substrate (cathode) and, as such, may be referred to as a high resistance virtual anode (HRVA) in some contexts. As noted above, this element may also be referred to as a channeled ionically resistive plate (CIRP). Normally, the CIRP **206** is placed in close proximity with respect to the wafer. In contrast, an anode in the same close-proximity to the substrate would be significantly less apt to supply a nearly constant current to the wafer, but would merely support a constant potential plane at the anode metal surface, thereby allowing the current to be greatest where the net resistance from the anode plane to the terminus (e.g., to peripheral contact points on the wafer) is smaller. So while the channeled ionically resistive element **206** has been referred to as a high-resistance virtual anode (HRVA), this does not imply that electrochemically the two are interchangeable. Under the best operational conditions, the CIRP **206** would more closely approximate and perhaps be better described as a virtual uniform current source, with nearly constant current being sourced from across the upper plane of the CIRP **206**. While the CIRP is certainly viewable as a "virtual current source", i.e. it is a plane from which the current is emanating, and therefore can be considered a "virtual anode" because it can be viewed as a location or source from which anodic current emanates, it is the relatively high-ionic-resistance of the CIRP **206** (with respect to the electrolyte) that leads the nearly uniform current across its face and to further advantageous, generally superior wafer uniformity when compared to having a metallic anode located at the same physical location. The plate's resistance to ionic current flow increases with increasing specific resistance of electrolyte contained within the various channels of the plate **206** (often but not always having the same or nearly similar resistance of the catholyte), increased plate thickness, and reduced porosity (less fractional cross sectional area for current passage, for example, by having fewer holes of the same diameter, or the same number of holes with smaller diameters, etc.).

Structure

The CIRP **206** contains micro size (typically less than 0.04") through-holes that are spatially and ionically isolated from each other and do not form interconnecting channels within the body of CIRP, in many but not all implementations. Such through-holes are often referred to as non-communicating through-holes. They typically extend in one dimension, often, but not necessarily, normal to the plated surface of the wafer (in some embodiments the non-communicating holes are at an angle with respect to the wafer which is generally parallel to the CIRP front surface). Often the through-holes are parallel to one another. Often the holes are arranged in a square array. Other times the layout is in an offset spiral pattern. These through-holes are distinct from 3-D porous networks, where the channels extend in three dimensions and form interconnecting pore structures, because the through-holes restructure both ionic current flow and fluid flow parallel to the surface therein, and straighten the path of both current and fluid flow towards the wafer surface. However, in certain embodiments, such a porous plate, having an interconnected network of pores, may be used in place of the 1-D channeled element (CIRP). When the distance from the plate's top surface to the wafer is small (e.g., a gap of about 1/10 the size of the wafer radius, for example less than about 5 mm), divergence of both current flow and fluid flow is locally restricted, imparted and aligned with the CIRP channels.

One example CIRP **206** is a disc made of a solid, non-porous dielectric material that is ionically and electrically resistive. The material is also chemically stable in the plating solution of use. In certain cases the CIRP **206** is made of a ceramic material (e.g., aluminum oxide, stannic

oxide, titanium oxide, or mixtures of metal oxides) or a plastic material (e.g., polyethylene, polypropylene, polyvinylidene difluoride (PVDF), polytetrafluoroethylene, polysulphone, polyvinyl chloride (PVC), polycarbonate, and the like), having between about 6,000-12,000 non-communicating through-holes. The disc **206**, in many embodiments, is substantially coextensive with the wafer (e.g., the CIRP disc **206** has a diameter of about 300 mm when used with a 300 mm wafer) and resides in close proximity to the wafer, e.g., just below the wafer in a wafer-facing-down electroplating apparatus. Preferably, the plated surface of the wafer resides within about 10 mm, more preferably within about 5 mm of the closest CIRP surface. To this end, the top surface of the channeled ionically resistive plate **206** may be flat or substantially flat. Often, both the top and bottom surfaces of the channeled ionically resistive plate **206** are flat or substantially flat.

Another feature of the CIRP **206** is the diameter or principal dimension of the through-holes and its relation to the distance between the CIRP **206** and the substrate. In certain embodiments, the diameter of each through-hole (or of a majority of through-holes, or the average diameter of the through-holes) is no more than about the distance from the plated wafer surface to the closest surface of the CIRP **206**. Thus, in such embodiments, the diameter or principal dimension of the through holes should not exceed about 5 mm, when the CIRP **206** is placed within about 5 mm of the plated wafer surface.

As above, the overall ionic and flow resistance of the plate **206** is dependent on the thickness of the plate and both the overall porosity (fraction of area available for flow through the plate) and the size/diameter of the holes. Plates of lower porosities will have higher impinging flow velocities and ionic resistances. Comparing plates of the same porosity, one having smaller diameter 1-D holes (and therefore a larger number of 1-D holes) will have a more micro-uniform distribution of current on the wafer because there are more individual current sources, which act more as point sources that can spread over the same gap, and will also have a higher total pressure drop (high viscous flow resistance).

In certain cases, however, the ionically resistive plate **206** is porous, as mentioned above. The pores in the plate **206** may not form independent 1-D channels, but may instead form a mesh of through holes which may or may not interconnect. It should be understood that as used herein, the terms channeled ionically resistive plate and channeled ionically resistive element (CIRP) are intended to include this embodiment, unless otherwise noted.

In a number of embodiments, the CIRP **206** may be modified to include (or accommodate) an edge flow element. The edge flow element may be an integral part of the CIRP **206** (e.g., the CIRP and edge flow element together form a monolithic structure), or it may be a replaceable part installed on or near the CIRP **206**. The edge flow element promotes a higher degree of cross-flow, and hence shear on the substrate surface, near the edge of the substrate (e.g., near an interface between the substrate and the substrate holder). Without an edge flow element, an area of relatively low cross-flow may develop near the interface of the substrate and substrate holder, for example due to the geometry of substrate and substrate holder, and the direction of electrolyte flow. The edge flow element may act to increase cross-flow in this area, thereby promoting more uniform plating results across the substrate. Further details related to the edge flow element are presented below.

Vertical Flow through the Through-Holes

The presence of an ionically resistive but ionically permeable element (CIRP) **206** close to the wafer substantially reduces the terminal effect and improves radial plating uniformity in certain applications where terminal effects are operative/relevant, such as when the resistance of electrical current in the wafer seed layer is large relative to that in the catholyte of the cell. The CIRP **206** also simultaneously provides the ability to have a substantially spatially-uniform impinging flow of electrolyte directed upwards at the wafer surface by acting as a flow diffusing manifold plate. Importantly, if the same element **206** is placed farther from the wafer, the uniformity of ionic current and flow improvements become significantly less pronounced or non-existent.

Further, because non-communicating through-holes do not allow for lateral movement of ionic current or fluid motion within the CIRP, the center-to-edge current and flow movements are blocked within the CIRP **206**, leading to further improvement in radial plating uniformity. In the embodiment shown in FIG. **9**, the CIRP **206** is a perforated plate having approximately 9000 uniformly spaced one-dimensional holes acting as microchannels and arranged in a square array (i.e., the holes are arranged in columns and rows) over the face of the plate (e.g., over a substantially circular area having a diameter of about 300 mm in the case of plating a 300 mm wafer) and with an effective average porosity of about 4.5%, and an individual microchannel hole size of about 0.67 mm (0.026 inches) in diameter. Also shown in FIG. **9** are the flow distribution adjustment rods **270**, which may be used to preferentially direct flow to enter the cross flow manifold **226** either through the CIRP manifold **208** and up through the holes in the CIRP **206**, or in through the cross flow injection manifold **222** and cross flow showerhead **242**. The cross flow confinement ring **210** is fitted on top of the CIRP, which is supported by the membrane frame **274**.

It is noted that in some embodiments, the CIRP plate **206** can be used primarily or exclusively as an intra-cell electrolyte flow resistive, flow controlling and therebyflow shaping element, sometimes referred to as a turboplate. This designation may be used regardless of whether or not the plate **206** tailors radial deposition uniformity by, for example, balancing terminal effects and/or modulating the electric field or kinetic resistances of plating additives coupled with the flow within the cell. Thus, for example, in TSV and WLP electroplating, where the seed metal thickness is generally large (e.g. >1000 Å thick) and metal is being deposited at very high rates, uniform distribution of electrolyte flow is very important, while radial non-uniformity control arising from ohmic voltage drop within the wafer seed may be less necessary to compensate for (at least in part because the center-to-edge non-uniformities are less severe where thicker seed layers are used). Therefore the CIRP plate **206** can be referred to as both an ionically resistive ionically permeable element, and as a flow shaping element, and can serve a deposition-rate corrective function by either altering the flow of ionic current, altering the convective flow of material, or both.

Distance between Wafer and Channeled Plate

In certain embodiments, a wafer holder **254** and associated positioning mechanism hold a rotating wafer very close to the parallel upper surface of the channeled ionically resistive element **206**. During plating, the substrate is generally positioned such that it is parallel or substantially parallel to the ionically resistive element (e.g., within about 10°). Though the substrate may have certain features thereon, only the generally planar shape of the substrate is

considered in determining whether the substrate and ionically resistive element are substantially parallel.

In typical cases, the separation distance is about 0.5-10 millimeters, or about 2-8 millimeters. In some cases, the separation distance is about 2 mm or less, for example about 1 mm or less. This small plate to wafer distance can create a plating pattern on the wafer associated with proximity "imaging" of individual holes of the pattern, particularly near the center of wafer rotation. In such circumstances, a pattern of plating rings (in thickness or plated texture) may result near the wafer center. To avoid this phenomenon, in some embodiments, the individual holes in the CIRP **206** (particularly at and near the wafer center) can be constructed to have a particularly small size, for example less than about $\frac{1}{5}^{th}$ the plate to wafer gap. When coupled with wafer rotation, the small pore size allows for time averaging of the flow velocity of impinging fluid coming up as a jet from the plate **206** and reduces or avoids small scale non-uniformities (e.g., those on the order of micrometers). Despite the above precaution, and depending on the properties of the plating bath used (e.g. particular metal deposited, conductivities, and bath additives employed), in some cases deposition may be prone to occur in a micro-non-uniform pattern (e.g., forming center rings) as the time average exposure and proximity-imaging-pattern of varying thickness (for example, in the shape of a "bulls eye" around the wafer center) and corresponding to the individual hole pattern used. This can occur if the finite hole pattern creates an impinging flow pattern that is non-uniform and influences the deposition. In this case, introducing lateral flow across the wafer center, and/or modifying the regular pattern of holes right at and/or near the center, have both been found to largely eliminate any sign of micro-non-uniformities otherwise found there.

Porosity of Channeled Plate

In various embodiments, the channeled ionically resistive plate **206** has a sufficiently low porosity and pore size to provide a viscous flow resistance backpressure and high vertical impinging flow rates at normal operating volumetric flow rates. In some cases, about 1-10% of the channeled ionically resistive plate **206** is open area allowing fluid to reach the wafer surface. In particular embodiments, about 2-5% the plate **206** is open area. In a specific example, the open area of the plate **206** is about 3.2% and the effective total open cross sectional area is about 23 cm^2.

Hole Size of Channeled Plate

The porosity of the channeled ionically resistive plate **206** can be implemented in many different ways. In various embodiments, it is implemented with many vertical holes of small diameter. In some cases the plate **206** does not consist of individual "drilled" holes, but is created by a sintered plate of continuously porous material. Examples of such sintered plates are described in U.S. Pat. No. 6,964,792, which is herein incorporated by reference in its entirety. In some embodiments, drilled non-communicating holes have a diameter of about 0.01 to 0.05 inches. In some cases, the holes have a diameter of about 0.02 to 0.03 inches. As mentioned above, in various embodiments the holes have a diameter that is at most about 0.2 times the gap distance between the channeled ionically resistive plate **206** and the wafer. The holes are generally circular in cross section, but need not be. Further, to ease construction, all holes in the plate **206** may have the same diameter. However this need not be the case, and both the individual size and local density of holes may vary over the plate surface as specific requirements may dictate.

As an example, a solid plate **206** made of a suitable ceramic or plastic material (generally a dielectric insulating and mechanically robust material), having a large number of small holes provided therein, e.g. at least about 1000 or at least about 3000 or at least about 5000 or at least about 6000 (9465 holes of 0.026 inches diameter has been found useful). As mentioned, some designs have about 9000 holes. The porosity of the plate **206** is typically less than about 5 percent so that the total flow rate necessary to create a high impinging velocity is not too great. Using smaller holes helps to create a large pressure drop across the plate as compared to larger holes, aiding in creating a more uniform upward velocity through the plate.

Generally, the distribution of holes over the channeled ionically resistive plate **206** is of uniform density and non-random. In some cases, however, the density of holes may vary, particularly in the radial direction. In a specific embodiment, as described more fully below, there is a greater density and/or diameter of holes in the region of the plate that directs flow toward the center of the rotating substrate. Further, in some embodiments, the holes directing electrolyte at or near the center of the rotating wafer may induce flow at a non-right angle with respect to the wafer surface. Further, the hole patterns in this region may have a random or partially random distribution of non-uniform plating "rings" to address possible interaction between a limited number of holes and the wafer rotation. In some embodiments, the hole density proximate an open segment of a flow diverter or confinement ring **210** is lower than on regions of the channeled ionically resistive plate **206** that are farther from the open segment of the attached flow diverter or confinement ring **210**.

Edge Flow Element

In many implementations, electroplating results may be improved through the use of an edge flow element and/or a flow insert. Generally speaking, an edge flow element affects the flow distribution near the periphery of the substrate, proximate the interface between the substrate and substrate holder. In some embodiments, the edge flow element may be integral with a CIRP. In some other embodiments, the edge flow element may be integral with a substrate holder. In yet other embodiments, the edge flow element may be a separate piece that can be installed on a CIRP or substrate holder. The edge flow element may be used to tune the flow distribution near the edge of the substrate, as is desired for a particular application. Advantageously, the flow element promotes a high degree of cross-flow near the periphery of the substrate, thereby promoting more uniform (from center to edge of the substrate), high quality electroplating results. An edge flow element is typically positioned, at least partially, radially inside of the inner edge of the substrate holder/the periphery of the substrate. In some cases, an edge flow element may be at least partially positioned at other locations, for example under the substrate holder and/or radially outside of the substrate holder, as described further below. In a number of drawings herein, the edge flow element is referred to as the "flow element."

The edge flow element may be made of various materials. In some cases, the edge flow element may be made of the same material as the CIRP and/or the substrate holder. Generally speaking, it is desirable for the material of the edge flow element to be electrically insulating.

Another method for improving cross-flow near the periphery of the substrate is to use a high rate of substrate rotation. However, fast substrate rotation presents its own set of disadvantages, and in various embodiments may be avoided. For example, where the substrate is rotated too quickly, it

can prevent formation of an adequate cross-flow across the substrate surface. In certain embodiments, therefore, the substrate may be rotated at a rate between about 50-300 RPM, for example between about 100-200 RPM. Similarly, cross-flow near the periphery of the substrate can be promoted by using a relatively smaller gap between the CIRP and the substrate. However, smaller CIRP-substrate gaps result in electroplating processes that are more sensitive and have tighter tolerance ranges for process variables.

FIG. **13**A presents experimental results showing bump height vs. radial position on the substrate for patterned substrates electroplated without an edge flow element. FIG. **13**B presents experimental results showing within-die non-uniformity vs. radial position on the substrate for the patterned substrates described in relation to FIG. **13**A. Notably, the bump height decreases toward the edge of the substrate. Without wishing to be bound by theory or mechanism of action, it is believed that this low bump height is a result of relatively low electrolyte flow near the periphery of the substrate. The poor convection conditions near the substrate-substrate holder interface lead to a lower local metal concentration, which leads to a reduced plating rate. Further, photoresist is often thicker near the edge of a substrate, and this increased photoresist thickness leads to deeper features, for which it is more difficult to achieve adequate convection, thereby leading to a lower plating rate at the edge of the substrate. As shown in FIG. **13**B, this decreasing plating rate/decreased bump height near the edge of the substrate corresponds with an increase in within-die non-uniformity. The within-die non-uniformity was calculated as the ((max bump height in a die)−(min bump height in the die))/(2*average bump height in the die).

FIG. **14**A depicts the structure of an electroplating apparatus near the periphery of the substrate **1400** at the outlet side of the apparatus. Electrolyte exits the cross-flow manifold **1402** by flowing over the CIRP **1404** and under the substrate **1400**, and out under the substrate holder **1406**, as shown by the arrows. In this example, the CIRP **1404** has a substantially flat portion that sits under the substrate **1400**. At the edge of this region, near the interface between the substrate **1400** and substrate holder **1406**, the CIRP **1404** angles downward, then flattens out again. FIG. **14**B depicts a graph presenting modeling results related to the flow distribution between the substrate **1400** and the CIRP **1404** in the region shown in FIG. **14**A.

The modeling results show the predicted shear velocity at a location 0.25 mm from the surface of the substrate. Notably, the shear flow decreases dramatically near the edge of the substrate.

FIG. **15** depicts experimental results related to bump height vs. radial position on the substrate, and modeling results showing the shear flow vs. radial position on the substrate (on the electrolyte outlet side). In this example, the substrate was not rotated during plating. The experimental bump height results followed the same trend as the predicted shear velocity, indicating that the lower shear velocity likely plays a role in low edge bump height.

FIG. **16**A depicts experimental results showing within-die non-uniformity vs. radial position on the substrate. FIG. **16**B depicts experimental results showing the thickness of photoresist vs. radial position on the substrate. Together, FIGS. **16**A and **16**B suggest there is a strong correlation between photoresist thickness and within-die non-uniformity, with higher resist thickness and non-uniformity being found near the edge of the substrate.

FIG. **17**A illustrates a cross-sectional view of an electroplating cell having an edge flow element **1710** installed therein. The edge flow element **1710** is situated under the edge of the substrate **1700**, proximate the interface between the substrate **1700** and substrate holder **1706**. In this example, the CIRP **1704** is shaped to include a raised plateau region which is nearly coextensive with the substrate **1700**. In certain embodiments, an edge flow element **1710** may be positioned, wholly or partially, radially outside of the raised portion of the CIRP **1704**. The edge flow element **1710** may also be positioned, wholly or partially, on the raised portion of the CIRP **1704**. Electrolyte flows through the cross-flow manifold **1702** as shown by the arrows. A flow diverter **1708** helps shape the path through which the electrolyte flows. The flow diverter **1708** is shaped differently at the inlet side (where the cross-flow originates) compared to the outlet side to promote cross-flow across the surface of the substrate.

As shown in FIG. **17**A, electrolyte enters the cross flow manifold **1702** on the inlet side of the electroplating cell. The electrolyte flows around the edge flow element **1710**, through the cross flow manifold **1702**, around the edge flow element **1710** a second time, and out through an outlet. As mentioned above, electrolyte also enters the cross flow manifold **1702** by traveling upwards through holes in the CIRP **1704**. One purpose of the edge flow element **1710** is to increase convection at the interface between the substrate **1700** and the substrate holder **1706**. This interface is shown in greater detail in FIG. **17**B. Without the use of an edge flow element **1710**, the convection in the region shown in the dotted circle is undesirably low. The edge flow element **1710** affects the flow path of electrolyte near the edge of the substrate **1700**, promoting greater convection in the region shown in the dotted circle. This helps overcome low convection and low plating rates near the substrate edge. This may also help combat differences that arise due to differing photoresist/feature height, as explained in relation to FIGS. **16**A and **16**B.

In certain embodiments, the edge flow element **1710** may be shaped such that the cross flow in the cross flow manifold **1702** is directed more favorably into the corner formed by the substrate **1700** and substrate holder **1706**. A variety of shapes may be used to achieve this purpose.

FIGS. **18**A-**18**C depict three available configurations for installing an edge flow element **1810** in an electroplating cell. Various other configurations may be used, as well. Regardless of the exact configuration, the edge flow element **1810** may be shaped like a ring or arc in many cases, though FIGS. **18**A-**18**C only show a cross-sectional view of one side of the edge flow element **1810**. In the first configuration (Type **1**, FIG. **18**A), the edge flow element **1810** is attached to the CIRP **1804**. The edge flow element **1810** in this example does not include any flow bypass for electrolyte to flow between the edge flow element **1810** and the CIRP **1804**. As such, all the electrolyte flows over the edge flow element **1810**. In the second configuration (Type **2**, FIG. **18**B), the edge flow element **1810** is attached to the CIRP **1804** and includes a flow bypass between the edge flow element and the CIRP. The flow bypass is formed by passages in the edge flow element **1810**. These passages permit some amount of electrolyte to flow through the edge flow element **1810** (between the upper corner of the edge flow element **1810** and the CIRP **1804**). In the third configuration (Type **3**, FIG. **18**C), the edge flow element **1810** is attached to the substrate holder **1806**. In this example, electrolyte may flow between the edge flow element **1810** and the CIRP **1804**. Further, passages in the edge flow element **1810** permit flow of electrolyte through the edge flow element **1810**, very near the interface between the substrate **1800** and the substrate holder **1806**. FIG. **18**D

presents a table summarizing some of the features of the edge flow elements shown in FIGS. **18**A-**18**C.

FIGS. **19**A-**19**E present examples for different methods of achieving adjustability in an edge flow element **1910**. In some embodiments, the edge flow element **1910** may be installed at a fixed location, e.g., on the CIRP **1904**, and have a fixed geometry, as shown in FIG. **19**A. However, in many other cases, there may be additional flexibility in the way the edge flow element is installed/used. For example, in some cases the position/shape of the edge flow element may be adjusted (manually or automatically), either between electroplating processes (e.g., to tune a particular plating process, as desired, compared to other plating processes), or within an electroplating process (e.g., to tune plating parameters over time within a single plating process).

In one example, shims may be used to adjust the position (and to some degree shape) of an edge flow element. For instance, a series of shims may be provided, with shims of various heights for different applications and desired flow patterns/characteristics. The shims may be installed between the CIRP and the edge flow element to raise the height of the edge flow element, thereby reducing the distance between the edge flow element and the substrate/substrate holder. In some cases, the shims may be used in an azimuthally asymmetric way, thereby achieving a different edge flow element height at different azimuthal locations. The same result can be achieved using screws (as shown by element **1912** in FIGS. **19**B and **19**C) or other mechanical features to position the flow shaping element. FIGS. **19**B and **19**C illustrate two embodiments where screws **1912** may be used to control the position of the edge flow element **1910**. As with the shims, the screws **1912** (located at different positions along the edge flow element **1910**) may be positioned in a way that results in azimuthally asymmetric positioning of the edge flow element **1910** (e.g., by positioning the screws **1912** at different heights). In each of FIGS. **19**B and **19**C, the edge flow element **1910** is shown at two different positions. In FIG. **19**B, the edge flow element changes between the two (or more) positions by rotating about a pivot point. In FIG. **19**C, the edge flow element changes between the two (or more) positions by moving the edge flow element in a linear manner. Additional screws or other positioning mechanisms may be provided for extra support.

In some implementations, the position and/or shape of the edge flow element **1910** may be dynamically adjusted during a plating process, for example using electric or pneumatic actuators. FIGS. **19**D and **19**E present embodiments where the edge flow element **1910** can by dynamically moved, even during an electroplating process, using a rotary actuator **1913** (FIG. **19**D) or a linear actuator **1915** (FIG. **19**E). Such adjustments allow for precise control of the electrolyte flow over time, thereby allowing a high degree of tunability and promoting high quality plating results.

Returning to FIG. **18**D, the first and second configurations shown in FIGS. **18**A and **18**B, respectively, allow for the edge flow element **1810** to be azimuthally asymmetric because the edge flow element **1810** is attached to the CIRP **1804** (which typically does not rotate during plating). The asymmetry may relate to differences in shape between portions of the edge flow element **1810** that are positioned near the inlet side of the electroplating cell vs. portions of the edge flow element that are positioned elsewhere, for example near the outlet side of the electroplating cell. Such azimuthal asymmetries may be used to combat non-uniformities that arise due to the way electrolyte cross-flows across the substrate surface during electroplating. Such asymmetry may relate to differences in a number of characteristics in the shape of the edge flow element **1810**, for example height, width, roundness/sharpness of edges, presence of flow bypass passages, vertical position, horizontal/radial position, etc. The third configuration shown in FIG. **18**C, being installed on the substrate holder **1806**, may also be azimuthally asymmetric. However, because in many embodiments the substrate **1800** and substrate holder **1806** rotate during electroplating, any asymmetry in the edge flow element **1810** would likely average-out due to the fact that the edge flow element **1810** rotates with the substrate **1800** during electroplating (at least in cases where the edge flow element is attached to the substrate holder **1806**, as in the embodiment of FIG. **18**C). As such, it is generally not as beneficial to have an azimuthally asymmetric edge flow element when the edge flow element is attached to, and rotates with, the substrate holder. For this reason, FIG. **18**D lists "No*" in relation to azimuthal asymmetry for the third configuration. All of the configurations described are considered to be within the scope of the present embodiments.

FIGS. **20**A-**20**C illustrate a number of ways in which the edge flow element **2010** may be azimuthally asymmetric. FIGS. **20**A-**20**C depict top views of an edge flow element **2010** positioned in an electroplating cell, for example on a CIRP **2004**. Other attachment methods may also be used, as discussed above. In each example, the cross-sectional shape of the edge flow element **2010** is shown. In FIG. **20**A, the edge flow element **2010** is azimuthally symmetric and extends around the entire perimeter of the substrate. Here, the edge flow element **2010** has a triangular cross-section, with the tallest portion positioned toward the inside edge of the edge flow element **2010**. In FIG. **20**B, the edge flow element is azimuthally asymmetric and extends around the entire perimeter of the edge flow element **2010**. Here, the azimuthal asymmetry results because the edge flow element has a first cross-sectional shape (e.g., triangular) near the electrolyte inlet, and a second cross-sectional shape (e.g., rounded pillar) near the electrolyte outlet (positioned opposite the inlet).

In similar embodiments, any combination of cross-sectional shapes may be used. Generally speaking, the cross-sectional shapes may be any shapes including, but not limited to, triangular, square, rectangular, circular, ellipsoidal, rounded, curved, pointed, trapezoidal, corrugated, hourglass shaped, etc. Flow through passages may or may not be provided through the edge flow element **2010** itself. In another similar embodiment, the cross-sectional shapes may be similar, but of varying sizes around the periphery, thus introducing the azimuthal asymmetry. Likewise, the cross-sectional shapes may be the same or similar, but positioned at different vertical and/or horizontal locations with respect to the substrate/substrate holder and/or CIRP **2004**. The transition to different cross-sectional shapes may be abrupt or gradual. In FIG. **20**C, the edge flow element **2010** is only present at certain azimuthal locations. Here, the edge flow element **2010** is only present on the downstream (outlet) side of the plating cell. In a similar embodiment, the edge flow element may only be present on the upstream (inlet) side of the plating cell. Azimuthally asymmetric edge flow elements may be particularly advantageous for tuning electroplating results to overcome any asymmetries that may arise as a result of cross-flowing electrolyte. This helps promote uniform, high quality plating results. As should be apparent, the azimuthal asymmetry may result from azimuthal variations in edge flow element shape, dimensions (e.g., height and/or width), position with respect to the substrate edge, bypass region presence or configuration, and the like.

With respect to FIG. **20**C, in certain embodiments an arc-shaped edge flow element **2010** may extend at least about 60°, at least about 90°, at least about 120°, at least about 150°, at least about 180°, at least about 210°, at least about 240°, at least about 270°, or at least about 300° proximate the periphery of the substrate. In these or other embodiments, the arc-shaped edge flow element may extend no more than about 90°, no more than about 120°, no more than about 150°, no more than about 180°, no more than about 210°, no more than about 240°, no more than about 270°, no more than about 300°, or no more than about 330°. The center of the arc may be positioned proximate the inlet area, the outlet area (opposite the inlet area), or at some other location offset from the inlet/outlet areas. In certain other embodiments where azimuthal asymmetries are used, the arc shapes described in this paragraph may correspond to the size of a region exhibiting such asymmetry. For example, a ring-shaped edge flow element may have an azimuthal asymmetry as a result of having different shim heights installed at different positions along the edge flow element, as explained with reference to FIG. **22** (further described below), for instance. In some such embodiments, a region having relatively thicker or thinner shims (thus resulting in a relatively taller or shorter edge flow element, respectively, after installation) may span an arc having any of the minimum and/or maximum dimensions described above. In one example, a region having relatively larger shims spans at least about 60°, and no more than about 150°. Any combination of the listed arc dimensions may be used, and the azimuthal asymmetry present may be any type of asymmetry described herein.

FIG. **21** depicts a cross-sectional view of an electroplating cell having an edge flow element **2110** installed therein. In this example, the edge flow element **2110** is positioned radially outside of the raised plateau portion of the CIRP **2104**. The shape of the edge flow element **2110** allows electrolyte near the inlet to travel upwards at an angle to reach the cross flow manifold **2102**, and similarly, allows electrolyte near the outlet to travel downwards at an angle to exit the cross flow manifold **2102**. As shown in FIGS. **19**A-**19**E, the uppermost portion of the edge flow element may extend above the plane of the raised portion of the CIRP. In other cases, the uppermost portion of the edge flow element may be flush with the raised portion of the CIRP **2104**. In some cases, the position of the edge flow element is adjustable, as described elsewhere herein. The shape and position of the edge flow element **2110** may promote a higher degree of cross-flow near the corner formed between the substrate **2100** and substrate holder **2106**.

FIG. **22**A illustrates a cross-sectional view of a CIRP **2204** and edge flow element **2210**. In this example, the edge flow element **2210** is a removable piece that fits into a groove **2216** in the CIRP **2204**. FIG. **22**B provides an additional view of the edge flow element **2210** and CIRP **2204** shown in FIG. **22**A. In this embodiment, the edge flow element **2210** is held in place on the CIRP **2204** using up to 12 screws, which provides 12 individual locations for tuning the height/position of the edge flow element **2210**. In similar embodiments, any number of screws/adjustment/attachment points may be used. The CIRP **2204** may include a second groove **2217**, which may provide an outlet for the electrolyte to exit from the cross-flow manifold, thereby promoting cross-flowing electrolyte. The edge flow element **2210** is secured into the groove **2216** in the CIRP **2204** using a series of screws (not shown in FIGS. **22**A and **22**B).

FIG. **22**C provides modeling results related to the x-direction velocity of cross-flow as electrolyte exits the cross-flow manifold. Also shown in FIG. **22**C, a series of shims **2218** may be used (in this example, shim washers that fit around the screws **2212** that secure the edge flow element **2210** into the groove **2216** in the CIRP **2204**) to adjust the height of the edge flow element **2210** at individual locations around the edge flow element **2210**. The height of the shim is labeled H. These heights may be adjusted independently to achieve an azimuthally asymmetric distance between the top of the edge flow element **2210** and the substrate (not shown). In this example, the edge flow element **2210** is positioned such that an inner edge of the edge flow element **2210** extends to a height/position that is above the raised portion of the CIRP **2204**, as shown in the black circle.

In some embodiments, the vertical distance between the uppermost part of an edge flow element and the uppermost portion of a CIRP may be between about 0-5 mm, for example between about 0-1 mm. In these or other cases, this distance may be at least about 0.1 mm, or at least about 0.25 mm, at one or more locations on the edge flow element. The vertical distance between the uppermost part of an edge flow element and the substrate may be between about 0.5-5 mm, in some cases between about 1-2 mm. In various embodiments, the distance between the uppermost part of an edge flow element and the uppermost portion of the CIRP is between about 10-90% of the distance between the raised portion of the CIRP and the substrate surface, in some cases between about 25-50%. The "uppermost portion of the CIRP" referenced in this paragraph excludes the edge flow element itself (e.g., in cases where the edge flow element is integral with the CIRP). Typically, the uppermost portion of the CIRP is an upper surface of the CIRP, positioned opposite the substrate in the cross-flow manifold. In various embodiments, as shown in FIG. **21**, the CIRP includes a raised plateau portion. The "uppermost portion of the CIRP" in such embodiments is the raised plateau portion of the CIRP. In embodiments where the CIRP includes a series of protuberances thereon, the top of the protuberances corresponds to the "uppermost portion of the CIRP." Only regions of the CIRP that are directly under the substrate are considered when determining what is the uppermost portion of the CIRP.

Returning to the embodiment of FIG. **22**C, without the shims **2218** (or with appropriately thin shims **2218**), the top of the edge flow element **2210** may be about coplanar with the raised portion of the CIRP **2204**. In one particular embodiment, the edge flow element **2210** is as shown in FIG. **22**C, and the shims **2218** are provided in an azimuthally asymmetric way such that near the inlet side of the electroplating cell, the top of the edge flow element **2210** is about coplanar with, or below, the raised portion of the CIRP **2204** (e.g., no shims, fewer shims, and/or thinner shims are provided near the inlet) and near the outlet side of the electroplating cell, the top of the edge flow element **2210** is above, though radially outside of, the raised portion of the CIRP **2204** (e.g., more shims and/or thicker shims are provided near the outlet compared to the inlet).

Notably, the flow in the corner formed between the substrate **2200** and the substrate holder **2206** is somewhat low, but is improved compared to the case where no edge flow element **2210** is provided.

FIG. **22**D depicts modeling results showing the x-direction velocity of cross-flow (i.e., flow in the horizontal direction) near the substrate vs. radial location on the substrate for several different shim thicknesses using the setup shown in FIG. **22**C. The height of the shim has a strong effect on the velocity of cross-flow near the edge of the substrate. Generally speaking, the thicker the shim, the

higher the velocity of cross-flow near the edge of the substrate. This increase in cross-flow near the periphery of the substrate may compensate for the low plating rate that is typically achieved near the substrate edge (e.g., as a result of apparatus geometry and/or photoresist thickness, as described above). These differences allow for the modulation/tunability of the edge flow profile by simply changing the height of the shims at relevant locations.

In certain embodiments, the edge flow element has a width (measured as the difference between the outer radius and the inner radius) between about 0.1-50 mm. In some such cases, this width is at least about 0.01 mm or at least about 0.25 mm. Typically, at least a portion of this width is positioned radially interior of the inner edge of the substrate holder. The height of the edge flow element depends in large part upon the geometry of the remaining parts of the electroplating apparatus, for example the height of the cross-flow manifold. Further, the height of the edge flow element depends on how this element is installed in an electroplating apparatus, and the accommodations made in other pieces of equipment (e.g., grooves machined into the CIRP). In certain implementations, an edge flow element may have a height that is between about 0.1-5 mm, or between about 1-2 mm. Where shims are used, they can be provided at a variety of thicknesses. These thicknesses are also dependent upon the geometry of the plating apparatus and the accommodations made in the CIRP or other portion of the apparatus for securing the edge flow element therein. For example, if the edge flow element fits into a groove in the CIRP, as shown in FIGS. **22**A and **22**B, relatively thicker shims may be needed if the groove in the CIRP is relatively deeper. In some embodiments, the shims may have thicknesses between about 0.25-4 mm, or between about 0.5-1.5 mm.

In terms of position, the edge flow element is typically positioned such that at least a portion of the edge flow element is radially interior of the inner edge of the substrate support. In many cases this means that the edge flow element is positioned such that at least a portion of the edge flow element is radially interior of the edge of the substrate itself. The horizontal distance by which the edge flow element extends inward from the inner edge of the substrate support may in certain embodiments be at least about 1 mm, or at least about 5 mm, or at least about 10 mm, or at least about 20 mm. In some embodiments, this distance is about 30 mm or less, for example about 20 mm or less, about 10 mm or less, or about 2 mm or less. In these or other embodiments, the horizontal distance by which the edge flow element extends radially outward from the inner edge of the substrate support may be at least about 1 mm, or at least about 10 mm. Generally, there is no upper limit for the distance by which the edge flow element extends radially outward from the inner edge of the substrate support, so long as the edge flow element can fit in the electroplating apparatus.

FIG. **23**A depicts modeling results for electrolyte flow where an edge flow element having a ramp-shape is used. In FIG. **23**A, the shaded area relates to the area through which electrolyte flows. The different shades indicate the rate at which electrolyte is flowing. The white space above the shaded area corresponds to the substrate and substrate holder (for example as labeled in FIG. **22**C). The white space below the shaded area corresponds to the CIRP and the edge flow element. For this example, the edge flow element may be any shape that, together with the CIRP, results in a flow path having the shape shown in FIG. **23**A. In some cases, the edge flow element may simply be the edge of the CIRP. In FIG. **23**A, the CIRP/edge flow element together result in a ramp shape near the interface between the substrate and substrate holder. The ramp has a ramp height, shown in the figure, which extends above the raised portion of the CIRP. The ramp has a maximum height that is located radially inside of the interface between the edge of the substrate and the substrate holder. In some embodiments, the ramp height may be between about 0.25-5 mm, for example between about 0.5-1.5 mm. A horizontal distance between the maximum height of the ramp and the inner edge of the substrate holder (labeled in FIG. **23**A as the "Ramp Inset from Cup") may be between about 1-10 mm, for example between about 2-5 mm. A horizontal distance between the inner edge of the substrate holder and the beginning of the ramp (labeled in FIG. **23**A as the "Inner Ramp Width" may be between about 1-30 mm, for example between about 5-10 mm. A horizontal distance between the beginning of the ramp and the end of the ramp (labeled in FIG. **23**A as the "Total Ramp Width" may be between about 5-50 mm, for example between about 10-20 mm. The average angle at which the ramp is inclined on the inner edge of the ramp may be between about 10-80 degrees. The average angle at which the ramp is declined on the outer edge of the ramp may be between about 10-80 degrees, for example between about 40-50 degrees. The top of the ramp may be a sharp angle, or it may be smooth, as shown.

FIG. **23**B depicts modeling results illustrating flow velocity vs. radial position on the substrate for different ramp heights. Higher ramp heights result in higher velocity flow. Higher ramp heights also correlate with more significant pressure drops.

FIG. **24**A depicts modeling results related to another type of edge flow element. In this example, the edge flow element (which, like the one in FIG. **23**A, may be a separate piece that attaches to the CIRP, or may be integral with the CIRP), includes a flow bypass that allows electrolyte to flow through passages in the edge flow element. The length of the flow bypass passage is labeled "Length," and the height of the flow bypass passage is labeled "Bypass height." The "Ramp Height" refers to the vertical distance between the top of the flow bypass passage and the top of the ramp. In certain embodiments, the flow bypass passage may have a minimum length of at least about 1 mm, or at least about 5 mm, and/or a maximum length of about 2 mm, or about 20 mm. The height of the flow bypass passage may be at least about 0.1 mm, or at least about 4 mm. In these or other cases the height of the flow bypass passage may be about 1 mm or less, or about 8 mm or less. In some embodiments, the height of the flow bypass passage may be between about 10-50% the distance between the CIRP (e.g., the raised portion of the CIRP, if present) and the substrate (this distance is also the height of the cross-flow manifold). Similarly, the height of the ramp may be between about 10-90% the distance between the CIRP and the substrate. This may correspond to a ramp height of at least about 0.2 mm, or at least about 4.5 mm in some cases. In these or other cases, the ramp height may be about 6 mm or less, for example about 1 mm or less.

FIG. **24**B depicts modeling results that were run using different values for the parameters labeled in FIG. **24**A. Notably, the results show that these geometrical parameters may be varied to tune the flow near the edge of the substrate, thereby achieving a desired flow pattern for any given application. It is not necessary to distinguish between the different cases shown in this graph. Instead, the results are relevant for showing that many different flow patterns may be achieved by varying the geometry of the edge flow element.

FIG. **25** presents flow modeling results related to an edge flow element **2510** that is positioned in the corner formed between the substrate **2500** and substrate holder **2506**. In this example, the edge flow element **2510** includes flow bypass passages to allow electrolyte to flow, as shown. Notably, electrolyte can flow between the CIRP **2504** and the edge flow element **2510**, and also between the edge flow element **2510** and the substrate **2500**/substrate holder **2506**. In one example, the edge flow element may be attached directly to the substrate holder, as described in relation to FIG. **18**C. In another example, the edge flow element may be attached directly to the CIRP, as described in relation to FIG. **18**B.

FIGS. **26**A-**26**D depict several examples of edge flow inserts according to various embodiments. Only a portion of the edge flow element is shown in each case. These edge flow elements may be installed in an electroplating cell by attaching them to the CIRP, for example within a groove as described in relation to FIG. **22**A. The edge flow elements shown in FIGS. **26**A-**26**D are fabricated to have different heights, different flow bypass passage heights, different angles, different degrees of azimuthal symmetry/asymmetry, etc. One type of asymmetry that is easily visible in the edge flow elements of FIGS. **26**A and **26**B is that at certain azimuthal positions, no flow bypass passages are present and the electrolyte must travel all the way over the uppermost portion of the edge flow element at these locations to exit the electroplating cell. At other positions on the edge flow element, flow bypass passages are present, allowing electrolyte to flow both over and under the uppermost portion of the edge flow element. In certain embodiments, an edge flow element includes portion(s) that have flow bypass passages and portion(s) that do not have flow bypass passages, the different portions being positioned at different azimuthal locations, as depicted in FIGS. **26**A and **26**B. The edge flow element may be installed in an electroplating apparatus such that the portion(s) having the flow bypass passages is aligned with either or both of the inlet/outlet areas of the electroplating cell. In some embodiments, the edge flow element may be installed in an electroplating apparatus such that the portion(s) lacking the flow bypass passages are aligned with either or both of the inlet/outlet areas of the electroplating cell.

Another way in which the edge flow element may be azimuthally asymmetric is by providing flow bypass passages of different dimensions at different locations on the edge flow element. For example, the flow bypass passages near the inlet and/or outlet may be wider or narrower, or taller or shorter, than flow bypass passages farther away from the inlet and/or outlet. Similarly, the flow bypass passages near the inlet may be wider or narrower, or taller or shorter, than flow bypass passages near the outlet. In these or other cases, the space between adjacent flow bypass passages may be non-uniform. In some embodiments, the flow bypass passages may be closer together (or farther apart) near the inlet and/or outlet regions, compared to regions that are farther away from the inlet and/or outlet. Similarly, the flow bypass passages may be closer together (or farther apart) near the inlet area compared to the outlet area. The shape of the flow bypass passages may also be azimuthally asymmetric, for example to promote cross-flow. One way to accomplish this in certain implementations may be to use flow bypass passages that are, to some degree, aligned with the direction of cross-flow. In some embodiments, the height of the edge flow element is azimuthally asymmetric. The relatively higher portions may be aligned with an inlet and/or outlet side of the electroplating apparatus in some embodiments. This same result can be accomplished using an edge flow element having an azimuthally symmetric height, installed onto a CIRP using shims of varying heights.

While it is understood that electrolyte may exit the electroplating cell at many positions, the "outlet area" of the electroplating cell is understood to be the area opposite the inlet (where the cross-flowing electrolyte originates, not considering electrolyte which enters the cross-flow manifold through holes in the CIRP). In other words, the inlet corresponds to the upstream area, where the cross-flow substantially originates, and the outlet corresponds to the downstream area that is opposite the upstream area.

FIGS. **27**A-**27**C present the experimental setup used for a number of experiments described in relation to FIGS. **28**-**30**. In this series of tests, an edge flow element **2710** was installed in a CIRP **2704** at varying heights at different positions. Four different setups were used, labeled in FIG. **27**A as A, B, C, and D. Shims of varying heights were used to position the edge flow element **2710** at the different heights. As shown in FIG. **27**A, the edge flow element **2710** was conceptually divided into an upstream portion **2710***a* (between about the 9 o'clock position and the 3 o'clock position) and a downstream portion **2710***b* (between about the 4 o'clock position and the 8 o'clock position). The upstream portion **2710***a* of the edge flow element **2710** was aligned with the inlet to the cross flow manifold (e.g., the center of the inlet was positioned at about the 12 o'clock position). The different setups tested are described in the table in FIG. **27**B. In FIG. **27**A, it should be understood that the CIRP **2710** is generally much longer/wider than shown in the bottom portion of the figure.

The table in FIG. **27**B describes three gap heights relevant to the experimental setup. The first gap height (the wafer-CIRP gap) corresponds to the distance between the substrate surface and the raised portion of the CIRP. This is the height of the cross-flow manifold. The second gap height (the upstream gap) corresponds to the distance between the substrate and the topmost portion of the edge flow element for the upstream portion of the edge flow element. Similarly, the third gap height (the downstream gap) corresponds to the distance between the substrate and the topmost portion of the edge flow element for the downstream portion of the edge flow element. In setup A, the upstream gap and downstream gap are each the same size as the substrate-CIRP gap. Here, the top of the edge flow element is flush with the raised portion of the CIRP. In setup B, the upstream and downstream gaps are equal, and are both smaller than the substrate-CIRP gap. In this example, the edge flow element extends to a position that is higher than the raised portion of the CIRP in an azimuthally symmetric way. In setup C, the upstream gap is the same size as the substrate-CIRP gap, while the downstream gap is smaller. In this example, the edge flow element is flush with the raised portion of the CIRP at the upstream locations on the edge flow element, and is higher than the raised portion of the CIRP at downstream locations of the edge flow element. Setup D is similar to setup C, with an even smaller downstream gap. Smaller gaps between the edge flow element and the substrate are a result of using larger shims between the edge flow element and the CIRP. FIG. **27**C depicts modeling results related to the cross-flow velocity of electrolyte at different locations. This figure shows geometry of the basic experimental setup in relation to FIGS. **27**A and **27**B.

FIG. **28** presents experimental results related to setups A and B described in relation to FIGS. **27**A-**27**C. For this experiment, the substrate was not rotated during electroplating. The graph in FIG. **28** illustrates plated bump height vs.

radial position on the substrate. The results indicate that setup B resulted in substantially more uniform bump height near the edge of the substrate compared to setup A. This suggests that raising the edge flow element above the plane of the raised portion of the CIRP can have substantial benefits on plating uniformity.

FIG. **29** presents experimental data related to setups A-D described in relation to FIGS. **27**A-**27**C. The graph illustrates within-die non-uniformity vs. radial position on the substrate. Lower degrees of non-uniformity are desired. In various embodiments, there may be a goal of <5% within-die non-uniformity. The D setup performed best (lowest non-uniformity). The B and C setups also performed better than the A setup. As such, it is believed that there are particular benefits to raising an edge flow element above the plane of the raised CIRP, particularly (but not necessarily exclusively) at downstream locations on the edge flow element.

FIG. **30** presents experimental results depicting plated bump height vs. radial position on the substrate for setups A-D described in relation to FIGS. **27**A-**27**C. Setup D resulted in the most uniform edge profile, with the lowest within-die non-uniformity. The "WiD" values shown in FIG. **30** relate to the within-die thickness non-uniformities that were observed on the substrates after plating.

It is to be understood that the configurations and/or approaches described herein are exemplary in nature, and that these specific embodiments or examples are not to be considered in a limiting sense, because numerous variations are possible. The specific routines or methods described herein may represent one or more of any number of processing strategies. As such, various acts illustrated may be performed in the sequence illustrated, in other sequences, in parallel, or in some cases omitted. Likewise, the order of the above described processes may be changed.

The subject matter of the present disclosure includes all novel and nonobvious combinations and sub-combinations of the various processes, systems and configurations, and other features, functions, acts, and/or properties disclosed herein, as well as any and all equivalents thereof.

Additional Examples

A few observations that suggest that improved cross flow through the cross flow manifold **226** is desirable are presented in this section. Throughout this section, two basic plating cell designs are tested. Both designs contain a confinement ring **210**, sometimes referred to as a flow diverter, defining a cross flow manifold **226** on top of the channeled ionically resistive plate **206**. Neither design includes an edge flow element, though such an element may be added to either setup, as desired. The first design, sometimes referred to as the control design and/or the TC1 design, does not include a side inlet to this cross flow manifold **226**. Instead, in the control design, all flow into the cross flow manifold **226** originates below the CIRP **206** and travels up through the holes in the CIRP **206** before impinging on the wafer and flowing across the face of the substrate. The second design, sometimes referred to as the second design and/or the TC2 design, includes a cross flow injection manifold **222** and all associated hardware for injecting fluid directly into the cross flow manifold **226** without passing through the channels or pores in the CIRP **206** (note that in some cases, however, the flow delivered to the cross flow injection manifold passes through dedicated channels near the periphery of the CIRP **206**, such channels being distinct/separate from the channels used to direct fluid from the CIRP manifold **208** to the cross flow manifold **226**).

FIGS. **10**A and **10**B through FIGS. **12**A and **12**B compare the flow patterns achieved using a control plating cell having no side inlet (**10**A, **11**A, and **12**A) vs. a second plating cell having a side inlet to the cross flow manifold **10**B, **11**B, and **12**B).

FIG. **10**A shows a top-down view of part of a control design plating apparatus. Specifically, the figure shows a CIRP **206** with a flow diverter **210**. FIG. **10**B shows a top-down view of part of the second plating apparatus, specifically showing the CIRP **206**, flow diverter **210** and cross flow injection manifold **222**/cross flow manifold inlet **250**/cross flow showerhead **242**. The direction of flow in FIGS. **10**A-**10**B is generally left to right, towards the outlet **234** on the flow diverter **210**. The designs shown in FIGS. **10**A-**10**B correspond to the designs modeled in FIGS. **11**A-**11**B through **12**A-**12**B.

FIG. **11**A shows the flow through the cross flow manifold **226** for the control design. In this case, all the flow in the cross flow manifold **226** originates from below the CIRP **206**. The magnitude of the flow at a particular point is indicated by the size of the arrows. In the control design of FIG. **11**A, the magnitude of the flow increases substantially throughout the cross flow manifold **226** as additional fluid passes through the CIRP **206**, impinges upon the wafer, and joins the cross flow. In the current design of FIG. **11**B, however, this increase in flow is much less substantial. The increase is not as great because a certain amount of fluid is delivered directly into the cross flow manifold **226** through the cross flow injection manifold **222** and associated hardware.

FIG. **12**A depicts the horizontal velocity across the face of a substrate plated in the control design apparatus shown in FIG. **10**A. Notably, the flow velocity starts at zero (at the position opposite the flow diverter outlet) and increases until reaching the outlet **234**. Unfortunately, the average flow at the center of the wafer is relatively low in the control embodiments. As a consequence, the jets of catholyte emitted from the channels of the channeled ionically resistive plate **206** predominate hydrodynamically in the center region. The problem is not as pronounced towards the edge regions of the work piece because the rotation of the wafer creates an azimuthally averaged cross flow experience.

FIG. **12**B depicts the horizontal velocity across the face of a substrate plated in the current design shown in FIG. **10**B. In this case, the horizontal velocity starts at the inlet **250** at a non-zero value due to the fluid injected from the cross flow injection manifold **222**, through the side inlet **250** and into the cross flow manifold **226**. Further, the flow rate at the center of the wafer is increased in the current design, as compared to the control design, thereby reducing or eliminating the region of low cross flow near the center of the wafer where the impinging jets may otherwise dominate. Thus, the side inlet substantially improves the uniformity of cross flow rates along the inlet-to-outlet direction, and will result in more uniform plating thickness.

Other Embodiments

While the above is a full description of the specific embodiments, various modifications, alternative constructions and equivalents may be used. Therefore, the above description and illustrations should not be taken as limiting the scope of the present invention which is defined by the appended claims.

What is claimed is:

1. An electroplating apparatus comprising:

(a) an electroplating chamber configured to contain an electrolyte and an anode while electroplating metal onto a substrate, the substrate being substantially planar;

(b) a substrate holder configured to hold the substrate such that a plating face of the substrate is separated from the anode during electroplating, wherein when the substrate is positioned in the substrate holder, a corner forms at the interface between the substrate and substrate holder, the corner defined on top by the plating face of the substrate and on the side by the substrate holder;

(c) an ionically resistive element including a substrate-facing surface that is separated from the plating face of the substrate by a gap of about 10 mm or less, wherein the ionically resistive element is at least coextensive with the plating face of the substrate during electroplating, the ionically resistive element adapted to provide ionic transport through the ionically resistive element during electroplating;

(d) an inlet to the gap for introducing electrolyte to the gap;

(e) an outlet to the gap for receiving electrolyte flowing in the gap; and

(f) an edge flow element configured to direct electrolyte into the corner at the interface between the substrate and the substrate holder, the edge flow element being arc-shaped or ring-shaped and positioned proximate a periphery of the substrate and at least partially radially inside of the corner at the interface between the substrate and the substrate holder,

wherein the inlet and outlet are positioned proximate azimuthally opposing perimeter locations on the plating face of the substrate during electroplating, and

wherein the inlet and outlet are adapted to generate cross-flowing electrolyte in the gap to create or maintain a shearing force on the plating face of the substrate during electroplating.

2. The apparatus of claim 1, wherein the edge flow element is configured to attach to the ionically resistive element and/or to the substrate holder.

3. The apparatus of claim 1, wherein the edge flow element is integral with the ionically resistive element and comprises a raised portion proximate the periphery of the ionically resistive element, the raised portion being raised with respect to a height of a remaining portion of the substrate-facing surface of the ionically resistive element, the remaining portion of the substrate-facing surface being positioned radially interior of the raised portion.

4. The apparatus of claim 2, wherein the ionically resistive element comprises a groove into which the edge flow element is installed.

5. The apparatus of claim 4, further comprising one or more shims positioned between the ionically resistive element and the edge flow element.

6. The apparatus of claim 5, wherein the one or more shims result in the edge flow element being positioned in a manner that is azimuthally asymmetric.

7. The apparatus of claim 1, wherein the edge flow element is azimuthally asymmetric with respect to one or more of (a) position (b) shape, and/or (c) presence or shape of flow bypass passages.

8. The apparatus of claim 7, wherein the edge flow element comprises at least a first portion and a second portion, the portions being defined based on an azimuthal asymmetry in the edge flow element, wherein the first portion is centered near the inlet to the gap or the outlet to the gap.

9. The apparatus of claim 1, wherein the edge flow element comprises flow bypass passages that allow electrolyte to flow through the edge flow element.

10. The apparatus of claim 1, wherein the edge flow element is ring-shaped.

11. The apparatus of claim 1, wherein the edge flow element is arc-shaped.

12. The apparatus of claim 1, wherein a position of the edge flow element with respect to the ionically resistive element is adjustable.

13. The apparatus of claim 12, further comprising shims and/or screws for adjusting the position of the edge flow element with respect to a position of the ionically resistive element.

14. The apparatus of claim 12, further comprising an actuator for adjusting the position of the edge flow element with respect to a position of the ionically resistive element, wherein the actuator permits the position of the edge flow element to be adjusted during electroplating.

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